Attention as Gauge-Theoretic Variational Inference

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Abstract

Transformer attention is variational inference over information sources. Modeling each token as a Gaussian agent on a statistical fiber bundle, we derive the generalized attention weight $\beta_{ij} = \mathrm{softmax}(-D_{\mathrm{KL}}[q_i \| \Omega_{ij} q_j]/\tau)$ from a single variational principle: minimizing the free energy of a mixture-of-sources generative model in which each agent infers, via gauge transport, which neighbor generated its state. The KL divergence arises exactly; the softmax follows from constrained optimization over the categorical source-selection posterior.

Under specific limits of this principle, standard transformer architectural choices are recovered as special cases of the variational geometry: the temperature scaling $1/\sqrt{d_k}$ from dimensional concentration of the KL divergence, layer normalization as the geometric condition for frame-independent inference, multi-head attention as a block-diagonal restriction of the gauge algebra, and causal masking and positional biases from non-uniform attention priors π_j. Two successive limits (isotropic covariances, flat gauge connection) recover the standard rule $\beta_{ij} \propto \mathrm{softmax}(Q_i K_j^\top/\sqrt{d_k})$, while the $\mathrm{GL}(K)$ invariance of KL divergence implies that W_Q, W_K are themselves gauge transformations with $W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$, suggesting that standard transformer attention can be interpreted as a degenerate case of gauge-theoretic attention operating in this limit.

We validate the unreduced theory by training a $\mathrm{GL}(K)$ gauge transformer on WikiText-103 using only KL divergences, gauge transport, and natural gradient dynamics—without learned attention projections, MLPs, or pointwise activation functions. The model achieves test perplexity 71.6, outperforming a standard transformer at matched embedding dimension by 1.66× (PPL 118.6 at $d_{\mathrm{model}} = 90$) while approaching a parameter-matched standard transformer (PPL 48.5 at $d_{\mathrm{model}} = 1{,}280$). The learned gauge frames develop categorical separation in a low-dimensional subspace with within-category diversification in the bulk. A separate consistency check on frozen BERT across 105 passages confirms quantitative agreement between KL-based and dot-product attention in the degenerate limit (grand mean $r = 0.804$, 95% CI $[0.771, 0.838]$). Both architectures operate in the flat bundle limit with trivially vanishing holonomy, a geometric regime we hypothesize is matched to the approximately path-independent semantic transport that compositional language requires.

Keywords: gauge theory, free energy principle, transformer attention, variational inference, information geometry, natural gradient, symmetry breaking, multi-agent systems, $\mathrm{GL}(K)$ invariance

1 Introduction

Language is a dynamic informational system: speakers and writers encode beliefs under uncertainty, listeners decode them, and the statistical regularities of this process reflect the structure of communication itself. Language modeling is a snapshot of this dynamic system, and the architectures that excel at it should reflect the mathematical structure of information exchange under uncertainty.

Recent advances in neuroscience and intelligent systems have independently converged on the idea that intelligent systems integrate perception, inference, and communication under the constraints of uncertainty (Bahdanau et al., 2014; Amari, 1998; Bronstein et al., 2021). Friston's Free Energy Principle (FEP) provides a general variational formulation of inference in cognitive systems (Friston, 2010; Parr et al., 2022; Friston et al., 2017; **?**), whereas the attention mechanism in modern machine learning architectures defines a powerful (although empirically derived) rule for token prediction (Clark et al., 2019; Vaswani et al., 2017). Despite their shared reliance on probabilistic inference and pairwise interaction, these two frameworks remain stubbornly separated. While several mathematical perspectives on attention have been developed (discussed below), a unified geometric framework from which the full suite of transformer architectural choices emerges as consequences of a single variational principle has been lacking.

Many varieties of transformer and attention architectures have been proposed, implemented, and studied in recent years (Fuchs et al., 2020; Thomas et al., 2018). Geometric deep learning approaches impose equivariance constraints using group-theoretic structure (Kondor and Trivedi, 2018; Finzi et al., 2020; Bronstein et al., 2021), but these typically engineer symmetry into architectures rather than deriving attention from a variational principle. Attempts at curved space token embeddings have produced mixed results (Bonnabel, 2013; Absil et al., 2008).

Several complementary mathematical perspectives on attention have been developed. Tsai et al. (2019) and Katharopoulos et al. (2020) interpret attention through the lens of kernel methods, showing that softmax attention approximates a kernel function and can be linearized. Ramsauer et al. (2021) demonstrate that attention implements the update rule of modern Hopfield networks, providing an energy-based perspective. The information bottleneck principle (Tishby and Zaslavsky, 2015; Shwartz-Ziv and Tishby, 2017) offers a related information-theoretic view of layer-wise representation learning. On the side of backpropagation-free learning, Hinton (2022) proposes the forward-forward algorithm as an alternative to backpropagation. In neuroscience, predictive coding (Rao and Ballard, 1999; Bogacz, 2017; Millidge et al., 2021) implements variational free energy minimization through local prediction errors, sharing the variational foundation of our framework but lacking the gauge-theoretic structure that connects to attention mechanisms. Each of these perspectives illuminates a different aspect of attention. Our work aims to unify them within a single geometric framework in which attention, temperature scaling, layer normalization, and training dynamics are all recovered as special cases or limits of one variational principle on a statistical fiber bundle.

In this report we propose a first-principles, gauge-equivariant framework that connects these domains. Our framework is based on a principal bundle geometry whereby each agent is modeled as a smooth local section of an associated bundle with statistical manifold fibers over a "noumenal" base manifold. Inter-agent communication arises naturally through a (generally) non-abelian gauge connection that defines parallel-transport operators between agents' local gauge frames. Within this geometry, attention emerges as a gauge-aligned Kullback–Leibler (KL) term derived directly from the variational free energy of a coupled multi-agent generative model based on mixture Gaussians.

We then show that in the flat-bundle, isotropic, delta-function limit, the gauge-covariant attention reduces to the standard transformer dot-product attention QK^T, establishing standard attention as a degenerate limit of the gauge-theoretic framework. We validate the unreduced theory by training a $\mathrm{GL}(K)$ gauge transformer on WikiText-103 using only KL divergences, gauge transport, and natural gradient dynamics—without learned attention projections, MLPs, or pointwise activation functions—demonstrating that gauge-covariant variational inference is a viable computational substrate for language modeling. We separately confirm the degenerate-limit predictions on frozen pretrained transformers (Supplementary Appendix E).

Finally, we show that gradient descent on the gauge-equivariant free energy, in the deterministic constant-gauge limit, recovers the standard transformer training update. This provides a variational interpretation of the training loss: the free energy functional maps onto the loss plus regularization. We emphasize that the chain rule is a general property of differentiable function composition; what the framework contributes is an interpretation of the loss function and its gradients as variational quantities, not a derivation of backpropagation itself.

2 Methods

We begin with a general geometric construction that naturally supports hierarchical emergence of meta-agents and scale interactions. The details of the most general geometry and framework can be found in the appendix and references (Nakahara, 2003; Kullback and Leibler, 1951; Sternberg, 1994; Fulton and Harris, 1991; Frankel, 2011; Baez and Muniain, 1994).

Briefly, in this current report, we describe agents as modeled by a local section of an associated bundle to a principal G fiber bundle whereby tokens are represented by agents occupying a single 0-dimensional base manifold. Although the base manifold is 0-dimensional and carries no intrinsic ordering or metric, the discrete index set of agents (tokens) may be endowed with additional structure such as a sequential ordering or a distance function that enters the theory only through the attention prior π_j (Section 3.4.5) and/or through position-dependent gauge frames (Section 4.5). This separation between the geometric base (which determines gauge connections and curvature) and the combinatorial index structure (which determines masking and positional biases) is important: causal masking and distance-dependent priors impose structure on which agents communicate, not on the underlying fiber geometry. We shall show that this framework enables a unified description of both intra-agent inference and inter-agent communication in terms of gauge frame transport and alignment via a generalized free energy principle.

Notation and glossary. Table 1 summarizes the principal symbols used throughout this work.

2.1 An intuitive simplification for non-geometers

For interdisciplinary readers, it may be helpful to visualize each agent in this framework as a vector and a matrix field defined over a finite spatial region.

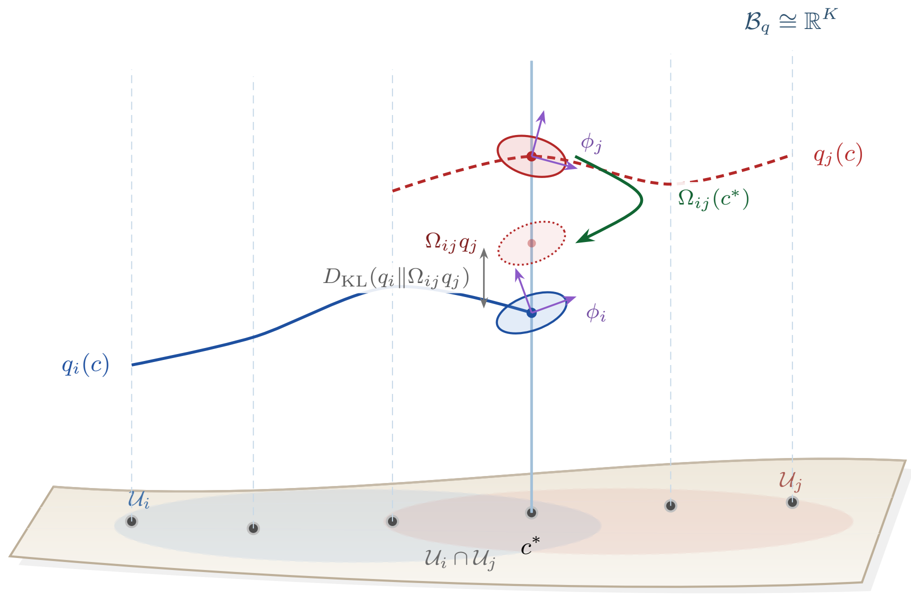


Figure 1: Agent belief sections over the associated bundle $\mathcal{E}_q = \mathcal{N} \times_{\rho_q} \mathcal{B}_q$. The shaded surface is the base manifold $\mathcal{C}$; vertical dashed lines are belief fibers $\mathcal{B}_q \cong \mathbb{R}^K$. Agents i and j each have a local open neighborhood on $\mathcal{C}$: $\mathcal{U}_i$ (blue, dashed) and $\mathcal{U}_j$ (red, dashed). Each agent's section is defined only over its own neighborhood: the blue section $q_i(c)$ exists over $\mathcal{U}_i$, the red section $q_j(c)$ over $\mathcal{U}_j$. In the overlap $\mathcal{U}_i \cap \mathcal{U}_j$, both sections coexist in the same fibers. At the highlighted fiber over $c^* \in \mathcal{U}_i \cap \mathcal{U}_j$, $q_i(c^*)$ and $q_j(c^*)$ reside in the same fiber but are expressed in different local gauge frames ϕ_i, ϕ_j (purple axes). The transport operator $\Omega_{ij}(c^*) = \exp(\phi_i)\exp(-\phi_j)$ (green arrow) acts within this fiber, rotating q_j from j's frame into i's frame to produce $\Omega_{ij} q_j$ (dotted red ellipse). The attention cost $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ is then the divergence between same-frame distributions within a single fiber. For transformers ($\dim \mathcal{C} = 0$), all tokens share one base point, the neighborhoods collapse to a single point, and the entire computation occurs within one fiber.

Concretely, at every point c in the base manifold, agent i carries a mean vector $\mu_i(c)$ and a covariance matrix $\Sigma_i(c)$ representing, respectively, its local expected state and uncertainty. Together these form a smooth Gaussian field—the agent's belief section of the statistical bundle.

We will see that the agent's local gauge frame $\phi_i(c)$ may be intuitively interpreted as the agent's subjective frame of reference; its internal coordinate system through which beliefs and models are represented and compared. In this sense, $\phi_i(c)$ captures an agent's "internal orientation" toward the world: it determines how external information is projected into the agent's internal representational space. Gauge transformations $\phi_i(c) \to \phi_i(c) + \xi(c)$ thus correspond to changes in perspective that leave the underlying informational content invariant, much like shifts of viewpoint in perception that preserve the same world model.

2.1.1 Gauge Group and Representations

The natural gauge group for KL-based attention is $G = \mathrm{GL}(K)$, the general linear group of invertible $K \times K$ matrices. This acts on the fibers via representations:

$$\rho_q : \mathrm{GL}(K) \to \mathrm{GL}(d_q), \qquad \rho_p : \mathrm{GL}(K) \to \mathrm{GL}(d_p). \tag{1}$$

For $\Omega \in \mathrm{GL}(K)$, the gauge action on a Gaussian state is:

$$\boxed{\begin{aligned} \rho_q(\Omega) \cdot (\mu_q, \Sigma_q) &= \big(\rho_q(\Omega)\mu_q, \rho_q(\Omega)\Sigma_q\rho_q(\Omega)^\top\big), \\ \rho_p(\Omega) \cdot (\mu_p, \Sigma_p) &= \big(\rho_p(\Omega)\mu_p, \rho_p(\Omega)\Sigma_p\rho_p(\Omega)^\top\big). \end{aligned}} \tag{2}$$

By Theorem 1, KL divergence is invariant under this action for any $\Omega \in \mathrm{GL}(K)$. The only requirement is invertibility ($\det \Omega \neq 0$), not orthogonality. (Since our transport operators are parameterized via the matrix exponential, $\det(\exp(\phi)) = \exp(\mathrm{tr}(\phi)) > 0$, so their determinants are always positive.)

2.1.2 Local Gauge Frames

Each agent i has a gauge frame field $\phi_i : \mathcal{U}_i \to \mathfrak{gl}(K)$ which may, in general, vary across the base manifold. On higher-dimensional base manifolds, a gauge frame field induces a local connection one-form $A_\mu^{(i)}(c) = U_i^{-1}(c)\partial_\mu U_i(c)$ (where $U_i = \exp(\phi_i)$) and associated field strength $F_{\mu\nu}^{(i)} = \partial_\mu A_\nu^{(i)} - \partial_\nu A_\mu^{(i)} + [A_\mu^{(i)}, A_\nu^{(i)}]$. However, a connection derived from a single-valued gauge function $\phi_i(c)$ in this way is pure gauge and has identically vanishing curvature $F_{\mu\nu} = 0$. Non-trivial curvature and holonomy require independent connection variables on edges or paths, transforming as $\Omega_{ij} \mapsto g_i \Omega_{ij} g_j^{-1}$ rather than being determined by vertex-local group elements (see Supplementary Appendix A for details).

In the 0-dimensional case, which we consider here, there exist no gauge connections A_μ or field strengths $F_{\mu\nu}$. What remains is the algebraic structure of group-valued frame transformations $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ acting on statistical fibers. Since these transports are constructed from vertex-local group elements, they satisfy the cocycle condition and have vanishing discrete holonomy (Lemma 2). This algebraic structure already suffices to recover transformer attention (as shown below), and already provides a functional language model when implemented without taking the degenerate limits (Section 5.3).

2.1.3 GL(K) Gauge Invariance of KL Divergence

The Kullback-Leibler divergence is invariant under simultaneous invertible linear transformations of both distributions. Specifically, the KL divergence (and indeed all f-divergences) is invariant under any common bi-jective change of variables (since the Jacobian factors cancel in the density ratio). By restricting to transformations that preserve the MVG family and act linearly on the sufficient statistics (μ, Σ) we find $\mathrm{GL}(K)$ as the symmetry group. This is a standard result in information geometry (see, e.g., Amari (2016)), but its implications for transformer attention have not been systematically developed. We state it here for completeness.

Theorem 1 **($\mathrm{GL}(K)$ Gauge Invariance)** *For any two Gaussian distributions $P = \mathcal{N}(\mu_P, \Sigma_P)$ and $Q = \mathcal{N}(\mu_Q, \Sigma_Q)$ on $\mathbb{R}^K$, and any invertible linear transformation $\Omega \in \mathrm{GL}(K)$, the KL divergence is invariant under simultaneous push-forward:*

$$D_{\mathrm{KL}}(\Omega_* P \| \Omega_* Q) = D_{\mathrm{KL}}(P \| Q), \tag{3}$$

where $\Omega_* P = \mathcal{N}(\Omega\mu_P, \Omega\Sigma_P\Omega^\top)$ denotes the pushforward of P under Ω.

Proof The KL divergence between Gaussians consists of three terms:

$$D_{\mathrm{KL}}(P\|Q) = \frac{1}{2}\left[\mathrm{Tr}(\Sigma_Q^{-1}\Sigma_P) + (\mu_P - \mu_Q)^\top \Sigma_Q^{-1}(\mu_P - \mu_Q) - K + \log\frac{\det\Sigma_Q}{\det\Sigma_P}\right]. \tag{4}$$

Under the pushforward $\Omega_* P = \mathcal{N}(\Omega\mu_P, \Omega\Sigma_P\Omega^\top)$ and $\Omega_* Q = \mathcal{N}(\Omega\mu_Q, \Omega\Sigma_Q\Omega^\top)$:
Trace term:

$$\mathrm{Tr}\big[(\Omega\Sigma_Q\Omega^\top)^{-1}(\Omega\Sigma_P\Omega^\top)\big] = \mathrm{Tr}\big[\Omega^{-\top}\Sigma_Q^{-1}\Omega^{-1}\Omega\Sigma_P\Omega^\top\big] = \mathrm{Tr}(\Sigma_Q^{-1}\Sigma_P). \tag{5}$$

Quadratic term:

$$\begin{aligned}&(\Omega\mu_P - \Omega\mu_Q)^\top(\Omega\Sigma_Q\Omega^\top)^{-1}(\Omega\mu_P - \Omega\mu_Q)\\ &= (\mu_P - \mu_Q)^\top\Omega^\top\Omega^{-\top}\Sigma_Q^{-1}\Omega^{-1}\Omega(\mu_P - \mu_Q) = (\mu_P - \mu_Q)^\top\Sigma_Q^{-1}(\mu_P - \mu_Q).\end{aligned} \tag{6}$$

Log-determinant term:

$$\log\frac{\det(\Omega\Sigma_Q\Omega^\top)}{\det(\Omega\Sigma_P\Omega^\top)} = \log\frac{(\det\Omega)^2\det\Sigma_Q}{(\det\Omega)^2\det\Sigma_P} = \log\frac{\det\Sigma_Q}{\det\Sigma_P}. \tag{7}$$

The $(\det\Omega)^2$ factors cancel identically. Since all three components are invariant, $D_{\mathrm{KL}}(\Omega_* P\|\Omega_* Q) = D_{\mathrm{KL}}(P\|Q)$. ■

The proof extends to all f-divergences: i.e. for any convex function f such that $f(1) = 0$,

$$D_f(\Omega_* P\|\Omega_* Q) = D_f(P\|Q), \quad \forall\Omega \in \mathrm{GL}(K), \tag{8}$$

since the factors $|\det\Omega|$ from the density transformation cancel in the ratio $p(x)/q(x)$. From a practical point of view, this implies that transport operators need only satisfy $\det(\Omega_{ij}) \neq 0$ (invertibility). Therefore, the full $\mathrm{GL}(K)$ acts as a gauge symmetry. It is important to point out that our exponential parameterization e^ϕ restricts to $\det > 0$ (since $\det(\exp(\phi)) = \exp(\mathrm{tr}(\phi)) > 0$), but this is merely the identity component. However, this choice is standard for gauge connections and does not present any imposition in our proof-of-principle experiments. Disconnected transformations are reserved for future study.

Finally, we shall demonstrate that the bilinear form $W_Q W_K^\top$ determining the attention logits in standard transformers can be identified with a gauge transformation. In the isotropic, flat-bundle limit, the constant gauge identification satisfies $W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$. This identification operates within each head's projected subspace (the per-head matrices W_Q^a, W_K^a are rectangular, not individually elements of $\mathrm{GL}(d_k)$; see Section 4.4).

Notation conventions. We use K (or K_q) for the belief fiber dimension (embedding dimension), which generally does not coincide with the representation dimension of the gauge group. For example, we may embed in 10 dimensions with two copies of the fundamental GL(5) group. When discussing the reduction to standard transformers, d_k denotes the per-head dimension ($d_k = K/H$ for H heads). We use q_i to denote both the distribution and its density $q_i(k_i)$, with the meaning clear from context. In what follows we take the gauge group dimension to be equal to the embedding dimension, a simplifying choice that does not restrict the generality of the derivations below.

3 Gauge-Covariant Variational Free Energy

We now construct the variational free energy for a multi-agent system coupled by gauge transport. All probability distributions are defined at a specific base manifold point $c = c^*$ unless otherwise noted. We label the fiber's latent coordinates as $k_i \in \mathcal{B}$ in order to define the necessary integrals.

3.1 Agent State Space

Each agent i maintains two distinct latent variables living in separate fiber bundles:

$$k_i \in \mathbb{R}^{K_q} \quad \text{(belief latent in } \mathcal{E}_q), \qquad m_i \in \mathbb{R}^{K_p} \quad \text{(model latent in } \mathcal{E}_p). \tag{9}$$

The full state of agent i is taken to be a multi-variate Gaussian (MVG) distribution over each latent:

$$\begin{aligned} q_i(k_i) &= \mathcal{N}(k_i; \mu_{q,i}, \Sigma_{q,i}), \quad \Sigma_{q,i} \succ 0, \\ s_i(m_i) &= \mathcal{N}(m_i; \mu_{p,i}, \Sigma_{p,i}), \quad \Sigma_{p,i} \succ 0. \end{aligned} \tag{10}$$

Each latent has an independent Gaussian base prior encoding agent-specific inductive biases:

$$p_i(k_i) = \mathcal{N}(k_i; \mu_{0,i}^{(q)}, \Sigma_{0,i}^{(q)}), \qquad r_i(m_i) = \mathcal{N}(m_i; \mu_{0,i}^{(p)}, \Sigma_{0,i}^{(p)}). \tag{11}$$

These priors are local in that they live within each agent's own gauge frame ϕ and need not be related across agents until transported via the gauge connection. The fiber dimensions K_q and K_p are independent dimensional parameters of the scaffold; the present work operates entirely on the belief channel $(q_i, p_i, \beta_{ij}, \Omega_{ij})$, so K_p plays no role in any equation of this section, and the model-channel scaffold is exercised in the companion treatment per the deferral following Eq. (17).

3.2 Gauge Transport

The gauge transport operators $\Omega_{ij}, \tilde{\Omega}_{ij}$ are pointwise gauge frame transformations:

$$\Omega_{ij}(c) = e^{\phi_i(c)} \cdot e^{-\phi_j(c)} \in \mathrm{GL}^+(K), \tag{12}$$

where $\phi_i : \mathcal{U}_i \to \mathfrak{gl}(K)$ is agent i's gauge frame field (an open local section of the Lie algebra bundle) and $\mathrm{GL}^+(K) = \{A \in \mathrm{GL}(K) : \det A > 0\}$ is the identity component of

$\mathrm{GL}(K)$. This vertex-frame parameterization restricts the framework to a globally trivial principal G-bundle (flat connection); the edge-relaxed non-flat extension (15) is deferred to the companion paper (Dennis, 2025) and discussed in Section 3.2.1 below. Under orthogonal projection, $\mathrm{GL}^+(K)$ maps to $\mathrm{SO}(K)$ rather than full $\mathrm{O}(K)$; the extension to the disconnected component via learnable reflections is discussed in Section 4.2. These act on the latent variables as

$$\begin{aligned} \Omega_{ij} k_j &:= \rho_q(\Omega_{ij}) k_j \in \mathbb{R}^{K_q}, \\ \tilde{\Omega}_{ij} m_j &:= \rho_p(\tilde{\Omega}_{ij}) m_j \in \mathbb{R}^{K_p}, \end{aligned} \tag{13}$$

mapping agent j's latent into agent i's frame.

3.2.1 Vanishing Holonomy and the Flat Bundle

Since the transport operators are constructed from vertex-local group elements, they satisfy a cocycle condition with immediate geometric consequences.

Lemma 2 (Cocycle Identity for Vertex-Frame Transport) *For gauge transport of the form $\Omega_{ij} = g_i g_j^{-1}$ with vertex-local group elements $g_i \in G$, the holonomy around any closed loop is identically the identity:*

$$H_{ijk} = \Omega_{ij}\Omega_{jk}\Omega_{ki} = g_i g_j^{-1} g_j g_k^{-1} g_k g_i^{-1} = I. \tag{14}$$

Equivalently, the cocycle condition $\Omega_{ij}\Omega_{jk} = \Omega_{ik}$ holds for all triples (i, j, k).

Proof Direct computation. The consecutive inverse-product pairs cancel identically: $g_j^{-1} g_j = I$ and $g_k^{-1} g_k = I$. ■

For $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$, the cocycle identity therefore holds regardless of the learned ϕ_i, and semantic transport from token i to token k is path-independent: the direct route via Ω_{ik} and the indirect route via $\Omega_{ij}\Omega_{jk}$ yield identical results. Non-trivial holonomy is recovered without abandoning the vertex-frame skeleton by inserting an edge-local connection δ_{ij} between the vertex factors,

$$\Omega_{ij} = \exp(\phi_i)\exp(\delta_{ij} \cdot G)\exp(-\phi_j), \tag{15}$$

which preserves the standard lattice gauge transformation law $\Omega_{ij} \mapsto g_i \Omega_{ij} g_j^{-1}$ under vertex-local actions and recovers the present theorem in the $\delta_{ij} = 0$ limit. In this parameterization the holonomy around a 3-cycle equals

$$H_{ijk} = U_i \exp(\delta_{ij} \cdot G)\exp(\delta_{jk} \cdot G)\exp(\delta_{ki} \cdot G) U_i^{-1}, \tag{16}$$

with gauge-invariant Wilson observable $W_{ijk} = \mathrm{Re}\,\mathrm{Tr}(H_{ijk})$ and Wilson-style action $S_{\mathrm{YM}} = \beta \sum_{ijk}(1 - W_{ijk}/K)$ (Wilson, 1974; Kogut and Susskind, 1975; Creutz, 1983). The companion paper Dennis (2025) develops this discrete Regime II structure in detail, including the gauge-covariance proof, the data-dependent connection caveat, and the DAG-versus-cycle distinction relevant to autoregressive versus bidirectional attention. Section 6.3 of the present paper continues with the implications of the flat-bundle limit.

3.3 Single-Agent Free Energy Principle

Before introducing multi-agent communication, we recall the standard variational free energy for a single agent i (Friston, 2010; Parr et al., 2022). Agent i maintains a belief $q_i(k_i)$ over its latent state, a prior $p_i(k_i)$ encoding expectations, and receives observations o_i. The free energy principle posits that the agent minimizes:

$$\mathcal{F}_i^{\text{single}} = D_{\mathrm{KL}}(q_i \| p_i) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)]. \tag{17}$$

The first term regularizes beliefs toward the prior and the second couples beliefs to observations. An analogous expression holds for a model channel (Supplementary Appendix G, Model-Channel Formalism); the symmetric two-channel form, with hyper-prior weight λ_h and the explicit attention/meta-attention entropy terms $\tau\beta_{ij}\log(\beta_{ij}/\pi_{ij})$ and $\tau\gamma_{ij}\log(\gamma_{ij}/\pi_{ij}^{(s)})$, is developed in the companion paper (Dennis, 2025); the form used here is its specialization to the single-channel limit with the inner β-minimization already performed (envelope/reduced form $-\tau\sum_i \log Z_i$). We declare here that the cross-fiber morphisms $\Phi_i, \tilde{\Phi}_i$ named in Table 1 do not enter (35): belief and model channels couple only through (i) the shared structure group G (with Ω_{ij} and $\tilde{\Omega}_{ij}$ acting as the same group element in two representations) and (ii) the iterative VFE updates; the cross-bundle morphisms are anticipated for future development and are not instantiated in the present implementation. We now answer the question: how do agents communicate?

3.4 Deriving Attention from a Mixture-of-Sources Model

We construct the agent-agent interaction terms and their softmax attention weights by positing each agent's local inference problem as an entropy-regularized soft-assignment functional over its gauge-transported neighbors, in the style of Cuturi (2013). The alignment free energy below is engineered to make the softmax form of β its exact KKT stationary point on the probability simplex, following the standard treatment of Boyd and Vandenberghe (2004).

3.4.1 Mixture-of-Sources Generative Model

Agent i (the query) seeks to update its belief $q_i(k)$ by attending to a set of source agents j (the keys). Agent i's generative model assumes its latent state k was drawn from one of the neighboring agents, selected by a categorical latent variable $z \in \{1, \ldots, N\}$:

$$P(k, z) = P(k \mid z)P(z), \tag{18}$$

where $P(z = j) = \pi_j$ is the prior probability of attending to agent j (the attention prior), and $P(k \mid z = j) = \mathcal{N}(k; \Omega_{ij}\mu_j, \Omega_{ij}\Sigma_j\Omega_{ij}^\top)$ is agent j's belief transported into agent i's gauge frame via Ω_{ij}.

This construction leads to a generative model consistent with our required gauge structure. Agent j's belief $q_j(k_j)$ lives in j's local frame, and $\Omega_{ij}q_j(k_j)$ is agent i's representation of what agent j believes, as seen by agent i. The categorical variable z endows the model with probabilistic source selection whereby attention is considered as posterior inference over which neighbor generated agent i's state.

The component distributions $P(k \mid z{=}j) = \Omega_{ij} q_j$ depend on the variational posteriors of other agents and are held fixed during the inner minimization over (q_i, β), with the $\{q_j\}_{j \neq i}$ updated by the variational steps of the other agents on a slower cycle. The alignment free energy $\sum_j \beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ is therefore an engineered consensus functional that penalizes belief disagreement after gauge transport, with the entropy term $\tau \beta_{ij} \log(\beta_{ij}/\pi_j)$ added to make the softmax form of β its exact stationary point on the simplex (Cuturi, 2013; Boyd and Vandenberghe, 2004). At the joint fixed point over all $\{q_i\}$, the pairwise objectives are mutually consistent.

3.4.2 Variational Posterior

Next, we approximate the true posterior $P(k, z \mid \text{data})$ using mean-field factorization:

$$Q(k, z) = q_i(k) \cdot \beta(z), \tag{19}$$

where $\beta(z = j) \equiv \beta_{ij} \geq 0$ are variational attention weights satisfying $\sum_j \beta_{ij} = 1$.

3.4.3 Alignment Free Energy

The free energy for the alignment component is the KL divergence between the variational posterior and the generative model:

$$\mathcal{F}_{\text{align}} = D_{\mathrm{KL}}[Q(k, z) \| P(k, z)] = \sum_j \beta_{ij} \int dk q_i(k) \log \frac{q_i(k) \beta_{ij}}{P(k \mid z{=}j) \pi_j}. \tag{20}$$

Decomposing the logarithm yields:

$$\mathcal{F}_{\text{align}} = \sum_j \beta_{ij} \left[\underbrace{\int dk q_i(k) \log \frac{q_i(k)}{P(k \mid z{=}j)}}_{= D_{\mathrm{KL}}[q_i \| \Omega_{ij} q_j]} + \log \beta_{ij} - \log \pi_j \right]. \tag{21}$$

The identification of the integral with $D_{\mathrm{KL}}[q_i \| \Omega_{ij} q_j]$ follows directly from the definition of KL divergence and the generative model $P(k \mid z{=}j) = (\Omega_{ij} q_j)(k)$, with no approximation.

Therefore, we may define an alignment energy $E_{ij} \equiv D_{\mathrm{KL}}[q_i \| \Omega_{ij} q_j]$ such that:

$$\mathcal{F}_{\text{align}} = \sum_j \beta_{ij} \big(E_{ij} + \log \beta_{ij} - \log \pi_j \big). \tag{22}$$

This has the usual "energy minus entropy" form $\mathcal{F}_{\text{align}} = \langle E \rangle_\beta - H(\beta) + \langle -\log \pi \rangle_\beta$, where $H(\beta) = -\sum_j \beta_{ij} \log \beta_{ij}$ is the entropy of the attention distribution. Under the uniform prior $\pi_j = 1/N$ the last term reduces to the constant $\log N$; under non-uniform π it enters the softmax solution as a prior bias on the logits, as Eq. (25) below makes explicit.

3.4.4 Minimization and the Softmax Solution

Next, we minimize $\mathcal{F}_{\text{align}}$ with respect to β_{ij} subject to the constraint $\sum_j \beta_{ij} = 1$ using Lagrange multipliers:

$$\mathcal{L} = \sum_j \beta_{ij}(E_{ij} + \log\beta_{ij} - \log\pi_j) - \lambda\left(\sum_j \beta_{ij} - 1\right). \tag{23}$$

The functional $f(\beta) = \sum_j \beta_{ij}(E_{ij} + \log\beta_{ij} - \log\pi_j)$ is strictly convex on the open simplex: its Hessian is the diagonal matrix $\mathrm{diag}(1/\beta_k)$ with strictly positive entries. The inequality constraints $\beta_j \geq 0$ are non-binding at any interior stationary point because $\log\beta_k \to -\infty$ at the boundary, so the interior KKT system reduces to the equality-Lagrangian above and any solution to it is the unique global minimum (Boyd and Vandenberghe, 2004; Wainwright and Jordan, 2008).

Requiring $\partial\mathcal{L}/\partial\beta_{ik} = 0$ produces:

$$E_{ik} + \log\beta_{ik} + 1 - \log\pi_k - \lambda = 0. \tag{24}$$

Solving and normalizing we find the softmax form, weighted by attention priors π_k:

$$\beta_{ik} = \frac{\pi_k \exp(-E_{ik})}{\sum_m \pi_m \exp(-E_{im})}. \tag{25}$$

For a uniform attention prior $\pi_k = 1/N$, the prior factors cancel, yielding:

$$\boxed{\beta_{ik} = \mathrm{softmax}_k(-E_{ik}) = \mathrm{softmax}_k\big(-D_{\mathrm{KL}}[q_i\|\Omega_{ik}q_k]\big).} \tag{26}$$

Without loss of generality we may include a temperature parameter $\tau > 0$ by rescaling the alignment energy $E_{ik} \to E_{ik}/\tau$. This rescaling, together with multiplication of $\mathcal{F}_{\mathrm{align}}$ by τ, is equivalent to writing the un-minimized alignment free energy in the canonical row-Lagrangian form

$$\mathcal{F}^{(\tau)}_{\mathrm{align}} = \sum_j \big[\beta_{ij}E_{ij} + \tau\beta_{ij}\log(\beta_{ij}/\pi_j)\big], \tag{27}$$

where τ couples explicitly to the attention entropy/prior term. The softmax stationary point $\beta^*_{ij} = \pi_j \exp(-E_{ij}/\tau)/Z_i$ is unchanged, while the substituted value becomes $\mathcal{F}^{(\tau)*}_{\mathrm{align}} = -\tau\log Z_i$, matching the $-\tau\log Z_i$ reduction used in (35). Additionally, as discussed in the appendix, the forward KL divergence is the unique f-divergence that preserves exponential-family closure under this linear coupling structure and yields a consistent dual interpretation for the attention weights (see Appendix A).

3.4.5 Non-Uniform Attention Priors: Masking and Positional Biases

The general solution (25) with non-uniform priors π_k provides a unified derivation of several architectural features historically introduced empirically into transformers.

Causal Masking. In auto-regressive language modeling, token i should not attend to future tokens $j > i$. Within our generative model, this constraint is a prior restriction on the availability of a source. Formally, setting the attention prior to

$$\pi_j^{(\mathrm{causal})} = \begin{cases} \frac{1}{i} & \text{if } j \leq i, \\ 0 & \text{if } j > i, \end{cases} \tag{28}$$

and substituting into (25) produces

$$\beta_{ij} = \frac{\mathbf{1}[j \leq i] \cdot \exp(-E_{ij}/\tau)}{\sum_{k \leq i} \exp(-E_{ik}/\tau)}, \tag{29}$$

which is precisely the masked attention used in GPT-family models (Radford et al., 2019).

Sliding Window Attention. Similarly, choosing the prior as a local window of size w around position i,

$$\pi_j^{(\text{window})} \propto \mathbf{1}[|i - j| \leq w], \tag{30}$$

produces the sliding window attention used in architectures such as Longformer (Beltagy et al., 2020) and Mistral (Jiang et al., 2023). In our gauge-theoretic framework, this encodes the prior belief that an agent's state was generated by a source within its "local neighborhood", a natural assumption for data with locality structure.

ALiBi: Linear Positional Bias. A distance-decaying attention prior of the form

$$\pi_j \propto \exp(-m|i - j|), \tag{31}$$

where $m > 0$ is a slope parameter, contributes an additive bias to the logits. Substituting into (25) we have:

$$\beta_{ij} = \frac{\exp(-E_{ij}/\tau - m|i - j|)}{\sum_k \exp(-E_{ik}/\tau - m|i - k|)} = \text{softmax}_j \left(-\frac{E_{ij}}{\tau} - m|i - j| \right). \tag{32}$$

This is exactly the "Attention with Linear Biases" (ALiBi) mechanism of Press et al. (2022). The slope m is typically set to a head-dependent constant (e.g., $m = 2^{-8/H} \cdot 2^{-h}$ for head h of H total heads). In the gauge-theoretic framework, ALiBi arises from a prior belief that nearby sources are exponentially more likely to have generated the agent's state.

Relative Position Biases. More generally, we can make the attention prior a learnable function of relative position,

$$\pi_j \propto \exp(b_{i-j}), \tag{33}$$

where $b_\delta \in \mathbb{R}$ is a (learnable) bias for relative offset $\delta = i - j$, leading to an additive bias b_{i-j} in the logits as:

$$\beta_{ij} = \text{softmax}_j \left(-\frac{E_{ij}}{\tau} + b_{i-j} \right). \tag{34}$$

This recovers relative position biases used in T5 (Raffel et al., 2020) and other architectures that parameterize attention with learnable position-dependent offsets. In a straightforward manner, each positioning architecture emerges as a special case of (25) operating via the same mathematical object. Although the choice of attention priors may be ad-hoc, their fundamental source and structure is determined from first principles within our framework.

3.5 Full Variational Free Energy

By combining the single-agent free energy (17) with the alignment terms derived above and summing over all agents, we obtain the global gauge-covariant variational free energy at $c^* \in \mathcal{C}$. Substituting the optimal attention weights $\beta^*_{ij} = \mathrm{softmax}_j(-E_{ij}/\tau + \log \pi_j)$ from (25) into the full alignment free energy $\mathcal{F}_{\mathrm{align}}$ (22) yields the reduced alignment energy $-\tau \log Z_i$, where $Z_i = \sum_j \pi_j \exp(-E_{ij}/\tau)$ is the partition function. The reduced (partially minimized) free energy is therefore:

$$\boxed{\mathcal{F}_{\mathrm{red}}[\{q_i\}] = \sum_i D_{\mathrm{KL}}(q_i \| p_i) - \tau \sum_i \log Z_i - \mathbb{E}_q[\log p(o|\{k_i\})],} \tag{35}$$

where $E_{ij} = D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ and $Z_i = \sum_j \pi_j \exp(-E_{ij}/\tau)$. The log-partition function $-\tau \log Z_i$ is the result of substituting β^*_{ij} into both the energy term $\sum_j \beta_{ij} E_{ij}$ and the entropy term $\tau \sum_j \beta_{ij} \log(\beta_{ij}/\pi_j)$ of $\mathcal{F}_{\mathrm{align}}$. In the working implementation the temperature is factorized as $\tau = \kappa\sqrt{K}$, with $\kappa > 0$ a learnable scalar and $\sqrt{K}$ the dimension scaling familiar from scaled dot-product attention; the standard-transformer recovery in Section 2.1.3 corresponds to the special case $\kappa = 1$.

The envelope theorem guarantees that the gradient of $\mathcal{F}_{\mathrm{red}}$ with respect to any belief parameter $x \in \{\mu_i, \Sigma_i, \phi_i\}$ takes the form:

$$\frac{d\mathcal{F}_{\mathrm{red}}}{dx} = \frac{\partial \mathcal{F}}{\partial x}\bigg|_{\beta=\beta^*} = (\text{prior terms}) + \sum_j \beta^*_{ij} \frac{\partial E_{ij}}{\partial x} - (\text{observation terms}), \tag{36}$$

with no explicit $\partial\beta^*/\partial x$ terms, since the cross-term $\sum_j (\partial\mathcal{F}/\partial\beta_{ij})(\partial\beta^*_{ij}/\partial x)$ vanishes at the optimal attention where $\partial\mathcal{F}/\partial\beta_{ij} = 0$.

Autograd versus reduced-free-energy gradients. In practice, implementations differentiate the composition $\sum_j \beta^*_{ij}(x) E_{ij}(x)$ via automatic differentiation, which applies the product rule and produces additional $(\partial\beta^*_{ij}/\partial x)E_{ij}$ terms. This computes the gradient of the attention-weighted energy $\langle E\rangle_{\beta^*} = \sum_j \beta^*_{ij} E_{ij}$, which differs from the reduced free energy $-\tau \log Z_i$ by the attention-entropy term $\tau \sum_j \beta^*_{ij} \log(\beta^*_{ij}/\pi_j)$. Using the softmax sensitivity $\partial\beta^*_{ij}/\partial x = -\tau^{-1}\beta^*_{ij}\big(\partial E_{ij}/\partial x - \langle \partial E/\partial x\rangle_{\beta^*}\big)$, the difference between the two gradients takes the covariance form

$$\nabla_x \langle E\rangle_{\beta^*} - \nabla_x \mathcal{F}_{\mathrm{red}} = -\tau^{-1} \mathrm{Cov}_{\beta^*}\big(E_{ij}, \partial E_{ij}/\partial x\big). \tag{37}$$

The two objectives therefore share critical points only where this covariance vanishes; this holds at the joint stationary point of $\mathcal{F}_{\mathrm{red}}$ in x and in the high-temperature limit $\tau \to \infty$, but not generically off-equilibrium. The $\partial\beta/\partial x$ terms, which vanish in the reduced-free-energy gradient by the envelope theorem, are present in the autograd gradient and provide a softmax-gradient nonlinearity (Section 4.6.1). Standard transformer training follows the autograd convention (differentiating through the softmax), and we adopt the same convention in the gradient expressions and algorithm below. The gradient expressions below are therefore gradients of the attention-weighted energy $\sum_j \beta^*_{ij} E_{ij}$ (equivalently, derivatives of $\mathcal{F}$ before the β-minimization is performed, evaluated at $\beta = \beta^*$), not of the reduced free energy $\mathcal{F}_{\mathrm{red}}$; the two differ by (37).

3.6 Interpretation

Each term in the variational free energy (35) encodes a distinct aspect of multi-agent inference:

(1) Belief Prior: $D_{\mathrm{KL}}(q_i \| p_i)$

The belief prior measures the deviation of agent i's current belief $q_i(k_i)$ from its local prior $p_i(k_i \mid m_i)$. This term regularizes beliefs towards its expected states. Without observations and inter-agent couplings, minimizing $\mathcal{F}$ would produce $q_i^* = p_i$ thereby recovering pure prior-based prediction.

(2) Belief Alignment: $\beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$

As discussed previously, belief alignment represents the discrepancy between agent i's belief and agent j's belief after gauge transport into agent i's local frame (i.e. agent i's interpretation of agent j's belief).

This term enforces an epistemic consensus whereby agents with high β_{ij} are driven to agree on their beliefs about the current state, modulo gauge transformations.

(3) Observation Likelihood: $-\mathbb{E}_q[\log p(o \mid \{k_i\})]$

This term is the expected negative log-likelihood of observations/data o given latent states $\{k_i\}$, averaged over the recognition distributions $\{q_i\}$. Without this term, the system is a pure vacuum theory where agents converge to their coupled priors without external input. Observations break this vacuum symmetry, forcing agents to specialize based on their sensory evidence (see Results).

3.7 State-Dependent Prior Precision

3.7.1 Precision as a Variational Parameter

In the variational free energy (35), the self-coupling term $D_{\mathrm{KL}}(q_i \| p_i)$ carries unit coefficient. This implicitly assumes that the prior precision is apriori known and fixed such that each agent trusts its prior with uniform strength regardless of context.

A more flexible treatment promotes the self-coupling weight from a fixed constant to a per-agent variational parameter $\alpha_i > 0$, subject to a regularizer that prevents degenerate solutions. The derivation below is a single-scale Lagrangian construction and does not invoke any hierarchical or multi-scale structure.

This augmented free energy for agent i is then:

$$\mathcal{F}_i = \alpha_i D_{\mathrm{KL}}(q_i \| p_i) + R(\alpha_i) + \sum_j \beta_{ij} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)], \tag{38}$$

where $R(\alpha_i)$ is a precision regularizer. We choose

$$R(\alpha_i) = b_0 \alpha_i - c_0 \log \alpha_i, \qquad b_0 > 0, c_0 > 0, \tag{39}$$

the negative log-density of a $\mathrm{Gamma}(\alpha_i; c_0 + 1, b_0)$ distribution — the conjugate prior for the precision parameter of a Gaussian likelihood (Bishop, 2006; Murphy, 2012). The first term penalizes arbitrarily large precision (preventing $\alpha_i \to \infty$), the second is a log-barrier that enforces $\alpha_i > 0$ and penalizes vanishing precision (preventing $\alpha_i \to 0$, which would trivially eliminate the self-coupling cost).

3.7.2 Optimal Precision

Optimizing $\mathcal{F}_i$ with respect to α_i (holding beliefs q_i fixed):

$$\frac{\partial \mathcal{F}_i}{\partial \alpha_i} = D_{\mathrm{KL}}(q_i \| p_i) + b_0 - \frac{c_0}{\alpha_i} = 0, \tag{40}$$

which gives the optimal state-dependent precision:

$$\boxed{\alpha_i^* = \frac{c_0}{b_0 + D_{\mathrm{KL}}(q_i \| p_i)}.} \tag{41}$$

This α_i^* is the MAP estimate of α_i under the Gamma prior $p(\alpha_i) \propto \alpha_i^{c_0} e^{-b_0 \alpha_i}$, with the linear-in-$\alpha$ self-coupling penalty $\alpha_i \cdot D_{\mathrm{KL}}(q_i \| p_i)$ playing the role of the sufficient statistic in the precision posterior.

For example, when an agent is close to its prior ($D_{\mathrm{KL}}(q_i \| p_i) \approx 0$), $\alpha_i \approx c_0/b_0$ takes a large value determined by the hyper-parameters such that the agent trusts its prior and resists displacement by the alignment term. Conversely, when an agent is far from its prior (large $D_{\mathrm{KL}}(q_i \| p_i)$), $\alpha_i \to 0$, the prior's influence is reduced and the alignment and observation terms dominate. The agent becomes less influenced by its prior allowing contextual information to influence its belief.

b_0 controls the sensitivity such that large b_0 yield approximately constant α_i, while small b_0 yield sharply state dependent α_i. The ratio c_0/b_0 sets the maximum precision (the value when beliefs match the prior exactly). Both hyper-parameters may be learned via empirical Bayes by introducing two scalar parameters.

In our experiments we study both $R(\alpha_i) \neq 0$ and $R(\alpha_i) = 0$ with $\alpha_i = \alpha = 1$.

Product-rule chain at autograd time. The closed-form Eq. (41) makes α_i^* a function of the belief through $D_{\mathrm{KL}}(q_i \| p_i)$. Differentiating with respect to any belief parameter $\theta \in \{\mu_i, \Sigma_i\}$ gives

$$\frac{\partial \alpha_i^*}{\partial \theta} = -\frac{(\alpha_i^*)^2}{c_0} \frac{\partial D_{\mathrm{KL}}(q_i \| p_i)}{\partial \theta}. \tag{42}$$

Applying the product rule to the self-coupling term $\alpha_i^* D_{\mathrm{KL}}(q_i \| p_i)$ gives

$$\frac{\partial}{\partial \theta}\big(\alpha_i^* D_{\mathrm{KL}}(q_i \| p_i)\big) = \alpha_i^* \frac{\partial D_{\mathrm{KL}}}{\partial \theta} - \frac{(\alpha_i^*)^2}{c_0} D_{\mathrm{KL}} \frac{\partial D_{\mathrm{KL}}}{\partial \theta} = (\alpha_i^*)^2 \frac{b_0}{c_0} \frac{\partial D_{\mathrm{KL}}}{\partial \theta}, \tag{43}$$

where the second equality uses $\alpha_i^*/c_0 = 1/(b_0 + D_{\mathrm{KL}})$ to collapse the bracket $1 - (\alpha_i^*/c_0) D_{\mathrm{KL}}$ to $b_0/(b_0 + D_{\mathrm{KL}})$. The effective autograd prefactor on $\partial D_{\mathrm{KL}}/\partial \theta$ is therefore $(\alpha_i^*)^2 b_0/c_0$ rather than the envelope-form α_i^* alone.

By the envelope theorem (Section 3.5) the explicit $\partial \alpha_i/\partial \theta$ contribution vanishes from the gradient of the reduced free energy $\mathcal{F}_{\mathrm{red}}$ when α_i is held at α_i^*. Autograd-based implementations that differentiate through $\alpha_i^*(\mu_i, \Sigma_i)$ along with D_{KL} pick up the correction in Eq. (43), which closes the chain rule on the $\partial \alpha_i/\partial \mu_i$ and $\partial \alpha_i/\partial \Sigma_i$ slots in Eq. (98). Our reference implementation evaluates the corrected form (`transformer/core/vfe_gradients.py`); the constant-α specialization $\alpha_i = 1$ collapses both the closed-form and the chain-rule correction.

3.8 Belief Dynamics

The belief dynamics follow gradient descent on the attention-weighted alignment energy at $\beta = \beta^*$, the autograd-convention form adopted at the close of Section 3.5:

$$\frac{\partial q_i}{\partial t} = -\eta_q \nabla_{q_i} \big[\textstyle\sum_j \beta^*_{ij} E_{ij} + \alpha_i D_{\mathrm{KL}}(q_i \| p_i) - \mathbb{E}_{q_i}[\log p(o_i \mid k_i)] \big]. \tag{44}$$

This differs from the gradient of the reduced free energy $\mathcal{F}_{\mathrm{red}}$ (Eq. 35) by the covariance term $-\tau^{-1}\mathrm{Cov}_{\beta^*}(E_{ij}, \partial E_{ij}/\partial q_i)$ of Eq. 37; the two vector fields coincide at joint stationary points and differ off-equilibrium by the softmax-gradient nonlinearity identified at Section 4.6.1.

The most general formulation includes additional model-channel terms (s_i, r_i, γ_{ij}) representing a slow meta-learning timescale (Supplementary Appendix G, Model-Channel Formalism). In the present work we focus exclusively on the belief (fast) subsystem, studying how beliefs $\{q_i\}$ evolve under alignment. We posit that standard transformers operate entirely in this regime: they perform inference (q_i updates) with frozen priors during a forward pass. As discussed in the appendix, the framework naturally supports hierarchical emergence of meta-agents through coarse-graining when groups of agents reach belief consensus (Supplementary Appendix A).

Note that in the full theory with general, possibly curved, base manifolds we would necessarily integrate over agent supports and overlaps in order to obtain the complete variational free energy (which we needn't consider for our present transformer derivations). This then introduces extensive and rich geometry including Riemann curvature, gauge curvature, Fischer curvature, induced gauge connections, belief/model transport between and within agents and fibers, non-trivial vertical and horizontal holonomies, and much, much more.

3.9 Symmetry Breaking

In the absence of observations ($p(o|\cdot) = \mathrm{const}$), the free energy (35) possesses a reparameterization degeneracy under the gauge group G (which is $\mathrm{GL}(K)$ in general, or a compact subgroup such as $\mathrm{SO}(K)$). The vacuum state corresponds to perfect alignment: $q_i = \Omega_{ij} q_j$ and $p_i = \tilde{\Omega}_{ij} p_j$ for all agent pairs (i, j), meaning that all agents maintain gauge-equivalent beliefs that differ only by frame transformations. In this regime, the dynamics drive the system toward a degenerate manifold of ground states parameterized by the gauge orbit. Agents synchronize over gradient descent towards $\|\mu_i(c)\| = \mu^*$, but the absolute orientations remain arbitrary.

A terminological note is warranted. In physics, local gauge symmetries are more precisely understood as redundancies of description rather than physical symmetries, and there are well-known subtleties in applying the language of spontaneous symmetry breaking to gauge redundancies (Elitzur's theorem forbids spontaneous breaking of local gauge symmetries in lattice gauge theory (Weinberg, 1996)). Our gauge group acts as a reparameterization symmetry of the KL divergence (Theorem 1), and the degeneracy of the vacuum manifold is a consequence of this reparameterization freedom rather than of a global physical symmetry.

Observations destroy this degeneracy by coupling agents to external data through the likelihood terms. Each agent's data stream o_i acts as an external field that selects a preferred orientation in its fiber. This is *explicit* symmetry breaking: the observation/likelihood term $-\mathbb{E}_q[\log p(o|k)]$ is manifestly not gauge-invariant, and it is this term that forces agents into distinct configurations. The system transitions from the symmetric vacuum to a phase where agents develop distinct specializations such that $\|\mu_i(c)\| \neq \|\mu_j(c)\|$ with the diversity driven by observations/data. The frame transformations Ω_{ij} then encode how these specialized representations relate geometrically. This observation-induced specialization is what enables non-trivial multi-agent coordination in that agents must actively maintain geometric relationships between their specialized frames rather than simply coexisting in a gauge-symmetric configuration. Whether the vacuum manifold supports approximate zero modes of the Hessian (corresponding to directions of near-zero curvature along the gauge orbit) is a question we leave to future work.

3.10 Summary

The gauge-covariant variational free energy (35) has been derived from two principles: (i) the standard free energy principle for single agents (prior regularization + observation likelihood), and (ii) the mixture-of-sources model for inter-agent communication (KL-based alignment with softmax attention weights). Both belief alignment and model alignment arise from variational inference over gauge-transported sources. The forward KL divergence $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ emerges as the alignment energy from the mixture generative model, and the softmax form of attention weights $\beta_{ij} = \mathrm{softmax}(-E_{ij}/\tau)$ follow from Lagrange optimization of the alignment free energy. Non-uniform attention priors π_k provide multiple theoretically grounded mechanisms for positional relationships. Furthermore, a context-dependent self-coupling weight α_i can be derived as a state-dependent precision through variational optimization with a log-barrier regularizer (Section 3.7), with the fixed-α case recovered as a limit.

Multi-agent communication within this gauge-covariant formulation allows the FEP to be satisfied. There is no necessitation of any informational substrate (such as neural networks, transmission lines, etc) within our formalism. Next, we show that this formulation connects attention, transformers, and architecture to variational inference without neural networks, MLPs, and other standard ad-hoc architectures.

4 Reduction to Transformer Attention

In this section we demonstrate that standard transformer self-attention emerges as a set of limiting cases of our gauge-theoretic framework. This establishes neural network training as a limit of a first-principles information-geometric minimization of a variational free energy over a bundle geometry.

4.1 Setup: Agents as Gaussian Beliefs in Local Frames

In our general formulation, each agent i (token) maintains a local MVG belief:

$$q_i = \mathcal{N}(\mu_i, \Sigma_i), \tag{45}$$

where $\mu_i \in \mathbb{R}^{d_k}$ is the agent's mean representation in its local gauge frame, and $\Sigma_i \in \mathbb{R}^{d_k \times d_k}$ is its belief covariance (symmetric positive definite). Note that we are embedding in $K = d_k$ dimensions in order to make contact with standard notation and, for now, the gauge group will be the fundamental $\mathrm{GL}(K = d_k)$.

Communication between agents i and j is mediated by a gauge frame transport operator

$$\Omega_{ij} \in \mathrm{GL}(d_k), \tag{46}$$

which maps representations from agent j's local frame into agent i's frame.

The value-aggregation output received by agent i is the posterior mean of the mixture variational model under responsibilities β_{ij} (derived formally in Section 4.2.3):

$$\hat{\mu}_i = \sum_j \beta_{ij} \Omega_{ij} \mu_j, \tag{47}$$

the gauge-theoretic analog of the attention aggregation $\sum_j \alpha_{ij} V_j$ in standard transformers.

4.2 Connection to Standard Transformers

The gauge-covariant attention $\beta_{ij} = \mathrm{softmax}(-D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)/\tau)$ derived in the preceding section is already a complete attention mechanism. Standard transformer attention is recovered as a special case along either of two equivalent routes. The first route (Section 4.2.1) retains $\Omega_{ij} = U_i U_j^{-1}$ non-trivially throughout and exposes untied query-key projections $Q_i = U_i^{-1} \mu_i$, $K_j = U_j^{\top} \Sigma_j^{-1} \mu_j$ directly from the transported KL, with the implicit pair-dependent bilinear ranging over $\mathrm{GL}(d_k)$ as the per-token frames and precisions vary; this route is structurally minimal in the sense that no object beyond the existing belief and frame fields is introduced. The second route (Section 4.2.2) takes three successive limits: isotropic covariances, shared frames giving trivial pairwise transport $\Omega_{ij} = I$, and a learned shared bilinear compatibility $M = W_Q W_K^{\top}$ acting on the means. This yields the familiar $QK^{\top}/\sqrt{d_k}$ form with M as a separate object that fills the cross-coupling slot vacated by the trivial transport. The two routes coincide on standard scaled dot-product attention; the first additionally captures the rotary positional structure of modern transformers without introducing extraneous fields. Our experiments (Section 5.3) implement the general gauge model without these limits.

Scale-Absorbed Reparameterization (Implicit-Variance Form). We consider a single-point base space $\mathcal{C}$ with a finite set of agents $\{1, \ldots, N\}$ (tokens) with no particular ordering.

The deterministic limit requires nuance. Naively setting $\Sigma_i \to 0$ yields Dirac delta beliefs $q_i(k_i) \to \delta(k_i - \mu_i)$, but the KL divergence between distinct Diracs is infinite:

$$D_{\mathrm{KL}}(\delta(k - \mu_i) \| \delta(k - \mu_j)) = +\infty \quad \text{for } \mu_i \neq \mu_j. \tag{48}$$

This divergence reflects the absolute continuity requirement in the definition of KL: $D_{\mathrm{KL}}(P \| Q)$ is finite only when P is absolutely continuous with respect to Q (i.e., $P(A) > 0$ implies $Q(A) > 0$ for all measurable A). Distinct Dirac measures have disjoint support and therefore violate this condition.

We may remedy this by taking a joint scaling limit where the belief variance σ^2 (discussed below) remains finite but is absorbed into learned parameters.

Next, we assume isotropic covariances $\Sigma_i = \Sigma_j = \sigma^2 I$ with $\sigma^2 > 0$. This implies that the log-determinant terms cancel ($\log \frac{|\Sigma_j|}{|\Sigma_i|} = 0$) and the trace term simplifies to $\mathrm{Tr}(\Sigma_j^{-1}\Sigma_i) = \mathrm{Tr}(I) = d_k$.

With gauge transport $\Omega_{ij} \in \mathrm{GL}(d_k)$, the transported Gaussian $\Omega_{ij} q_j$ has mean $\Omega_{ij}\mu_j$ and covariance $\Omega_{ij}\Sigma_j\Omega_{ij}^\top = \sigma^2(\Omega_{ij}\Omega_{ij}^\top)$. Therefore the full KL divergence is:

$$D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) = \frac{1}{2}\Big[\log\det(\Omega_{ij}\Omega_{ij}^\top) + \mathrm{Tr}\big((\Omega_{ij}\Omega_{ij}^\top)^{-1}\big) - d_k\Big] + \frac{1}{2\sigma^2}(\mu_i - \Omega_{ij}\mu_j)^\top(\Omega_{ij}\Omega_{ij}^\top)^{-1}(\mu_i - \Omega_{ij}\mu_j). \tag{49}$$

The Mahalanobis quadratic form can be simplified using the identity $(\Omega_{ij}\Omega_{ij}^\top)^{-1} = \Omega_{ij}^{-\top}\Omega_{ij}^{-1}$:

$$(\mu_i - \Omega_{ij}\mu_j)^\top(\Omega_{ij}\Omega_{ij}^\top)^{-1}(\mu_i - \Omega_{ij}\mu_j) = \|\Omega_{ij}^{-1}(\mu_i - \Omega_{ij}\mu_j)\|^2 = \|\Omega_{ij}^{-1}\mu_i - \mu_j\|^2. \tag{50}$$

This identity has the interpretation that rather than transporting the key belief into the query's frame via Ω_{ij} and measuring with the induced metric, the KL divergence equivalently transports the query into the key's frame via Ω_{ij}^{-1} and measures the Euclidean distance there. Either perspective is appropriate. The Mahalanobis metric automatically accounts for the distortion introduced by gauge transport.

Generalization to Rényi α-divergence. The construction above uses the KL divergence as the alignment functional. The softmax structure of β extends without change to the one-parameter Rényi family $D_\alpha(q\|p)$ with order $\alpha \in (0,\infty) \setminus \{1\}$, since the row-Lagrangian KKT argument of Section 3.4 depends only on the entropy term $\tau \sum_j \beta_{ij} \log(\beta_{ij}/\pi_j)$ and on linearity of the alignment energy in β_{ij}, not on the specific form of the divergence. The $\alpha \to 1$ limit recovers the KL. For independent diagonal Gaussians with $q = \mathcal{N}(\mu_q, \mathrm{diag}(\sigma_q))$ and $p = \mathcal{N}(\mu_p, \mathrm{diag}(\sigma_p))$, the Rényi divergence admits the closed form

$$D_\alpha(q\|p) = \frac{1}{2}\left[\alpha \sum_k \frac{(\mu_p^k - \mu_q^k)^2}{\tilde{\sigma}_k} + \frac{1}{\alpha - 1}\sum_k \Big((1-\alpha)\log\sigma_q^k + \alpha\log\sigma_p^k - \log\tilde{\sigma}_k\Big)\right], \tag{51}$$

with the blended variance $\tilde{\sigma}_k = (1-\alpha)\sigma_q^k + \alpha\sigma_p^k$. The full-covariance generalization follows by replacing the per-coordinate blend with the matrix blend $\tilde{\Sigma} = (1-\alpha)\Sigma_q + \alpha\Sigma_p$ and reading the log-determinant and Mahalanobis terms off the analogous closed form. The transported-key version is obtained by substituting $(\mu_p, \Sigma_p) \to (\Omega_{ij}\mu_j, \Omega_{ij}\Sigma_j\Omega_{ij}^\top)$, exactly as in the KL case.

The attention construction is unchanged in form:

$$\beta_{ij}^{(\alpha)} = \mathrm{softmax}_j\left(-\frac{D_\alpha(q_i\|\Omega_{ij}q_j)}{\tau}\right), \tag{52}$$

with the KL row $\alpha = 1$ as the standard limit and other values selecting different mass/-coverage trade-offs (e.g., $\alpha = 1/2$ recovers the Bhattacharyya coefficient up to a sign). All gradient derivations above carry over by replacing $\partial D_{\mathrm{KL}}/\partial\theta$ with $\partial D_\alpha/\partial\theta$; the additional product-rule term in Eq. (43) retains its form with $D_{\mathrm{KL}} \to D_\alpha$. Our reference implementation exposes the order α through the per-block configuration option (`alpha_divergence` in `transformer/core/block_config.py`); the diagonal closed form Eq. (51) is the kernel evaluated inside the attention path and the iterative belief updates, with a Taylor branch near $\alpha = 1$ for numerical stability. The extension is to the softmax form of β; the closed-form geometric-mean stationary belief $q_i^* \propto p_i^{1/2} \prod_j (\Omega_{ij} q_j)^{\beta_{ij}/2}$ derived for KL in Supplementary Appendix H is specific to $\alpha = 1$ and does not generalize to $\alpha \neq 1$, where the stationarity condition for q_i^* becomes transcendental.

Isotropic attention with general gauge transport. The structural terms $\log\det(\Omega_{ij}\Omega_{ij}^\top)$, $\mathrm{Tr}((\Omega_{ij}\Omega_{ij}^\top)^{-1})$, and d_k depend on the gauge transport but not on the belief means μ_i, μ_j. Defining the *geometric bias*

$$S(\Omega_{ij}) := \frac{1}{2}\Big[\log\det(\Omega_{ij}\Omega_{ij}^\top) + \mathrm{Tr}\big((\Omega_{ij}\Omega_{ij}^\top)^{-1}\big) - d_k\Big], \tag{53}$$

the KL divergence under isotropic beliefs with general transport takes the form

$$D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) = S(\Omega_{ij}) + \frac{1}{2\sigma^2}\|\Omega_{ij}^{-1}\mu_i - \mu_j\|^2. \tag{54}$$

This is already a fully specified attention mechanism. The attention weight

$$\beta_{ij} = \mathrm{softmax}_j\left(-\frac{S(\Omega_{ij}) + \frac{1}{2\sigma^2}\|\Omega_{ij}^{-1}\mu_i - \mu_j\|^2}{\tau}\right) \tag{55}$$

combines a content-dependent term (the Mahalanobis distance between belief means, measured in the key's frame) with a pair-dependent geometric bias $S(\Omega_{ij})$ that encodes the distortion introduced by gauge transport. When Ω_{ij} is orthogonal, $S(\Omega_{ij}) = 0$ identically; for general $\mathrm{GL}(K)$ transport, $S(\Omega_{ij}) \geq 0$ with equality if and only if $\Omega_{ij} \in \mathrm{O}(K)$. The geometric bias therefore penalizes transport operators that distort volume or stretch directions, providing a geometric regularizer intrinsic to the KL divergence. This is the natural form of attention under a single simplifying limit (isotropic covariances); standard transformer attention requires a further specialization.

Achieving $\mathrm{O}(K)$ transport via learnable reflections. The condition $S(\Omega_{ij}) = 0$ is equivalent to $\Omega_{ij} \in \mathrm{O}(K)$, but orthogonality serves a second, equally important role: the transported covariance $\sigma^2(\Omega_{ij}\Omega_{ij}^\top)$ equals $\sigma^2 I$ if and only if $\Omega_{ij}\Omega_{ij}^\top = I$, i.e., $\Omega_{ij} \in \mathrm{O}(K)$. Thus orthogonal transport is the *unique* condition under which isotropic covariances are preserved under gauge transport. For $\Omega_{ij} \in \mathrm{GL}(K) \setminus \mathrm{O}(K)$, a belief that is isotropic in its own frame becomes anisotropic when transported to another frame, and the geometric bias $S(\Omega_{ij}) > 0$ measures precisely this induced anisotropy.

However, the matrix exponential parameterization $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j) \in \mathrm{GL}^+(K)$ (Eq. 12) restricts transport to the identity component. Under orthogonal projection (e.g., Newton-Schulz iteration), $\mathrm{GL}^+(K)$ maps to $\mathrm{SO}(K)$, the special orthogonal group with det $= +1$. The full orthogonal group $\mathrm{O}(K) = \mathrm{SO}(K) \rtimes \mathbb{Z}_2$ includes a second connected component of orientation-reversing transformations (det $= -1$) that is topologically inaccessible to the exponential map. To extend to full $\mathrm{O}(K)$, one introduces per-agent diagonal reflection matrices $R_i = \mathrm{diag}(s_i)$ with $s_i \in \{-1, +1\}^K$, yielding the combined transport

$$\Omega_{ij} = R_i M_{ij} R_j, \qquad M_{ij} = \exp(\phi_i)\exp(-\phi_j) \in \mathrm{GL}^+(K), \tag{56}$$

When ϕ_i is restricted to $\mathfrak{so}(K)$ (skew-symmetric matrices), $M_{ij} \in \mathrm{SO}(K)$ automatically; for general $\phi_i \in \mathfrak{gl}(K)$, orthogonal projection (e.g., Newton-Schulz iteration, as described above) maps M_{ij} to $\mathrm{SO}(K)$. With $M_{ij} \in \mathrm{SO}(K)$, the combined transport covers both components of $\mathrm{O}(K)$ since $\det(\Omega_{ij}) = (\prod_k s_i^{(k)})(\prod_k s_j^{(k)}) \det(M_{ij})$ ranges over ± 1. Orthogonality $\Omega_{ij}\Omega_{ij}^\top = R_i M_{ij} \underbrace{R_j R_j}_{I} M_{ij}^\top R_i = R_i \underbrace{M_{ij} M_{ij}^\top}_{I} R_i = \underbrace{R_i^2}_{I} = I$ follows from the fact that each factor is independently orthogonal, ensuring $S(\Omega_{ij}) = 0$ and preservation of isotropic covariance under transport. The Mahalanobis distance decomposes cleanly: since $\Omega_{ij}^{-1} = R_j M_{ij}^\top R_i$ and each R is its own inverse,

$$\|\Omega_{ij}^{-1}\mu_i - \mu_j\|^2 = \|M_{ij}^\top (s_i \odot \mu_i) - (s_j \odot \mu_j)\|^2, \tag{57}$$

so the reflections are absorbed into the embeddings via elementwise sign flips $\mu_i \mapsto s_i \odot \mu_i$, requiring no modification to the $\mathrm{SO}(K)$ attention computation. Gradient flow through the discrete signs can be maintained via the straight-through estimator, treating the sign function as identity in the backward pass while sampling from $\{-1, +1\}$ in the forward pass. The $\mathrm{O}(K)$ extension is therefore a minimal modification: the continuous gauge dynamics on $\mathrm{SO}(K)$ are unchanged, and the discrete reflections provide the topologically necessary second component.

The rescaled KL divergence $\widetilde{D}_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) := \sigma^2 \cdot D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) \to \frac{1}{2}\|\Omega_{ij}^{-1}\mu_i - \mu_j\|^2$ as $\sigma \to 0$ remains finite and equals half the squared distance between means as measured in the key's reference frame. This should be understood as the regime where σ is small compared to characteristic scales of μ_i, such that σ^{-2} will be absorbed into the learned projections (see below).

Constant gauge specialization ($\Omega_{ij} = \Omega$). Standard transformer attention emerges when we further assume a single global frame shared by all agents, $\Omega_{ij} = \Omega$ for all i, j, corresponding to a flat connection with no position-dependent frame misalignment. The geometric bias $S(\Omega)$ becomes constant across all agent pairs and cancels under softmax, while Ω^{-1} is absorbed into learned projections.

4.2.1 Untied Query-Key Carving from Per-Token Frames

The trivial-frame reduction below is one realization of the standard attention structure. A second, complementary reduction retains $\Omega_{ij} = U_i U_j^{-1}$ non-trivially throughout and exhibits a structural identification with standard scaled dot-product attention's $Q^\top K$ inner-product form, with untied query-key projections derived directly from the gauge-covariant KL and a content-dependent key-side bias absorbed into the additive prior slot. This is a generalization of Vaswani et al. (2017) §3.2.1, which uses uniform prior; the gauge framework's key-side bias is the absorbed j-only terms of the transported KL decomposition (Eq. 58) and is structurally analogous to the additive prior mechanisms (causal masking, ALiBi) discussed in Section 3.4.5.

Under arbitrary SPD covariances Σ_i (independent of the frames U_i), the transported Gaussian KL admits the exact decomposition

$$D_{\mathrm{KL}}(q_i \| \Omega_{ij\#} q_j) = \frac{1}{2}\Big[\log\det\Sigma_j - \log\det\Sigma_i + 2\log|\det U_i| - 2\log|\det U_j| - d_k + \mathrm{tr}(H_j P_i) + x_i^\top H_j x_i - 2x_i^\top k_j + r_j\Big], \tag{58}$$

with the common-frame quantities

$$x_i = U_i^{-1}\mu_i, \qquad P_i = U_i^{-1}\Sigma_i U_i^{-\top}, \qquad H_j = U_j^\top \Sigma_j^{-1} U_j, \qquad k_j = U_j^\top \Sigma_j^{-1}\mu_j, \qquad r_j = \mu_j^\top \Sigma_j^{-1}\mu_j. \tag{59}$$

Equation (58) is the unique inhomogeneous form taken by the gauge-covariant Gaussian KL when the covariance field is unconstrained and the frames are general $\mathrm{GL}(d_k)$ elements; it has been verified symbolically against the direct Gaussian KL to machine precision.

The cross term $-2x_i^\top k_j$ is the only coupling between query and key indices through the belief means. It carves cleanly as a dot product between per-token vectors,

$$x_i^\top k_j = (U_i^{-1}\mu_i)^\top (U_j^\top \Sigma_j^{-1}\mu_j) = Q_i^\top K_j, \tag{60}$$

where the gauge-derived query and key projections are

$$\boxed{Q_i = U_i^{-1}\mu_i, \qquad K_j = U_j^\top \Sigma_j^{-1}\mu_j.} \tag{61}$$

For the same token, the two projections satisfy $W_Q^{(i)} = U_i^{-1}$ and $W_K^{(i)} = U_i^\top \Sigma_i^{-1}$, with the inverse-transpose tying broken by the precision factor Σ_i^{-1}. The construction is therefore genuinely untied: W_Q and W_K are different functions of the per-token belief, not the same projection up to symmetry.

The implicit pair-dependent bilinear is

$$M_{ij} := \Omega_{ij}^{-\top}\Sigma_j^{-1} = U_i^{-\top} U_j^\top \Sigma_j^{-1}. \tag{62}$$

As (U_i, U_j) ranges over $\mathrm{GL}(d_k)^2$ and Σ_j over $\mathrm{SPD}(d_k)$, polar decomposition shows that for each pair (i,j) the bilinear M_{ij} can be realized as any element of $\mathrm{GL}(d_k)$. The expressive power of the gauge-derived bilinear is therefore identical to that of an independently learned $W_Q W_K^\top \in \mathrm{GL}(d_k)$ in standard attention. The image is per-pair: the family $\{M_{ij}\}_{i,j}$ is constrained by the shared factorization through the per-token $\{U_i, \Sigma_j\}$, exactly as $\{W_Q\mu_i \cdot W_K\mu_j\}$ is constrained by shared W_Q, W_K in standard attention. Generically M_{ij} is non-symmetric, recovering the asymmetry of $W_Q^\top W_K$ that the symmetric C^{-1} closure of the trivial-frame route does not. Although M_{ij} depends on the pair (i,j), its evaluation $\mu_i^\top M_{ij}\mu_j$ factorizes through per-token vectors via Eqs. (60)–(61), exactly as in standard attention.

Reduction to the standard form. The remaining terms in Eq. (58) fall into two groups under the row-softmax over j. Terms depending on j alone, namely r_j, $\log\det\Sigma_j$, and $\log|\det U_j|$, contribute a key-side prior bias that is absorbed into $\log\pi_{ij}$. Terms coupling i and j through the belief covariances rather than the means, namely $x_i^\top H_j x_i$ and $\mathrm{tr}(H_j P_i)$, are gauge-theoretic uncertainty corrections beyond vanilla attention; they vanish under the closure $\Sigma_j = U_j C U_j^\top$ for shared SPD C, which collapses H_j to the constant C^{-1}. Under this closure the carving in Eq. (61) becomes the symmetric form $Q_i = C^{-1/2}U_i^{-1}\mu_i$, $K_j = C^{-1/2}U_j^{-1}\mu_j$ with a single shared bilinear C^{-1}. The resulting attention rule is structurally $\mathrm{softmax}_j(Q_i^\top K_j/\sqrt{d_k} + b_{ij})V_j$ with the absorbed key-side bias $b_{ij} = -r_j/(2\tau) + \mathrm{const}$ and $r_j = \|C^{-1/2}U_j^{-1}\mu_j\|^2 = \|K_j\|^2$ the squared key norm. The framework therefore recovers the structural form of standard attention modulo this content-dependent additive bias; the strict Vaswani et al. (2017) §3.2.1 form with uniform prior is obtained only when the key-norm variation across j is approximately constant, e.g., under the layer-normalization or high-dimensional-concentration conditions discussed in Section 4.2.2.

Identification with rotary positional structure. The natural identification of the per-token frame U_i with a real transformer architecture is the per-position rotational frame of rotary positional embeddings (Su et al., 2024), in which $U_i \in \mathrm{O}(d_k)$ is a block-diagonal rotation depending on token position. In that case $U_i^{-1} = U_i^\top$ and the closure $\Sigma_j = U_j C U_j^\top$ holds automatically for any C commuting with the rotation block structure. The carving in Eq. (61) reduces to $Q_i = U_i^\top \mu_i$ and $K_j = U_j^\top \mu_j$, which is the rotation-modulated query-key projection used in rotary attention, with μ_i playing the role of the position-independent content vector. The recovery in this case is to RoPE-style attention, which is itself an extension of Vaswani et al. (2017) §3.2.1 with position-dependent gauge structure rather than the original uniform-prior base form. The trivial-frame specialization $U_i = U$ for all i removes the per-token positional variation; in that limit $\Omega_{ij} = I$ and the analysis of Section 4.2.2 below applies, with the learned bilinear M replacing the structural role played by per-token frame variation.

4.2.2 Derivation of Dot-Product Attention

Expanding the KL Divergence. Under constant gauge, the KL divergence reduces to

$$s_{ij} = D_{\mathrm{KL}}(q_i \| \Omega q_j) = \frac{1}{2\sigma^2}\|\Omega^{-1}\mu_i - \mu_j\|^2 + C, \tag{63}$$

where $C = S(\Omega)$ is independent of the belief means and cancels under softmax. Expanding the squared norm we have:

$$s_{ij} = \frac{1}{2\sigma^2}\|\Omega^{-1}\mu_i\|^2 + \frac{1}{2\sigma^2}\|\mu_j\|^2 - \frac{1}{\sigma^2}\mu_i^\top \Omega^{-\top}\mu_j + C. \tag{64}$$

Softmax Cancellations. For a fixed query agent i, the softmax over keys j is:

$$\beta_{ij} = \frac{\exp(-s_{ij}/\tau)}{\sum_k \exp(-s_{ik}/\tau)}. \tag{65}$$

The i-terms $\frac{1}{2\sigma^2}\|\Omega^{-1}\mu_i\|^2 + C$ are independent of j and therefore cancel between numerator and denominator of the softmax yielding:

$$\beta_{ij} \propto \exp\left(\frac{1}{\tau}\left[\frac{1}{\sigma^2}\mu_i^\top \Omega^{-\top}\mu_j - \frac{1}{2\sigma^2}\|\mu_j\|^2\right]\right). \tag{66}$$

Thus, the effective logit is:

$$\tilde{s}_{ij} = \frac{1}{\sigma^2}\mu_i^\top \Omega^{-\top}\mu_j - \frac{1}{2\sigma^2}\|\mu_j\|^2. \tag{67}$$

Key Bias Cancellation. The second term $-\frac{1}{2\sigma^2}\|\mu_j\|^2$ however, depends on key j and is a key-dependent bias on the raw embedding norms. This bias involves $\|\mu_j\|^2$ rather than $\|\Omega\mu_j\|^2$ since the Mahalanobis metric (50) already accounts for the gauge transport's effect on norms via the $(\Omega\Omega^\top)^{-1}$ factor in the distance measure. This residual bias arises purely from the embedding space itself. Two well-trodden mechanisms ensure its approximate cancellation:

(1) High-Dimensional Concentration: For embeddings in $\mathbb{R}^{d_k}$ with approximately independent components of per-component variance σ_0^2 (e.g., $\mu_j^{(i)} \sim \mathcal{N}(0, \sigma_0^2)$; here σ_0 denotes the embedding-component scale, distinct from the belief variance σ), the concentration of measure implies:

$$\|\mu_j\|^2 = \sum_{i=1}^{d_k} (\mu_j^{(i)})^2 = d_k \sigma_0^2 \pm O(\sigma_0^2 \sqrt{d_k}), \tag{68}$$

with relative fluctuations $O(1/\sqrt{d_k}) \to 0$ as d_k increases.

(2) Layer Normalization: Modern transformer architectures enforce constant key-norms explicitly via layer normalization, which normalizes $\|\mu_j\|$ to be constant across tokens. This makes the key bias:

$$-\frac{1}{2\sigma^2}\|\mu_j\|^2 \approx (\text{constant in } j), \tag{69}$$

which cancels under softmax.

However, on the gauge theory side, the algebraic origin of this cancellation can be understood partly through the Lie algebraic decomposition of the gauge group. The Lie algebra $\mathfrak{gl}(K)$ splits as a direct sum:

$$\mathfrak{gl}(K) = \mathfrak{sl}(K) \oplus \mathbb{R}, \tag{70}$$

where $\mathfrak{sl}(K)$ consists of traceless matrices (volume-preserving transformations) and $\mathbb{R}$ is the trace component (overall scaling). The key-norm bias $\|\mu_j\|^2$ depends on the magnitude of the embedding vector. Layer normalization, which centers and rescales μ_j to have constant norm, effectively quotients out the positive scaling degree of freedom from the embedding space, projecting representations onto a sub-manifold of constant norm. This eliminates the source of key-norm bias at its algebraic origin. Combined with the Mahalanobis metric (which handles the gauge transport's effect on norms through $(\Omega\Omega^\top)^{-1}$), this ensures that attention depends on belief content rather than representation magnitude.

Factorization into Query-Key Dot Product. The leading term in $\tilde{s}_{ij}$ is:

$$\frac{1}{\sigma^2}\mu_i^\top \Omega^{-\top} \mu_j. \tag{71}$$

We identify the product of W_Q, W_K with the gauge transport:

$$W_Q W_K^\top = \frac{1}{\sigma^2}\Omega^{-\top} \in \mathrm{GL}(d_k), \tag{72}$$

where d_k denotes the per-head dimension d_{head} under the multi-head construction of Section 4.4; in the single-head case $d_k = d_{\text{head}}$ trivially.

Such a factorization always exists for any invertible Ω: any $M \in \mathrm{GL}(d_k)$ can be written as $M = AB^\top$ for suitable $A, B \in \mathrm{GL}(d_k)$ (e.g., via SVD: $M = U\Lambda V^\top = (U\Lambda^{1/2})(V\Lambda^{1/2})^\top$, where invertibility of M ensures all singular values are positive, so A and B are themselves invertible).

This identification concerns the bilinear form $M = W_Q W_K^\top$ that determines the attention logits, not the individual matrices W_Q and W_K separately. In standard transformer implementations, the per-head projection matrices $W_Q^a, W_K^a \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ are rectangular (with $d_{\text{model}} \gg d_{\text{head}}$), and neither is individually invertible in the d_{model}-dimensional space. The ambient-space logit kernel for head a is the low-rank matrix $W_Q^a (W_K^a)^\top \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ with rank $\leq d_{\text{head}}$, which is not an element of $\mathrm{GL}(d_{\text{model}})$. To identify the gauge-theoretic $\mathrm{GL}(d_{\text{head}})$ structure, one factors each rectangular map via thin SVD as $W_Q^a = U_Q^a A_Q^a$ and $W_K^a = U_K^a A_K^a$, where $U_Q^a, U_K^a \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ have orthonormal columns (isometric subspace embeddings) and $A_Q^a, A_K^a \in \mathrm{GL}(d_{\text{head}})$ are invertible head-space maps. The invertible head-space kernel is then

$$M_h^a := A_Q^a (A_K^a)^\top \in \mathrm{GL}(d_{\text{head}}), \tag{73}$$

and the actual transformer logit kernel is its low-rank lift $W_Q^a (W_K^a)^\top = U_Q^a M_h^a (U_K^a)^\top$. The gauge-theoretic identification $\sigma^{-2}\Omega^{-\top} \leftrightarrow M_h^a$ operates at the level of the invertible head-space factor, not the ambient low-rank kernel. In the block-diagonal multihead gauge model (Section 4.4), the isometric factors U_Q^a, U_K^a can be interpreted as selecting head subspaces, consistent with a $\mathrm{GL}(d_{\text{head}})^H \subset \mathrm{GL}(d_k)$ decomposition. The identification is value-level on the head subspace: the bilinear form $\mu_i^\top M_h^a \mu_j$ equals the standard attention logit on the head subspace, and the SVD identity $W_Q^a (W_K^a)^\top = U_Q^a M_h^a (U_K^a)^\top$ is exact for any rectangular factorization. It does not assert parameter-level identity between the gauge framework's learned (σ, Ω) and the standard transformer's atomically-learned W_Q^a, W_K^a, which are different parameterizations of the same bilinear form under different optimizer geometries (natural gradient on the gauge manifold versus stochastic gradient on the rectangular projection space).

Scope of the head-space identification. The gauge framework's identification covers the invertible per-head bilinear M_h^a on the head subspace and the value aggregation up to the W_O absorption documented in §4.2.3. It does *not* derive the rectangular subspace embeddings U_Q^a, U_K^a, U_V^a, which select which d_{head}-dimensional subspace of the d_{model}-dimensional embedding space each head reads from. Subspace selection is a structural design choice with no analog in the (σ, Ω) parameterization and is not predicted by the framework. The block-diagonal $\mathrm{GL}(d_{\text{head}})^H$ structure of the head-space kernels is derived; the choice of rectangular embedding factors is structural and orthogonal to the gauge derivation.

Next, rather than completely taking $\sigma \to 0$ we recognize that σ^{-2} and $\Omega^{-\top}$ always appear together in the combination $\sigma^{-2}\Omega^{-\top}$. The learned matrices W_Q, W_K can then be considered to parametrize this combined quantity directly, rendering σ an implicit (finite) scale factor absorbed into the learned weights. Therefore, the full limit need not be taken.

Next, we define query and key vectors as:

$$Q_i \equiv \mu_i^\top W_Q \in \mathbb{R}^{1 \times d_k}, \qquad K_j \equiv \mu_j^\top W_K \in \mathbb{R}^{1 \times d_k}. \tag{74}$$

Then:

$$Q_i K_j^\top = \mu_i^\top W_Q W_K^\top \mu_j = \frac{1}{\sigma^2} \mu_i^\top \Omega^{-\top} \mu_j. \tag{75}$$

Thus the logit term reduces to a standard dot product as:

$$\tilde{s}_{ij} = Q_i K_j^\top + O(1), \tag{76}$$

where $O(1)$ represents the approximately constant key bias (controlled by layer normalization).

Temperature Scaling. Since σ^{-2} is absorbed into the learned projections W_Q, W_K, the effective logit $\tilde{s}_{ij} = Q_i K_j^\top$ carries no residual dependence on σ. The softmax $\beta_{ij} = \text{softmax}_j(\tilde{s}_{ij}/\tau)$ must therefore set τ to account only for the dimensional scaling of the dot product.

In high-dimensional spaces, dot products $Q_i K_j^\top$ have standard deviation $O(\sqrt{d_k})$ (for unit-variance components), while key-bias fluctuations are $O(1)$ after layer normalization. Normalizing pre-softmax logits to $O(1)$ variance therefore requires:

$$\tau = \sqrt{d_k}. \tag{77}$$

Combining all simplifications we have:

$$\boxed{\beta_{ij} = \text{softmax}_j\left(\frac{Q_i K_j^\top}{\sqrt{d_k}}\right)}, \tag{78}$$

recovering the standard transformer attention weighting rule, up to the key-norm bias term that cancels under the layer normalization or high-dimensional concentration conditions established above.

4.2.3 Value Aggregation

The mixture-of-sources generative model (Section 3.4) specifies that the component distribution for source j in agent i's frame has mean $\Omega_{ij}\mu_j$. The posterior mean under mixture responsibilities β_{ij} is therefore:

$$\hat{\mu}_i = \sum_j \beta_{ij}\Omega_{ij}\mu_j, \tag{79}$$

which is the standard mixture-of-Gaussians expectation. This is the aggregation consistent with the generative model.

A separate geometric object arises in the attention weight computation, where the Mahalanobis identity (50) introduces $\Omega^{-\top}$ as the natural transform for the dual (covector) structure of the inner product. Specifically, $\Omega^{-\top}$ appears in the logit $\mu_i^\top \Omega^{-\top}\mu_j$ because the KL divergence measures distance in the transported covariance metric $(\Omega\Omega^\top)^{-1}$, and $\Omega^{-\top}$ is the correct transform for covectors (dual vectors) under a linear map. To be clear, this dual transform belongs to the attention scoring geometry, not to the value aggregation.

Next, we define the value projection:

$$V_j \equiv W_V^\top \mu_j, \qquad W_V \in \mathbb{R}^{d_k \times d_k}, \tag{80}$$

where d_k here denotes the per-head dimension d_{head} under the multi-head construction of Section 4.4, paralleling the convention used at Eq. (73) above.

We absorb the gauge transport Ω into the learned matrix W_V at the head-space level. The thin-SVD lift documented for the query and key projections at Eq. (73) applies to the value projection by parallel construction: the standard transformer's rectangular $W_V^a \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ factors as $W_V^a = U_V^a A_V^a$ with $U_V^a \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ an isometric subspace embedding and $A_V^a \in \text{GL}(d_{\text{head}})$ the invertible head-space map, where the isometric factor is absorbed into the per-head slice of the multi-head output projection W_O under the standard concat-then-W_O construction.

Then:

$$\boxed{\hat{\mu}_i = \sum_j \beta_{ij} V_j}, \tag{81}$$

identical to the standard transformer attention update $z_i = \sum_j \alpha_{ij} V_j$.

4.2.4 Complete Attention Formula

Standard scaled dot-product attention is recovered under three preconditions established in the preceding paragraphs: constant gauge $\Omega_{ij} = \Omega$ for all agent pairs, isotropic covariance $\Sigma_i = \sigma^2 I$ with the joint factor $\sigma^{-2}\Omega^{-\top}$ absorbed into the learned projections so that $W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$, and key-norm constancy supplied either by layer normalization or by high-dimensional concentration of $\|\mu_j\|^2$. The first two preconditions are exact algebraic specializations of the general framework: constant gauge renders the geometric bias $S(\Omega)$ pair-independent and cancels it under softmax shift invariance, and the $\sigma^{-2}\Omega^{-\top}$ absorption is a literal change of parameterization rather than an analytic $\sigma \to 0$ limit. The third precondition is the chain's one approximate step, recorded with $\approx$ at the layer-normalization equation derived above; it is an exact softmax-shift cancellation only when key-norm fluctuations across the keys vanish, and approximate otherwise.

Under these three preconditions, the gauge-theoretic message communication

$$\beta_{ij} \propto \exp\left(-\frac{1}{\tau} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)\right), \quad \hat{\mu}_i = \sum_j \beta_{ij} \Omega_{ij} \mu_j, \tag{82}$$

reduces to

$$\boxed{\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V}, \tag{83}$$

the canonical scaled dot-product attention of Vaswani et al. (2017). The gauge identification operates on the invertible $\mathrm{GL}(d_{\mathrm{head}})$ head-space factor M_h^a of the rectangular per-head projections (Eq. (73)); the ambient transformer kernel $W_Q^a (W_K^a)^\top$, of rank at most d_{head}, is the low-rank lift of this head-space kernel through the isometric subspace embeddings U_Q^a, U_K^a.

The limits are deliberately aggressive. They collapse the statistical manifold into a basic Euclidean space and absorb the gauge parameters into the learned matrices.

Each limit discards a specific geometric degree of freedom and each can be independently relaxed to produce a generalization of standard attention. The gauge-theoretic framework provides a taxonomy of design choices whereby standard attention corresponds to the three-limit simplification, while much richer architectures (such as non-isotropic beliefs, learned gauge transport, position-dependent connections, etc) occupy intermediate points which may have unique and interesting applications.

Our $GL(K)$ experiments (Section 5.3) utilize the full generality (at least for a 0 dimensional manifold) and demonstrate that meaningful language modeling is possible using the complete gauge-theoretic variational free energy alone without incorporating standard neural network architectures.

4.3 Training as Variational Free Energy Minimization

Having established that attention emerges from the variational free energy, we now show that, in the deterministic constant-gauge limit, gradient descent on this free energy recovers structures analogous to standard transformer training dynamics, providing a variational interpretation of each component.

An important distinction must be maintained between two operations that the free energy framework treats uniformly but that standard transformers implement differently. In variational inference, minimizing $\mathcal{F}$ with respect to the variational density q constitutes inference (the E-step), while updating generative model parameters constitutes learning (the M-step). In a standard transformer, the forward pass computes per-token hidden states via deterministic layer maps (analogous to amortized inference whereby a single learned function replaces iterative optimization of q), while gradient descent on weights across data batches constitutes learning. The gradient expressions below describe the structure of both operations since they share the same functional form. However, what differs is in which variables are updated and on what timescale. The forward pass corresponds to a single step of belief inference with frozen parameters, and weight training corresponds to slow parameter updates that reduce the expected free energy across data.

4.3.1 Observation Likelihood as Loss Function.

In our gauge theory, observations by agents act as a source term that breaks the vacuum degeneracy. The vacuum free energy (without observations), written with β_{ij} at its optimal softmax value, takes the reduced form (cf. Eq. 35):

$$\mathcal{F}_{\text{vacuum}}[\{q_i\}] = \sum_i \alpha_i D_{\text{KL}}(q_i \| p_i) - \tau \sum_i \log Z_i. \tag{84}$$

Introducing per-agent observations o_i adds the likelihood term:

$$\mathcal{F}[\{q_i\}] = \mathcal{F}_{\text{vacuum}}[\{q_i\}] - \sum_i \mathbb{E}_{q_i}[\log p(o_i \mid k_i)]. \tag{85}$$

In the deterministic limit $q_i \to \delta(k_i - \mu_i)$, this becomes:

$$-\mathbb{E}_{q_i}[\log p(o_i \mid k_i)] \to -\log p(o_i \mid \mu_i), \tag{86}$$

the standard negative log-likelihood.

For Gaussian observations $p(o \mid \mu) = \mathcal{N}(o \mid \mu, \Sigma)$:

$$-\log p(o \mid \mu) = \frac{1}{2}(o - \mu)^\top \Sigma^{-1}(o - \mu) + \text{const}. \tag{87}$$

$$\implies \quad \mathcal{L}_{\text{obs}} = \frac{1}{2}\|o - \mu\|^2_\Sigma \quad \text{(Mean-squared error)}. \tag{88}$$

For categorical observations $p(o \mid \mu) = \text{Categorical}(\text{softmax}(\mu))$:

$$-\log p(o \mid \mu) = -\sum_k o_k \log(\mathrm{softmax}(\mu)_k) \quad \text{(Cross-entropy loss)}. \tag{89}$$

These are standard loss functions in machine learning.

Deterministic Free Energy. In the deterministic limit with flat bundle ($\Omega_{ij} = \Omega$), each belief becomes a narrow Gaussian $q_i = \mathcal{N}(\mu_i, \epsilon I)$ with $\epsilon \to 0^+$, and each prior is isotropic: $p_i = \mathcal{N}(\mu_{\text{prior}}, \sigma_p^2 I)$. The self-coupling term from the adaptive free energy (38) decomposes as:

$$\alpha_i D_{\text{KL}}(\mathcal{N}(\mu_i, \epsilon I) \| \mathcal{N}(\mu_{\text{prior}}, \sigma_p^2 I)) = \frac{\alpha_i}{2\sigma_p^2} \|\mu_i - \mu_{\text{prior}}\|^2 + C(\epsilon), \tag{90}$$

where $C(\epsilon)$ collects μ_i-independent terms (involving ϵ/σ_p^2 and $\log(\sigma_p^2/\epsilon)$) that diverge as $\epsilon \to 0$ but do not contribute to gradients with respect to the belief parameters. The effective gradient contribution of the prior KL is therefore determined by the μ_i-dependent part $\frac{\alpha_i}{2\sigma_p^2}\|\mu_i - \mu_{\text{prior}}\|^2$. Using this μ_i-dependent part in the state-dependent precision (41) gives:

$$\alpha_i = \frac{c_0}{b_0 + \dfrac{1}{2\sigma_p^2}\|\mu_i - \mu_{\text{prior}}\|^2}. \tag{91}$$

Defining the per-token regularization strength $\lambda_{p,i} \equiv \alpha_i/\sigma_p^2$, the full deterministic free energy is:

$$\mathcal{F}[\{\mu_i\}] = \underbrace{\sum_i \frac{\lambda_{p,i}}{2}\|\mu_i - \mu_{\text{prior}}\|^2}_{\text{Adaptive prior regularization}} + \underbrace{\sum_{i,j} \frac{\beta_{ij}}{2\sigma^2}\|\Omega^{-1}\mu_i - \mu_j\|^2}_{\text{Alignment coupling}} - \underbrace{\sum_i \log p(o_i \mid \mu_i)}_{\text{Observation likelihood}}. \tag{92}$$

As previously discussed this form constitutes an adaptive weight decay whereby tokens whose beliefs have drifted far from the prior ($\|\delta_i\|$ large) experience weaker regularization ($\lambda_{p,i}$ small, gate open), while tokens near the prior are strongly anchored ($\lambda_{p,i}$ large, gate closed).

In the limit where $\alpha_i \to \alpha = \text{const}$ and $\lambda_{p,i} \to \lambda_p \equiv \alpha/\sigma_p^2$ we recover uniform L2 regularization:

$$\mathcal{F}[\{\mu_i\}] = \underbrace{\sum_i \frac{\lambda_p}{2}\|\mu_i - \mu_{\text{prior}}\|^2}_{\text{L2 regularization (weight decay)}} + \underbrace{\sum_{i,j} \frac{\beta_{ij}}{2\sigma^2}\|\Omega^{-1}\mu_i - \mu_j\|^2}_{\text{Alignment coupling}} - \underbrace{\sum_i \log p(o_i \mid \mu_i)}_{\text{Observation likelihood}}. \tag{93}$$

This identifies the standard weight decay coefficient λ_p as the product of two quantities that the general theory treats independently: the prior precision α and the inverse prior variance $1/\sigma_p^2$. In the full theory, both are generally state-dependent.

4.3.2 Gradient Descent Dynamics.

In the limit $\lambda_{p,i} = \lambda_p$ the gradient of the free energy with respect to μ_i is:

$$\boxed{\begin{aligned}\frac{\partial \mathcal{F}}{\partial \mu_i} &= \lambda_p \left(\mu_i - \mu_{\text{prior}}\right) \\ &\quad + \sum_j \left[\frac{\beta_{ij}}{\sigma^2}(\Omega\Omega^\top)^{-1}\left(\mu_i - \Omega\mu_j\right) + \frac{1}{2\sigma^2}\frac{\partial \beta_{ij}}{\partial \mu_i}\left\|\Omega^{-1}\mu_i - \mu_j\right\|^2\right] \\ &\quad - \frac{\partial \log p(o_i \mid \mu_i)}{\partial \mu_i}.\end{aligned}} \tag{94}$$

The first term, $\lambda_p(\mu_i - \mu_{\text{prior}})$, as discussed, matches weight decay or L2 regularization. The second term, $\sum_j \frac{\beta_{ij}}{\sigma^2}(\Omega\Omega^\top)^{-1}(\mu_i - \Omega\mu_j)$, is the gradient of the attention-weighted alignment energy with respect to μ_i: it pulls μ_i toward its gauge-transported neighbors $\Omega_{ij}\mu_j$, and under natural-gradient preconditioning together with the covariance-alignment fixed point (Eq. 98 and the discussion following) the resulting step direction targets the mixture-posterior aggregation $\hat{\mu}_i$ of Eq. 79. The third term, $\sum_j \frac{\partial \beta_{ij}}{\partial \mu_i}\|\Omega^{-1}\mu_i - \mu_j\|^2/(2\sigma^2)$, represents gradient flow through the attention weights themselves—a manifestly non-linear "activation" that arises from differentiating the attention-weighted energy $\sum_j \beta_{ij}E_{ij}$ via the product rule (the autograd gradient). This term is absent from the gradient of the reduced free energy $\mathcal{F}_{\text{red}}$ by the envelope theorem (Section 3.5), but is present in standard autograd-based implementations that differentiate through the softmax. The fourth term, $-\frac{\partial \log p(o_i|\mu_i)}{\partial \mu_i}$, is the gradient of the loss function (i.e. the supervised learning signal).

Since beliefs q_i live on the Gaussian statistical manifold, the geometrically correct dynamics uses the natural gradient. The Euclidean gradient must be pre-conditioned by the inverse Fisher information metric which, for MVGs is $G_\mu^{-1} = \Sigma_i$ (see Supplementary Appendix D for derivation). The natural gradient flow is:

$$\frac{d\mu_i}{dt} = -\eta\Sigma_i \frac{\partial \mathcal{F}}{\partial \mu_i}. \tag{95}$$

In the isotropic limit $\Sigma_i = \sigma^2 I$, the Fisher preconditioning absorbs the σ^{-2} factors in the alignment terms. Defining the effective learning rate $\tilde{\eta} \equiv \eta\sigma^2$ and discretizing yields:

$$\boxed{\begin{aligned}\mu_i^{(t+1)} = \mu_i^{(t)} - \tilde{\eta}\Bigg[&\lambda_p\left(\mu_i - \mu_{\text{prior}}\right) + \sum_j \left(\beta_{ij}(\Omega\Omega^\top)^{-1}\left(\mu_i - \Omega\mu_j\right) + \frac{1}{2}\frac{\partial \beta_{ij}}{\partial \mu_i}\left\|\Omega^{-1}\mu_i - \mu_j\right\|^2\right) \\ &- \frac{\partial \log p(o_i \mid \mu_i)}{\partial \mu_i}\Bigg].\end{aligned}} \tag{96}$$

The natural gradient recovers the standard transformer update rule under a chain of limits. Fisher preconditioning with $\Sigma_i = \sigma^2 I$ cancels the scalar σ^{-2} factors in the boxed update (Eq. 102 above), leaving $\beta_{ij}(\Omega\Omega^\top)^{-1}(\mu_i - \Omega\mu_j)$. Absorbing $(\Omega\Omega^\top)^{-1}$ into the learned key/query projection $W_Q W_K^\top$ (Limit 2 of Section 4.2) and the constant gauge Ω into W_V

(Limit 3) further reduces the step direction to $\sum_j \beta_{ij} V_j$, identifying it with the value aggregation $\sum_j \alpha_{ij} V_j$. This is consistent with the E-step algorithm we implement, which applies Fisher preconditioning $\tilde{\nabla}_\mu \leftarrow \Sigma_i \cdot \nabla_\mu \mathcal{F}$ without imposing the additional projection-absorption limits.

General (non-isotropic) mean gradient. The preceding expressions assumed isotropic covariances $\Sigma_i = \sigma^2 I$ and a shared gauge connection Ω. We now state the fully general gradient, valid for arbitrary covariances Σ_i, edge-dependent gauge connections Ω_{ij}, general prior precision $\Sigma_{p,i}^{-1}$, and state-dependent prior coupling α_i (Section 3.7).

Differentiating the attention-weighted energy $\sum_j \beta_{ij} E_{ij}$ (the autograd objective; cf. Section 3.5) with respect to μ_i and applying the product rule through both the attention weights $\beta_{ij}(\mu_i)$ and the adaptive prior coupling $\alpha_i(\mu_i)$ gives:

$$\boxed{\begin{aligned}\frac{\partial \mathcal{F}}{\partial \mu_i} &= \alpha_i \Sigma_{p,i}^{-1} (\mu_i - \mu_{\text{prior}}) + \frac{\partial \alpha_i}{\partial \mu_i} D_{\text{KL}}(q_i \| p_i) \\ &\quad + \sum_j \left[\beta_{ij} \left(\Omega_{ij} \Sigma_j \Omega_{ij}^\top \right)^{-1} (\mu_i - \Omega_{ij} \mu_j) + \frac{\partial \beta_{ij}}{\partial \mu_i} D_{\text{KL}}(q_i \| \Omega_{ij} q_j) \right] \\ &\quad - \frac{\partial \log p(o_i \mid \mu_i)}{\partial \mu_i}.\end{aligned}} \tag{97}$$

The structure parallels the isotropic case term-by-term, with four generalizations: (i) the prior precision $\lambda_p I$ is replaced by the full matrix $\alpha_i \Sigma_{p,i}^{-1}$, where α_i is the state-dependent prior coupling; (ii) the term $(\partial \alpha_i / \partial \mu_i) D_{\text{KL}}(q_i \| p_i)$ represents gradient flow through the adaptive gate whereby agents far from their prior (large $\|\mu_i - \mu_{\text{prior}}\|$) have $\partial \alpha_i / \partial \mu_i < 0$, which reduces the effective regularization and allows beliefs to drift further, reinforcing the gating behavior described in Section 3.7; (iii) the isotropic transported precision $\sigma^{-2}(\Omega \Omega^\top)^{-1}$ is replaced by the edge-dependent transported precision $(\Omega_{ij} \Sigma_j \Omega_{ij}^\top)^{-1}$, which accounts for both the neighbor's covariance structure and the geometry of the gauge transport along each edge; and (iv) the quadratic mean-mismatch $\frac{1}{2\sigma^2} \|\Omega^{-1} \mu_i - \mu_j\|^2$ is replaced by the full KL divergence $D_{\text{KL}}(q_i \| \Omega_{ij} q_j)$, which additionally couples to the covariance mismatch between Σ_i and the transported covariance $\Omega_{ij} \Sigma_j \Omega_{ij}^\top$. The full KL cannot be reduced to a purely mean-dependent expression because the covariance contributions (trace and log-determinant terms) are j-dependent through Ω_{ij} and Σ_j, and therefore do not cancel under the constraint $\sum_j \partial \beta_{ij} / \partial \mu_i = 0$. As noted in Section 3.5, the $(\partial \beta_{ij} / \partial \mu_i) D_{\text{KL}}$ terms arise from differentiating the attention-weighted energy and are absent from the gradient of the reduced free energy $\mathcal{F}_{\text{red}}$ by the envelope theorem.

When $\alpha_i = \alpha = \text{const}$, the gate gradient vanishes ($\partial \alpha_i / \partial \mu_i = 0$) and the prior term reduces to $\alpha \Sigma_{p,i}^{-1} (\mu_i - \mu_{\text{prior}})$, recovering the uniform regularization case.

The corresponding natural gradient flow is:

$$
\boxed{\begin{aligned}
\frac{d\mu_i}{dt} &= -\eta \Sigma_i \frac{\partial \mathcal{F}}{\partial \mu_i} \\
&= -\eta \Bigg[\Sigma_i \Sigma_{p,i}^{-1}(\mu_i - \mu_{\text{prior}}) + \sum_j \Bigg(\beta_{ij} \Sigma_i \left(\Omega_{ij} \Sigma_j \Omega_{ij}^\top \right)^{-1} (\mu_i - \Omega_{ij}\mu_j) \\
&\qquad + \frac{\partial \beta_{ij}}{\partial \mu_i} \Sigma_i D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j) \Bigg) - \Sigma_i \frac{\partial \log p(o_i \mid \mu_i)}{\partial \mu_i} \Bigg].
\end{aligned}} \tag{98}
$$

When the covariance alignment condition $\Sigma_i \approx \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ holds (as enforced by the covariance gradient dynamics below), the Fisher-preconditioned alignment term simplifies to $\beta_{ij}(\mu_i - \Omega_{ij}\mu_j)$, recovering attention-weighted message passing with geometric transport (the non-isotropic generalization of the value aggregation $\sum_j \alpha_{ij} V_j$). This covariance alignment condition is precisely the fixed point of the Σ-gradient (Eq. 100), demonstrating internal consistency whereby the mean and covariance dynamics cooperate to produce clean message passing at equilibrium.

Covariance gradient. Similarly, the complete theory also includes gradient dynamics for the covariance Σ_i and gauge frame ϕ_i. The covariance gradient reveals a novel uncertainty channel absent in standard transformers. Differentiating the attention-weighted energy with respect to Σ_i and applying the product rule through both the attention weights and the adaptive prior coupling $\alpha_i(\Sigma_i)$ (neglecting the observation term, which contributes an $O(1)$ correction that is small compared to the $O(\Sigma_i^{-1})$ alignment terms in the high-precision regime; see Supplementary Appendix B for the full derivation) yields:

$$
\boxed{\begin{aligned}
\frac{\partial \mathcal{F}_i}{\partial \Sigma_i} &= \frac{1}{2} \left[-(1+\alpha_i)\Sigma_i^{-1} + \alpha_i \Sigma_{p,i}^{-1} + \sum_j \beta_{ij} (\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1} \right] \\
&\qquad + \frac{\partial \alpha_i}{\partial \Sigma_i} D_{\mathrm{KL}}(q_i \| p_i) + \sum_j \frac{\partial \beta_{ij}}{\partial \Sigma_i} D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j).
\end{aligned}} \tag{99}
$$

This parallels the mean gradient term-by-term: $\alpha_i \Sigma_{p,i}^{-1}$ is the prior precision scaled by the adaptive coupling (analogous to weight decay for μ); the $(\partial\alpha_i/\partial\Sigma_i) D_{\mathrm{KL}}(q_i\|p_i)$ term is gradient flow through the adaptive gate (the covariance analogue of the $(\partial\alpha_i/\partial\mu_i) D_{\mathrm{KL}}$ term in Eq. 97); $\sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}$ is attention-weighted precision aggregation that pulls agent i's precision toward its gauge-transported neighbors; and $\sum_j (\partial\beta_{ij}/\partial\Sigma_i) D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ is gradient flow through the attention weights (present in the autograd gradient but absent from the gradient of $\mathcal{F}_{\text{red}}$ by the envelope theorem; cf. Section 3.5). In the general case (Eq. 41), $\alpha_i = c_0/(b_0 + D_{\mathrm{KL}}(q_i\|p_i))$ depends on Σ_i through the trace and log-determinant terms of the full KL divergence, so $\partial\alpha_i/\partial\Sigma_i \neq 0$. This gate gradient term vanishes only in the deterministic limit (Eq. 91), where α_i reduces to a function of μ_i alone. The coefficient $-(1+\alpha_i)$ on Σ_i^{-1} arises because the entropy of q_i enters both the prior KL (weighted by α_i) and each neighbor KL (weighted by β_{ij}, summing to 1).

Since $\sum_j \partial\beta_{ij}/\partial\Sigma_i = 0$ (the attention weights sum to unity), the attention weight correction vanishes whenever the KL divergences $D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ are uniform across neighbors.

It also vanishes in both the hard-attention ($\tau \to 0$, where β_{ij} concentrates on a single neighbor) and uniform-attention ($\tau \to \infty$, where $\partial\beta_{ij}/\partial\Sigma_i \propto 1/\tau \to 0$) limits. Setting this correction aside, the equilibrium condition $\partial\mathcal{F}_i/\partial\Sigma_i = 0$ yields the fixed-point equation:

$$\Sigma_i^{-1} = \frac{1}{2}\left[\Sigma_{p,i}^{-1} + \sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}\right]. \tag{100}$$

Each agent's precision is thus determined self-consistently as a β-weighted combination of its prior precision and its neighbors' transported precisions. When the state-dependent prior coupling is weak ($\alpha_i \ll 1$; cf. Section 3.7), the prior term becomes negligible. Simultaneously, the entropy contribution from the prior KL vanishes, changing the coefficient on Σ_i^{-1} from -2 to -1 in the gradient. The fixed point then reduces to $\Sigma_i^{-1} \approx \langle(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}\rangle_\beta$, the attention-weighted average of transported neighbor precisions. More generally, with state-dependent precision α_i (adaptive weight decay), the fixed-point interpolates as:

$$\Sigma_i^{-1} = \frac{\alpha_i}{1+\alpha_i}\Sigma_{p,i}^{-1} + \frac{1}{1+\alpha_i}\sum_j \beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}, \tag{101}$$

a convex combination that continuously interpolates between prior-dominated ($\alpha_i \gg 1$, precision $\approx \Sigma_{p,i}^{-1}$) and alignment-dominated ($\alpha_i \ll 1$, precision $\approx \langle(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}\rangle_\beta$) regimes. Since α_i is itself state-dependent (Eq. 91), agents whose beliefs have drifted far from the prior (large $\|\mu_i - \mu_{\text{prior}}\|$) have small α_i and therefore defer to their neighbors' precisions, while agents near the prior retain strong prior anchoring. This then is the covariance analogue of the adaptive weight decay described in Section 3.7. This fixed point is an attractor of the gradient dynamics (Supplementary Appendix B), so covariance alignment $\Sigma_i \approx \Omega_{ij}\Sigma_j\Omega_{ij}^\top$ emerges from the variational dynamics itself rather than as an imposed constraint. This uncertainty channel where precision propagates through the attention graph has no counterpart in standard transformers and constitutes a concrete prediction of the general theory.

Gauge frame gradient. The gauge frame gradient $\partial\mathcal{F}/\partial\phi_i$ drives learning of the parallel transport operators $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$. Its structure involves the differential of the matrix exponential map and differs for compact ($\mathrm{SO}(N)$) and general ($\mathrm{GL}(K)$) gauge groups; the full derivation is given in Supplementary Appendix C. In practice, automatic differentiation through the matrix exponential handles these gradients implicitly however we derive them in detail.

4.3.3 Residual Connections as Variational Gradient Flow.

The residual (skip) connection is a defining architectural feature of modern transformers, where each sublayer's output is added to its input: $x_{\ell+1} = x_\ell + f(x_\ell)$. This structure emerges directly from the variational gradient flow. Identifying each transformer layer ℓ with one step of natural-gradient descent on the free energy gives the Euler discretization:

$$\mu_i^{(\ell+1)} = \mu_i^{(\ell)} - \tilde{\eta}\Sigma_i \frac{\partial\mathcal{F}}{\partial\mu_i}\bigg|_{\mu_i^{(\ell)}}, \tag{102}$$

where the Fisher preconditioning Σ_i and effective learning rate $\tilde{\eta} = \eta\sigma^2$ are as derived above. This inherently has the form new = old + correction. The agent-based variational model was derived from first principles, and this residual structure emerged as a consequence of the gradient flow before the connection to transformer architecture was recognized. That the same structure appears as a deliberate architectural choice in transformers (motivated by gradient propagation considerations) suggests both frameworks capture the same underlying computational principle. The backward gradient flow through the Jacobians $\partial\mu_j^{(\ell+1)}/\partial\mu_i^{(\ell)}$ corresponds to variational message-passing, where each layer refines its beliefs based on downstream prediction errors.

The strength of the identity path relative to the correction is governed by the state-dependent precision α_i (Section 3.7). When α_i is large (agent near its prior), the identity path dominates and the belief is strongly anchored; when α_i is small (agent far from prior), the correction terms dominate. In the constant-α limit, this reduces to a fixed-strength residual connection with uniform regularization, as in standard transformers where the skip connection weight is implicitly unity. The adaptive α_i thus generalizes the fixed residual connection to a state-dependent gating mechanism.

The full variational flow for all agent parameters takes the form:

$$\theta_i^{(\ell+1)} = \theta_i^{(\ell)} - \eta_\theta \tilde{\nabla}_{\theta_i} \mathcal{F}\Big|_{\theta_i^{(\ell)}}, \qquad \theta_i \in \{\mu_i, \Sigma_i, \phi_i\}, \tag{103}$$

where $\tilde{\nabla}_{\theta_i}$ denotes the natural gradient on each channel's parameter manifold (Fisher-Rao for μ_i and Σ_i; for $\phi_i \in \mathfrak{gl}(K) = \mathfrak{sl}(K) \oplus \mathbb{R}\cdot I$ the Killing form is degenerate on the central direction, so we use the Cartan-involution-modified bilinear form $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(X\theta(Y))$ with $\theta(X) = -X^\top$, which is positive-definite on the full algebra; see Supplementary Appendix C). Each channel possesses its own residual structure and effective learning rate. This multi-channel residual flow corresponds to the residual stream interpretation of transformers (Elhage et al., 2021), where each layer reads from and writes to a shared residual pathway. In the gauge-theoretic framework, this residual stream is the belief state $(\mu_i, \Sigma_i, \phi_i)$, accumulating incremental updates from successive layers of variational refinement.

Layers as VFE iterations. The identification $\ell \leftrightarrow$ gradient step makes the notion of "depth" a statement about how many steps of free energy minimization are performed. A standard L-layer transformer corresponds to L natural-gradient steps, each with independently parameterized attention weights $\beta_{ij}^{(\ell)}$ (since $W_Q^{(\ell)}, W_K^{(\ell)}, W_V^{(\ell)}$ differ across layers). Within a single block one may also iterate the belief update T times with shared weights, recomputing β_{ij} after each step; this gives T inner VFE iterations per layer and LT effective gradient steps in total. The standard transformer is the $T = 1$ case. The opposite extreme, $L = 1$ with $T \to \infty$, recovers a deep equilibrium model (Bai et al., 2019) in which the forward pass iterates to a fixed point of the variational gradient. Intermediate regimes interpolate between fast feedforward evaluation ($T = 1$, many layers) and slow iterative inference ($T \gg 1$, few layers), with the total depth of belief refinement governed by the product LT.

4.4 Multi-Head Attention as Block-Diagonal $\mathrm{GL}(K)$ Structure

Standard transformer architectures employ multi-head attention, partitioning the d_k-dimensional embedding space into H independent heads (Vaswani et al., 2017):

$$\mu_i = [h_i^1, h_i^2, \ldots, h_i^H], \quad h_i^a \in \mathbb{R}^{d_{\text{head}}}, \quad d_k = H \times d_{\text{head}}. \tag{104}$$

Each head computes attention independently using separate projection matrices (W_Q^a, W_K^a, W_V^a), and the results are concatenated. Under the gauge-theoretic identification, this corresponds to a block-diagonal restriction within the full $\mathrm{GL}(d_k)$ gauge group.

4.4.1 Block-Diagonal Gauge Structure in $\mathrm{GL}(d_k)$.

Recall from Section 2.1.3 that, in the isotropic flat-bundle limit, standard transformer attention can be interpreted as $\mathrm{GL}(d_k)$ gauge-theoretic attention with the effective gauge transport identified as $\Omega = (\sigma^2 W_K W_Q^\top)^{-1} \in \mathrm{GL}(d_k)$. Multi-head attention restricts this to a block-diagonal subgroup:

$$\Omega = \begin{pmatrix} \Omega^1 & 0 & \cdots & 0 \\ 0 & \Omega^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Omega^H \end{pmatrix}, \quad \Omega^a = (\sigma^2 W_K^a (W_Q^a)^\top)^{-1} \in \mathrm{GL}(d_{\text{head}}), \tag{105}$$

where each head a learns its own $\mathrm{GL}(d_{\text{head}})$ gauge transformation. This block structure means that components in different heads do not mix under gauge transport. Each head implements an independent gauge-theoretic attention mechanism. Each per-head transport $\Omega^a \in \mathrm{GL}(d_{\text{head}})$ is full rank and invertible within its block, so the Mahalanobis metric correction $(\Omega^a)^{-\top}$ from Eq. (50) remains well-defined. The rectangular shape of the standard per-head projection matrices $W_Q^a \in \mathbb{R}^{d_k \times d_{\text{head}}}$ can be factored via thin SVD as $W_Q^a = U_Q^a A_Q^a$, where U_Q^a is an isometric subspace embedding and $A_Q^a \in \mathrm{GL}(d_{\text{head}})$ is the invertible head-space map (cf. Eq. 73). In the gauge-theoretic interpretation, the isometric factor selects the head subspace (analogous to the block projection), while the invertible factor carries the $\mathrm{GL}(d_{\text{head}})$ gauge transport.

The full multi-head gauge group is thus the direct product:

$$G_{\text{multi-head}} = \mathrm{GL}(d_{\text{head}})^H \subset \mathrm{GL}(d_k), \tag{106}$$

a $(H \cdot d_{\text{head}}^2)$-dimensional subgroup of the d_k^2-dimensional $\mathrm{GL}(d_k)$.

Per-head holonomy. The vertex-frame parameterization adopted throughout the main paper places the model in Regime I, where the holonomy vanishes identically by Lemma 2 and the discussion below is vacuous as written. The companion paper (Dennis, 2025) develops the edge-relaxed Regime II construction in detail; under that extension, the transport (15) factorizes across heads. Each head a would carry its own connection field $\delta_{ij}^{(a)}$ acting in its own irrep block, and the holonomy would decompose as a direct sum

$$H_{ijk} = \bigoplus_{a=1}^{H} H_{ijk}^{(a)}, \qquad H_{ijk}^{(a)} = \exp(\delta_{ij}^{(a)} \cdot G_a) \exp(\delta_{jk}^{(a)} \cdot G_a) \exp(\delta_{ki}^{(a)} \cdot G_a), \tag{107}$$

with the Wilson observable factorizing additively, $W_{ijk} = \sum_{a=1}^{H} W_{ijk}^{(a)}$. Each per-head term is independently gauge-invariant under the per-head $\mathrm{GL}(d_{\mathrm{head}})$ subgroup. Aggregate diagnostics that compute a single $\|H_{ijk} - I\|_F$ on the full block-diagonal representation report the sum of per-head deviations and discard the per-head decomposition; explicit per-head measurements require slicing the connection field by head before evaluating the holonomy.

Per-head temperature from the belief covariance. Since each head operates on an independent irrep block with its own $\mathrm{GL}(d_{\mathrm{head}})$ gauge transport, each head possesses its own belief covariance $\Sigma^{(a)}$ which controls the scale of the KL divergence and thereby the effective attention temperature for that head. In the isotropic limit $\Sigma^{(a)} = \sigma_a^2 I$, the KL between beliefs in head a reduces to $D_{\mathrm{KL}}^{(a)} = \frac{1}{2\sigma_a^2}\|\cdot\|^2$, so the per-head covariance σ_a^2 *is* the per-head temperature. This is not an additional degree of freedom appended to the theory: it is intrinsic to the multi-head decomposition, since each $\mathrm{GL}(d_{\mathrm{head}})$ block carries its own covariance structure.

In the implementation, we factor the effective temperature into a dimensional normalization $\sqrt{d_{\mathrm{head}}}$ and a scalar κ_a that parameterizes the isotropic component of the per-head covariance:

$$\beta_{ij}^{(a)} = \mathrm{softmax}_j \left(-\frac{D_{\mathrm{KL}}(q_i^{(a)} \| \Omega_{ij}^{(a)} q_j^{(a)})}{\kappa_a \sqrt{d_{\mathrm{head}}}} \right), \tag{108}$$

where $q_i^{(a)}$ denotes the belief projected onto head a's subspace. In the isotropic limit, $\kappa_a \propto \sigma_a^2$: the attention temperature and the belief covariance are the same quantity. In the full non-isotropic theory where $\Sigma^{(a)}$ is a learned $d_{\mathrm{head}} \times d_{\mathrm{head}}$ matrix, the covariance provides per-head, per-token, and per-direction temperature control; κ_a then serves as a convenient global scalar handle on the attention sharpness that does not require modifying $\Sigma^{(a)}$ (which also couples to the self-alignment term $\alpha_i D_{\mathrm{KL}}(q_i \| p_i)$ and gradient dynamics).

In standard transformers, σ_a^{-2} is absorbed into the learned projections $W_Q^{(a)}, W_K^{(a)}$, making the per-head temperature implicit in the weight scales $\|W_Q^{(a)}\|, \|W_K^{(a)}\|$. Heads with larger-norm projections produce higher-variance logits (lower effective temperature, sharper attention), while heads with smaller-norm projections produce lower-variance logits (higher effective temperature, softer attention). The gauge framework retains the per-head covariance structure that standard transformers absorb.

4.4.2 Geometric Interpretation via Lie Algebra Decomposition.

The Lie algebra $\mathfrak{gl}(d_k)$ has dimension d_k^2, with generators spanning all $d_k \times d_k$ matrices. The block-diagonal restriction to $\mathfrak{gl}(d_{\mathrm{head}})^H$ projects out all off-diagonal blocks thereby retaining only $H \cdot d_{\mathrm{head}}^2$ generators.

Therefore, each head's Lie algebra $\mathfrak{gl}(d_{\mathrm{head}})$ decomposes as:

$$\mathfrak{gl}(d_{\mathrm{head}}) = \mathfrak{sl}(d_{\mathrm{head}}) \oplus \mathbb{R}, \tag{109}$$

where $\mathfrak{sl}(d_{\mathrm{head}})$ consists of traceless matrices (volume-preserving transformations) and $\mathbb{R}$ is the trace (overall scaling). This means each head independently controls the overall magnitude of attention in that subspace (scaling) and how different dimensions within the head interact (shearing and rotation).

4.4.3 Off-Diagonal Gauge Mixing: The Projected-Out Generators.

The block-diagonal restriction projects out a substantial portion of the full gauge algebra. Decomposing $\mathfrak{gl}(d_k)$ into block-diagonal and off-diagonal components:

$$\mathfrak{gl}(d_k) = \underbrace{\bigoplus_{a=1}^{H} \mathfrak{gl}(d_{\text{head}})}_{\text{block-diagonal (retained)}} \oplus \underbrace{\mathfrak{m}}_{\text{off-diagonal (projected out)}}, \tag{110}$$

the complement $\mathfrak{m}$ has codimension $d_k^2 - H \cdot d_{\text{head}}^2$ within $\mathfrak{gl}(d_k)$. For typical architectures (e.g., $d_k = 512$, $H = 8$, $d_{\text{head}} = 64$), this is $512^2 - 8 \times 64^2 = 229{,}376$ generators out of 262,144 total: the multi-head architectural choice projects out 87.5% of the framework's $\mathfrak{gl}(d_k)$ generators, retaining only the block-diagonal subalgebra $\bigoplus_a \mathfrak{gl}(d_{\text{head}})$. This codimension is measured relative to the framework's full $\mathrm{GL}(d_k)$ reference, not relative to the standard transformer's parameter manifold $\big(3 \cdot H \cdot d_{\text{model}} \cdot d_{\text{head}} + d_{\text{model}}^2\big)$, which has a different dimension and is not a subset of $\mathfrak{gl}(d_k)$. These off-diagonal generators are the inter-head mixing transformations. They are elements $X \in \mathfrak{m}$ that have non-zero entries only in off-diagonal blocks X^{ab} $(a \neq b)$, each of which maps $\mathbb{R}^{d_{\text{head}}}$ (head b) into $\mathbb{R}^{d_{\text{head}}}$ (head a).

Restoring these would yield a gauge transport with off-diagonal blocks:

$$\Omega_{ij} = \begin{pmatrix} \Omega_{ij}^{11} & \Omega_{ij}^{12} & \cdots & \Omega_{ij}^{1H} \\ \Omega_{ij}^{21} & \Omega_{ij}^{22} & \cdots & \Omega_{ij}^{2H} \\ \vdots & \vdots & \ddots & \vdots \\ \Omega_{ij}^{H1} & \Omega_{ij}^{H2} & \cdots & \Omega_{ij}^{HH} \end{pmatrix} \in \mathrm{GL}(d_k), \tag{111}$$

where the off-diagonal blocks $\Omega_{ij}^{ab} \in \mathbb{R}^{d_{\text{head}} \times d_{\text{head}}}$ $(a \neq b)$ implement cross-head gauge transport. When transporting the belief from token j to the frame of token i, head b's representation at j is rotated into head a's subspace at i. The VFE alignment gradient $\partial \mathcal{F} / \partial \Omega_{ij}^{ab}$ for the off-diagonal blocks would then drive the system to learn these cross-head correlations, encoding when one pattern appears in head b, it should be transported into head a's frame for comparison.

This is a qualitatively different mechanism from the standard output projection $W_O \in \mathbb{R}^{d_k \times d_k}$ in transformers. The output projection mixes head outputs after attention. After each head has independently computed its attention-weighted aggregation $\sum_j \beta_{ij}^a V_j^a$, the results are concatenated and linearly mixed. Off-diagonal gauge transport, however, mixes heads before attention: the cross-head blocks Ω_{ij}^{ab} enter the KL divergence that determines the attention scores themselves. This makes the attention weights cross-head aware whereby head a's attention pattern can depend on the content of head b through the transport, before any aggregation occurs at all. The full-$\mathrm{GL}(d_k)$ transport is therefore more expressive than the single-layer block-diagonal-plus-output-projection factorization, as it couples the attention computation across heads within one layer rather than only the output computation. Standard transformers recover cross-head awareness across multiple layers via W_O

followed by the next layer's projections, so the framework's expressiveness advantage is at the per-layer level rather than at the function-class level of the full multi-layer stack.

From the perspective of the VFE gradient flow, the block-diagonal restriction constrains the gradient $\nabla_\phi \mathcal{F}$ to lie within the block-diagonal sub-algebra $\bigoplus_a \mathfrak{gl}(d_{\text{head}})$. Restoring the off-diagonal components amounts to allowing the gradient to flow along the full $\mathfrak{gl}(d_k)$, utilizing all d_k^2 generators. The projected-out gradient components $\nabla_\phi \mathcal{F}\big|_{\mathfrak{m}}$ represent cross-head alignment forces that are present in the full VFE but zeroed out by the architectural constraint. Whether these forces carry any useful signal is an empirical question with significant implications.

4.5 Rotary Position Embeddings as Position-Dependent Gauge Frames

The prior-based positional mechanisms discussed in Section 3.4.5 (masking, ALiBi, relative biases) all operate through the attention prior π_j, entering as additive biases in the pre-softmax logits. A complementary class of positional encodings operates instead through the gauge transport Ω_{ij} itself, making the gauge frame position-dependent. Rotary Position Embeddings (RoPE) (Su et al., 2024) are the most widely adopted example. Here we derive RoPE from our framework.

4.5.1 RoPE as a Gauge Subgroup Restriction.

RoPE applies position-dependent rotations to the query and key vectors before computing the dot product:

$$Q_i \to R(\theta_i) Q_i, \qquad K_j \to R(\theta_j) K_j, \tag{112}$$

where $R(\theta_m) \in \mathrm{SO}(2)^{d_k/2}$ is a block-diagonal rotation matrix with $d_k/2$ independent 2×2 rotation blocks:

$$R(\theta_m) = \operatorname{diag}\left(\begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 \\ \sin m\theta_1 & \cos m\theta_1 \end{pmatrix}, \ldots, \begin{pmatrix} \cos m\theta_{d_k/2} & -\sin m\theta_{d_k/2} \\ \sin m\theta_{d_k/2} & \cos m\theta_{d_k/2} \end{pmatrix} \right), \tag{113}$$

with frequencies $\theta_n = 10000^{-2n/d_k}$. The attention logits become:

$$(R(\theta_i)Q_i)^\top (R(\theta_j)K_j) = Q_i^\top R(\theta_i)^\top R(\theta_j) K_j = Q_i^\top R(\theta_{j-i}) K_j, \tag{114}$$

where $R(\theta_{j-i}) = R(\theta_i)^\top R(\theta_j)$ depends only on the relative position $j - i$.

In the gauge-theoretic framework, this has a straightforward interpretation. Recall from Section 2.1.3 that $W_Q W_K^\top = \sigma^{-2} \Omega^{-\top}$. With RoPE, the rotation is applied to query and key vectors after projection, so the effective logit kernel becomes:

$$\sigma^{-2} (\Omega_{ij}^{(\text{RoPE})})^{-\top} = W_Q R(\theta_{j-i}) W_K^\top, \tag{115}$$

where $R(\theta_{j-i}) \in \mathrm{SO}(2)^{d_k/2}$ is sandwiched between the learned projections because it acts in the projected d_k-dimensional space. This partially relaxes Limit 2 (flat bundle): the gauge connection acquires a position-dependent component, but it is restricted to the compact (Abelian) subgroup $\mathrm{SO}(2)^{d_k/2} \subset \mathrm{GL}(d_k)$.

4.5.2 Gauge-Theoretic Interpretation.

The factorization (115) identifies the position-dependent gauge frame as:

$$\phi_i^{(\mathrm{pos})} = \mathrm{diag}(i\theta_1 J, i\theta_2 J, \ldots, i\theta_{d_k/2} J) \in \mathfrak{so}(2)^{d_k/2} \subset \mathfrak{gl}(d_k), \tag{116}$$

where $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is the SO(2) generator and θ_n are the RoPE frequencies. The position-dependent rotation is then $R(\theta_i) = \exp(\phi_i^{(\mathrm{pos})})$, and the inter-agent rotation $R(\theta_{j-i}) = \exp(-\phi_i^{(\mathrm{pos})})\exp(\phi_j^{(\mathrm{pos})})$ has exactly the form of a gauge transport operator (12) restricted to the $\mathrm{SO}(2)^{d_k/2}$ subgroup. The opposite-sign placement of $\phi^{(\mathrm{pos})}$ relative to the general definition $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ differs by a transpose: on the abelian subgroup the two forms reduce to $\exp(\phi_i)\exp(-\phi_j) = \exp(\phi_i - \phi_j) = R(\theta_i - \theta_j)$ and $\exp(-\phi_i)\exp(\phi_j) = \exp(\phi_j - \phi_i) = R(\theta_j - \theta_i) = R(\theta_i - \theta_j)^\top$, which are inverses (transposes for an orthogonal rotation) rather than equal. The transpose is absorbed into the asymmetric $W_Q \neq W_K$ convention via the symmetric identity $Q_i^\top M K_j = K_j^\top M^\top Q_i$, so either sign convention recovers RoPE at the level of attention logits.

The multi-frequency structure of RoPE ($\theta_n = 10000^{-2n/d_k}$) corresponds to different components of the gauge frame oscillating at different frequencies across positions: low-frequency components (θ_n small, n large) encode coarse positional structure and remain nearly aligned across distant positions, while high-frequency components (θ_n large, n small) encode fine-grained local position and de-correlate rapidly.

4.5.3 Interpolation Between Flat and Full Gauge.

RoPE occupies an intermediate position in the gauge-theoretic framework:

$$\underbrace{\sigma^{-2}\exp(-\phi_j)^\top \exp(\phi_i)^\top}_{\text{Full GL}(K)\text{ gauge (this work)}} \supset \underbrace{W_Q R(\theta_{j-i}) W_K^\top}_{\text{RoPE}} \supset \underbrace{W_Q W_K^\top}_{\text{Standard attention (flat bundle)}}, \tag{117}$$

where each expression is the logit kernel $\sigma^{-2}\Omega_{ij}^{-\top}$ appearing in $\tilde{s}_{ij} = \mu_i^\top(\cdot)\mu_j$. Standard attention uses a constant, learned gauge with $W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$ (Limit 2 fully imposed). RoPE augments this with a fixed, position-dependent $\mathrm{SO}(2)^{d_k/2}$ rotation sandwiched between the projections (Limit 2 partially relaxed for positional information). The full gauge-theoretic framework allows the complete transport Ω_{ij} to be agent-pair-dependent and learned (Limit 2 fully relaxed). This hierarchy clarifies the design space such that positional encoding methods can be understood as choices about which subgroup of $\mathrm{GL}(K)$ carries position-dependent structure, and which components remain position-independent and learned from data.

An important distinction is that RoPE applies the position-dependent gauge only to the attention computation (queries and keys) and not to the value aggregation. In our framework, the full gauge transport Ω_{ij} mediates both the attention score $D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$ and the value aggregation $\hat{\mu}_i = \sum_j \beta_{ij}\Omega_{ij}\mu_j$. The asymmetry in RoPE with position-dependent attention but position-independent values corresponds to factoring the gauge transport into an attention gauge and a value gauge that need not coincide. This decomposition is supported (but not required) by the full framework.

4.6 Further Architectural Correspondences

Beyond the attention mechanism, several additional transformer components possess origins within the gauge-theoretic framework. These are components that are derived from the variational principle, components whose structural role is explained, and components that are accommodated by the framework without being uniquely predicted.

4.6.1 Feed forward Networks and Non-linear Activations.

Standard transformers interleave attention sublayers with feedforward networks (FFNs), typically two-layer MLPs with a nonlinear activation such as GeLU (Hendrycks and Gimpel, 2016). Within the gauge-theoretic framework, the FFN corresponds to the iterative VFE dynamics that adjust beliefs between attention computations. The nonlinear activation function is not an independent architectural choice but a natural consequence of the information-geometric structure: variational inference on Gaussian beliefs produces a Gated Linear Unit (GLU) (Dauphin et al., 2017) whose gate is a Boltzmann weight over KL energies. The neural network is the computational substrate of a deeper information-geometric dynamics; the activation functions used in practice (SiLU, GELU, ReLU) are specific instantiations of this dynamics under successive limits.

Message passing as a gated linear unit. Consider the per-edge contribution of neighbor j to the mean update (Eq. 97):

$$\Delta\mu_i\big|_j = -\tilde{\eta}\Sigma_i\beta_{ij}(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}\big(\mu_i - \Omega_{ij}\mu_j\big). \tag{118}$$

The residual $e_{ij} \equiv \mu_i - \Omega_{ij}\mu_j$ is linear in the beliefs. The attention gate $\beta_{ij} = \mathrm{softmax}_j\big(-D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)/\tau\big)$ is manifestly a Boltzmann weight: each neighbor's contribution is weighted by $\exp(-E_{ij}/\tau)/Z_i$, where the energy $E_{ij} = D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j)$ depends on the residual magnitude through the Gaussian KL as:

$$D_{\mathrm{KL}}(q_i\|\Omega_{ij}q_j) = \tfrac{1}{2}e_{ij}^\top(\Omega_{ij}\Sigma_j\Omega_{ij}^\top)^{-1}e_{ij} + (\text{covariance terms}). \tag{119}$$

Each per-edge contribution therefore has the general structure:

$$\boxed{\Delta\mu_i\big|_j \propto \underbrace{e_{ij}}_{\text{linear}} \cdot \underbrace{\frac{\exp\big(-\frac{1}{2\tau}\|e_{ij}\|^2_{\Sigma_j^{-1}} + \cdots\big)}{\sum_k \exp\big(-\frac{1}{2\tau}\|e_{ik}\|^2_{\Sigma_k^{-1}} + \cdots\big)}}_{\text{Boltzmann gate}},} \tag{120}$$

which is a Gated Linear Unit: a linear function of the input gated by a Boltzmann distribution over KL energies computed from the same input.

Binary limit: sigmoid as a special case. The general Boltzmann gate operates over all n_s sources simultaneously, where n_s denotes the number of neighbors contributing to agent i's update (distinct from the sequence length N defined in Table 1; in the full attention setting $n_s = N - 1$, but the gate structure holds for any n_s). In the special case where only $n_s = 2$ sources compete for influence on agent i, the Boltzmann distribution over two states reduces to the logistic sigmoid σ, since $\exp(-E_1/\tau)/(\exp(-E_1/\tau)+\exp(-E_2/\tau)) = \sigma((E_2-E_1)/\tau)$:

$$\beta_{i1} = \sigma\left(\frac{D_{\mathrm{KL},i2} - D_{\mathrm{KL},i1}}{\tau}\right), \qquad \Delta\mu_i\big|_1 \propto e_{i1}\cdot\sigma\left(\frac{\|e_{i2}\|^2 - \|e_{i1}\|^2}{2\sigma^2\tau} + \cdots\right). \tag{121}$$

This recovers the SiLU/Swish (Ramachandran et al., 2018) structure $f(x) = x\cdot\sigma(g(x))$, with the gate argument determined by relative KL energies. This binary case corresponds to the FFN sublayer, where the hidden representation is split into two competing pathways representing precisely two sources whose relative informational surprise determines the gate. The sigmoid is thus not the general activation but instead is the $n_s = 2$ limit of the Boltzmann gate.

Boltzmann gate, Gaussian family, and connection to GELU. The information-geometric origin of the gate determines its functional form. For example, in the isotropic limit $\Sigma_j = \sigma^2 I$, the un-normalized gate for neighbor j is $\exp(-E_{ij}/\tau) = \exp(-\|e_{ij}\|^2/2\sigma^2\tau)$, a Gaussian function of the prediction error. The Boltzmann distribution over KL energies is therefore a Gaussian mixture gate: each neighbor's weight is a Gaussian in prediction-error space, normalized by the partition function $Z_i = \sum_k \exp(-\|e_{ik}\|^2/2\sigma^2\tau)$. The activation functions used in practice belong to the same exponential family as this information-geometric gate, though they are not functionally identical to it:

$$\begin{aligned} \mathrm{SiLU}(x) &= x\cdot\sigma(x) && \text{logistic sigmoid gate } (n_s = 2), \\ \mathrm{GELU}(x) &= x\cdot\Phi(x) && \text{Gaussian CDF gate,} \\ \mathrm{VFE}(e) &= e\cdot\exp(-\|e\|^2/\tau')/Z && \text{Boltzmann gate (general } n_s\text{, Gaussian energy).} \end{aligned} \tag{122}$$

All three belong to the Gaussian family of gated linear units, but the correspondence is one of family membership rather than functional identity. The VFE gate uses the Gaussian PDF $\propto \exp(-x^2/2)$ as its un-normalized weight, while GELU uses the Gaussian CDF $\Phi(x) = \frac{1}{2}\operatorname{erfc}(-x/\sqrt{2})$, the integral of the same PDF. The logistic sigmoid $\sigma(x) = 1/(1+e^{-x})$ is the two-state Boltzmann distribution. These are related by the number of competing information sources ($n_s = 2$ for sigmoid, $n_s > 2$ for softmax) and the functional form of the energy ($E \propto x$ for SiLU, $E \propto x^2$ for VFE). The framework thus explains why gated activations from this family work well (they implement information-geometric gating), but does not uniquely predict GELU over SiLU or vice versa.

In the binary case ($n_s = 2$), ReLU is the zero-temperature ($\tau \to 0$) limit: the sigmoid concentrates into a step function Θ, giving $x\cdot\Theta(x) = \max(0, x)$.

Softmax-gradient correction. Beyond the GLU structure of the message-passing term itself, the autograd gradient of the entropy-suppressed surrogate $\sum_j \beta_{ij}E_{ij}$ contains an additional nonlinear contribution from differentiating β_{ij} with respect to μ_i (this term is absent from the gradient of the reduced free energy $\mathcal{F}_{\mathrm{red}}$ by the envelope theorem; cf. Section 3.5). The bare softmax sensitivity displayed below is the gradient channel that contributes to $\nabla_{\mu_i}\sum_j \beta_{ij}E_{ij}$; under the canonical entropy-augmented $\mathcal{F}$ at Eq. (35), this contribution cancels identically against $\tau\sum_j(\partial\beta_{ij}/\partial\mu_i)\log(\beta_{ij}/\pi_j)$ via the row-Lagrangian KKT identity at $\beta = \beta^*$, by the same cancellation that produces the $-\tau\log Z_i$ reduction:

$$\frac{\partial \beta_{ij}}{\partial \mu_i} = -\frac{\beta_{ij}}{\tau}\left[\frac{\partial D_{\mathrm{KL},ij}}{\partial \mu_i} - \sum_k \beta_{ik}\frac{\partial D_{\mathrm{KL},ik}}{\partial \mu_i}\right], \tag{123}$$

where $D_{\mathrm{KL},ij} \equiv D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$. This centered gradient is an additional gradient channel that contributes only to the gradient of the entropy-suppressed surrogate $\sum_j \beta_{ij} E_{ij}$ and vanishes identically in the gradient of the canonical entropy-augmented $\mathcal{F}$ (see Eq. (37) for the covariance form of the gap between the two gradients). It creates positive feedback whereby similar beliefs yield higher β, which in turn pulls beliefs closer together. For a single component, $\partial \beta_j / \partial z_j = \beta_j(1-\beta_j)$, which is itself a sigmoid-shaped function of the logit. This correction is unique to the VFE setting, where β depends on the same μ being updated. However, in standard transformers, the analogous derivative appears only during backpropagation, not in the forward pass.

The same softmax-gradient non-linearity also appears in the covariance and gauge frame channels. The Σ_i gradient (Eq. 99) contains the term $\sum_j (\partial \beta_{ij}/\partial \Sigma_i) D_{\mathrm{KL}}(q_i \| \Omega_{ij} q_j)$, which has the identical centered structure. Similarly, the gauge frame gradient (Eq. **??**) yields $\partial \beta_{ij}/\partial \phi_i = -(\beta_{ij}/\tau)[\partial K_{ij}/\partial \phi_i - \sum_k \beta_{ik} \partial K_{ik}/\partial \phi_i]$, the same self-normalizing form. Each variational channel (μ, Σ, ϕ) thus contributes its own GLU-type activation and the composition of these channels within each VFE iteration produces a richer non-linear map than any single channel alone.

FFN depth from VFE iterations. In the full VFE implementation, multiple iterations of gradient descent with recomputed β at each step constitute the analogue of the FFN sublayer. Each iteration applies the GLU map (120); under descent on the entropy-suppressed surrogate $\sum_j \beta_{ij} E_{ij}$, the per-iteration map is the GLU composed with the softmax-gradient channel (123), while under descent on the canonical entropy-augmented $\mathcal{F}$ the composition collapses to the bare GLU map by the cancellation identified above. The number of VFE iterations plays the role of FFN depth, as established in the "Layers as VFE iterations" paragraph (Section 4.3.3).

4.6.2 Token Embeddings as Prior Banks.

In standard transformers, token embeddings map discrete tokens to continuous vectors via a learned embedding matrix $W_{\mathrm{embed}} \in \mathbb{R}^{V \times d}$. In the gauge-theoretic framework, each token type $c \in \{1, \ldots, V\}$ is associated with a learned prior distribution:

$$p_c = \mathcal{N}(\mu_p^c, \Sigma_p^c), \qquad \phi_c \in \mathfrak{gl}(K), \tag{124}$$

comprising a prior mean μ_p^c, prior covariance Σ_p^c, and gauge frame coefficients ϕ_c. At the start of each sequence, agent i's belief is initialized to the prior of its token type: $q_i(0) = p_{\mathrm{token}(i)}$. The collection $\{(\mu_p^c, \Sigma_p^c, \phi_c)\}_{c=1}^V$ constitutes a prior bank, which is the generalization of the embedding matrix. In the deterministic, isotropic limit ($\Sigma_p^c \to \sigma^2 I$, ϕ_c absorbed), only the prior means survive and the prior bank reduces to $W_{\mathrm{embed}} = [\mu_p^1, \ldots, \mu_p^V]^\top$, recovering the standard embedding matrix.

4.7 Summary

Under the successive reductions (the first two are exact algebraic specializations, the third a literal change of parameterization at finite σ): (i) isotropic beliefs with σ^{-2} absorbed into learned projections, (ii) constant gauge ($\Omega_{ij} = \Omega$ for all pairs), and (iii) learned gauge via projections ($W_Q W_K^\top = \sigma^{-2}\Omega^{-\top} \in \mathrm{GL}(d_k)$), variational natural-gradient descent on the free energy (which is $\mathrm{GL}(K)$-equivariant in the unrestricted framework by Theorem 1) is:

$$\theta_{k+1} = \theta_k - \eta \tilde{\nabla}_\theta \mathcal{F}(\theta_k) \tag{125}$$

and recovers the standard training update:

$$\theta_{k+1} = \theta_k - \eta \tilde{\nabla}_\theta \mathcal{L}(\theta_k), \tag{126}$$

under the identifications: $\mathcal{F}$ (free energy) $\leftrightarrow$ $\mathcal{L}$ (loss + regularization); attention weights $\beta_{ij} \leftrightarrow$ transformer attention α_{ij}; observations $-\log p(o \mid \mu) \leftrightarrow$ loss function; and symmetry breaking $\leftrightarrow$ training.

Beyond the core attention formula, the gauge-theoretic framework provides a unified account of key transformer architectural choices (Table 3), each recovered under specific limits or identifications. Causal masking, ALiBi, relative position biases, and sliding window attention all emerge as special cases of the non-uniform attention prior π_k in the mixture-of-sources model (Section 3.4.5). Rotary position embeddings correspond to position-dependent gauge frames restricted to $\mathrm{SO}(2)^{d_k/2}$, partially relaxing the flat-bundle limit while retaining the learned constant gauge (Section 4.5). Layer normalization is a projection onto the $\mathfrak{sl}(K)$ subalgebra that eliminates key-norm bias at its algebraic origin. Residual connections are the inherent structure of iterative variational gradient descent (Section 4.3.3), with the state-dependent precision α_i generalizing the fixed skip connection to a context-dependent gate. The FFN sublayer corresponds to multiple VFE iterations with recomputed attention weights, and the message-passing term derives a Boltzmann-gated linear unit (Eq. 120) in the same exponential family as GELU and SiLU (Section 4.6.1). Weight decay is the prior coupling $\alpha D_{\mathrm{KL}}(q_i \| p_i)$ in the deterministic limit.

The $\mathrm{GL}(K)$ gauge invariance theorem (Theorem 1) is the equivariance statement for the unrestricted framework: the free energy is invariant under simultaneous push-forward of all agent representations by a common $\Omega \in \mathrm{GL}(K)$. After the constant-gauge specialization the per-pair Ω_{ij} collapse to a single learned element of $\mathrm{GL}(d_k)$, and the SVD identity at Eq. (73) places the head-space invertible factor in $\mathrm{GL}(d_{\mathrm{head}})$. The learned projections W_Q, W_K parameterize this head-space factor and thereby realize one orbit representative of the framework's gauge action, explaining why transformers can learn arbitrary (invertible) attention patterns on the head subspace. Each row of Table 3 represents a degree of freedom that the full gauge-theoretic framework retains and that standard transformers collapse into a simpler form. The framework thus provides not only an explanatory account of existing architectures but a taxonomy of generalizations, indexed by which simplifying limits are relaxed.

5 Simulations and Empirical Validation

5.1 Experimental Design

All experiments were performed on a Windows workstation with 64GB RAM, AMD Ryzen 9900x CPU, and an Nvidia RTX5090 GPU.

5.1.1 Transformer Validation Protocol (Summary)

To validate the degenerate-limit predictions, we compared the KL gauge attention $\beta_{ij}^{(\text{flat})} = \text{softmax}_j(-\|Q_i - K_j\|^2/\tau)$ against standard dot-product attention $\alpha_{ij} = \text{softmax}_j(Q_i K_j^\top/\sqrt{d})$ on frozen pretrained transformers. The full protocol and detailed results are presented in Supplementary Appendix E. Briefly, we extracted hidden states from all 144 heads of BERT-base-uncased across 105 diverse English passages, computed per-head Pearson correlations between α and β attention rows, and performed a temperature sweep, key-norm bias analysis, entropy matching, multi-model validation (five architectures), per-head temperature dispersion diagnostic, and sequence-length sensitivity analysis.

The principal findings are: (i) grand mean correlation $\bar{r} = 0.804$ (95% CI $[0.771, 0.838]$) at the theory-predicted optimal temperature $\tau = 19.0 \approx 2\sqrt{d_k}$ for $d_k = 64$, with cross-passage SE < 0.02 for all 144 heads; (ii) entropy matching $H(\beta)/H(\alpha) = 1.076$, confirming the temperature recovers distributional shape, not merely correlation; (iii) the correspondence generalizes across five architectures ($\bar{r} > 0.6$ for all, $\bar{r} = 0.851$ for ALBERT), with cross-model variation explained by per-head temperature dispersion rather than pretraining differences; and (iv) a Bayesian hierarchical model yields a shrinkage-corrected grand mean of $\bar{r}_{\text{post}} = 0.867$ (94% HDI $[0.808, 0.915]$).

An important caveat: the high correlation follows in large part from an algebraic identity. When key-norms are approximately constant (as enforced by layer normalization), $\|Q_i - K_j\|^2$ and $-2Q_i K_j^\top$ differ only by terms approximately constant under softmax. The non-trivial content lies in the quantitative predictions: the temperature scaling $\tau \approx 2\sqrt{d_k}$, the key-norm bias (Cohen's $d = 1.43$, 94% HDI $[1.08, 1.88]$), the per-head temperature structure, and their cross-model variation. We therefore treat the BERT validation as a *consistency check* confirming that the degenerate limit behaves as predicted, while the $\text{GL}(K)$ language model (next section) provides the stronger test of the unreduced framework.

5.1.2 GL(K) Language Modeling

To validate the complete gauge-theoretic framework beyond attention mechanism correspondence, we trained a $\text{GL}(K)$ gauge transformer on WikiText-103 language modeling (102M tokens, vocabulary size 50,257). These experiments test whether variational free energy minimization can drive actual language learning, with the gauge structure (μ, Σ, ϕ) evolving jointly during training. The gauge VFE architecture contains no MLPs, pointwise activation functions (ReLU, GELU, etc.), or learned attention projections (W_Q, W_K, W_V); only a linear output projection to vocabulary logits is retained. The model does employ nonlinear operations inherent to the framework: softmax for attention normalization and matrix exponentials for gauge transport. All other computation derives from KL divergences, parallel transport via $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$, and natural gradient descent on the Fisher-Rao manifold. Rotary Position Embeddings (RoPE) (Su et al., 2024) are applied to the belief means μ before KL-divergence computation in the attention sublayer, providing relative position information through the $\mathrm{SO}(2)^{K/2}$ subgroup of the gauge group (see Section **??**). The causal attention mask provides additional sequential structure by restricting each token's context to preceding positions, and the gauge transport operators Ω_{ij} can in principle learn position-dependent transformations through the token-specific gauge frames ϕ_i. The standard transformer baselines likewise use RoPE, ensuring a controlled comparison.

We report results for a GL(10) gauge VFE model with embedding dimension $K = 90$, decomposed into 9 fundamental irreps of dimension 10 (yielding 9 attention heads), context length $N = 128$, batch size 16, and 60,000 training steps (58.8M parameters). The model uses VFE-dynamic mode with evolving covariance (Σ) and gauge frame (ϕ), full (non-diagonal) covariance matrices, and learnable per-component learning rates. Standard transformer baselines with dot-product attention and MLP sublayers, trained on the same data with matched configurations, serve as comparisons (Table 4). Full configurations are documented in the code repository.

Variational Inference Structure (E-step/M-step). We utilize an expectation-maximization structure as follows:

E-step (Belief Inference): Given current parameters, compute the closed-form softmax $\beta = \beta^*$ at the current beliefs (the stationary point of $\mathcal{F}_{\text{align}}$ in β, Eq. 25), then update agent beliefs $\{q_i\}$ via natural gradient descent on the attention-weighted alignment energy at $\beta = \beta^*$ (the autograd-convention form, cf. Section 3.5):

$$q_i^{(t+1)} \leftarrow q_i^{(t)} - \eta_E \tilde{\nabla}_{q_i} \big[\textstyle\sum_j \beta^*_{ij} E_{ij} + \alpha_i D_{\mathrm{KL}}(q_i \| p_i) \big] \Big|_{q_i = q_i^{(t)}}, \tag{127}$$

where $\tilde{\nabla}$ denotes natural gradients projected via the Fisher-Rao metric. The autograd gradient differs from $\nabla \mathcal{F}_{\text{red}}$ by the covariance term of Eq. 37, coinciding at joint stationarity. This corresponds to variational inference within the forward pass.

M-step (Learning): After belief inference, gradients flow via backpropagation through the entire computation graph:

$$\theta \leftarrow \theta - \eta_M \nabla_\theta \mathcal{L}[\{q_i^*\}, \theta], \tag{128}$$

where $\theta = \{\mu_{\text{embed}}, \Sigma_{\text{embed}}, \phi_{\text{embed}}, W_{\text{out}}\}$. Within each E-step iteration, belief gradients are computed analytically from detached snapshots of μ, Σ, and ϕ, so the VFE dynamics are not differentiated through via autograd. However, the final beliefs $\{q_i^*\}$ retain a gradient path to the initial embeddings through the additive update chain $\mu^{(t+1)} = \mu^{(t)} + \Delta\mu^{(t)}$, acting as a straight-through estimator: the M-step learns embeddings whose initializations lead to good converged beliefs, without requiring second-order differentiation through the E-step trajectory.

5.1.3 Belief and Prior Initialization

Agent beliefs $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ represent context-dependent semantic interpretations that evolve through communication. Token priors $p_c = \mathcal{N}(\mu_p^c, \Sigma_p^c)$ and gauge frame coefficients $\phi_c^a \in \mathbb{R}$ are initialized from $\mathcal{N}(0, \sigma^2_{\text{init}})$ with $\sigma_{\text{init}} = 0.3$, serving as learnable token representations analogous to embedding matrices.

Algorithm 1 GL(K) Gauge Transformer Training Loop

Require: Token sequence $(x_1, \ldots, x_N)$; learnable parameters $\theta = \{\mu_{\text{embed}}, \Sigma_{\text{embed}}, \phi_{\text{embed}}, W_{\text{out}}\}$

Initialize beliefs from priors: $\mu_i \leftarrow \mu_{\text{embed}}^{x_i}$, $\Sigma_i \leftarrow \Sigma_{\text{embed}}^{x_i}$, $\phi_i \leftarrow \phi_{\text{embed}}^{x_i}$

— *E-step: Variational belief inference (within forward pass)* —

for $t = 1$ **to** T_{inner} **do**

  Compute transport: $\Omega_{ij} \leftarrow \exp(\phi_i)\exp(-\phi_j)$ ▷ Gauge parallel transport

  for each head h (irrep block) **do**

    $D_{ij}^{(h)} \leftarrow D_{\text{KL}}(q_i^{(h)} \| \Omega_{ij}^{(h)} q_j^{(h)})$ ▷ Pairwise KL divergences

    $\beta_{ij}^{(h)} \leftarrow \text{softmax}_j \Big(- D_{ij}^{(h)} / \kappa_h \Big)$ ▷ KL attention weights

  end for

  — *Mean gradient (autograd: direct + softmax nonlinearity, Eq. 123):*

  $$\nabla_\mu \mathcal{F} \leftarrow \underbrace{\alpha \Sigma_p^{-1}(\mu_i - \mu_p)}_{\text{prior}} + \sum_h \Bigg[\underbrace{\sum_j \beta_{ij}^{(h)} \nabla_\mu D_{ij}^{(h)}}_{\text{direct alignment}} + \underbrace{\sum_j D_{ij}^{(h)} \frac{\partial \beta_{ij}^{(h)}}{\partial \mu_i}}_{\text{softmax nonlinearity}} \Bigg] + \underbrace{\nabla_\mu \mathcal{L}_{\text{obs}}}_{\text{observation}}$$

  $\tilde{\nabla}_\mu \leftarrow \Sigma_i \cdot \nabla_\mu \mathcal{F}$ ▷ Natural gradient (Fisher preconditioning)

  $\mu_i \leftarrow \mu_i - \eta_E \tilde{\nabla}_\mu$ ▷ Trust-region-constrained update

  — *Covariance gradient (first line of Eq. 99; $\partial\alpha/\partial\Sigma$, $\partial\beta/\partial\Sigma$ corrections omitted):*

  $\nabla_\Sigma \mathcal{F} \leftarrow \frac{\alpha}{2}(\Sigma_p^{-1} - \Sigma_i^{-1}) + \frac{1}{2}\sum_h \sum_j \beta_{ij}^{(h)} \big[(\Omega_{ij}^{(h)} \Sigma_j^{(h)} \Omega_{ij}^{(h)\top})^{-1} - \Sigma_i^{(h)-1} \big]$

  $\Sigma_i \leftarrow \text{Retract}_{\text{SPD}}(\Sigma_i, -\eta_\Sigma \Sigma_i \nabla_\Sigma \mathcal{F} \Sigma_i)$ ▷ SPD retraction with Fisher metric

  — *Gauge frame gradient (differentiating through $\beta(\phi)$, Supplementary Eq. C.1):*

  $\nabla_\phi \mathcal{F} \leftarrow \nabla_{\phi_i} \Big[\sum_h \sum_j \beta_{ij}^{(h)}(\phi) D_{ij}^{(h)}(\phi) + \frac{\alpha_\phi}{2} \|\phi_i\|^2 \Big]$ ▷ Autograd through β

  $\phi_i \leftarrow \text{Retract}_{\mathfrak{g}}(\phi_i, -\eta_\phi \nabla_\phi \mathcal{F})$ ▷ Lie algebra retraction

end for

— *M-step: Parameter learning (backpropagation)* —

Compute loss: $\mathcal{L} = \mathcal{L}_{\text{CE}}(\mu_i^*, W_{\text{out}}) + \hat{\alpha} \sum_i D_{\text{KL}}(q_i^* \| p_i) + \frac{\alpha_\phi}{2} \|\phi\|^2$

$\theta \leftarrow \theta - \eta_M \nabla_\theta \mathcal{L}$ ▷ Adam on all embeddings and W_{out}

For GL(K), the gauge frame ϕ_c directly parameterizes elements of $\mathfrak{gl}(K)$, with transport operators $\Omega_i = \exp(\phi_i) \in \mathrm{GL}^+(K)$ computed via matrix exponential. A single matrix exponential $\exp : \mathfrak{gl}(K) \to \mathrm{GL}^+(K)$ is not surjective onto the identity component: matrices in $\mathrm{GL}^+(K)$ with negative real eigenvalues of odd Jordan-block multiplicity have no real logarithm (Culver, 1966). However, the pairwise transport $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ is a free product of two exponentials, and surjectivity holds in the existential sense: for every $A \in \mathrm{GL}^+(K)$ there exists at least one pair $(\phi_i, \phi_j) \in \mathfrak{gl}(K)^2$ with $\Omega_{ij} = A$. By the polar decomposition $A = PO$ with P symmetric positive-definite and $O \in \mathrm{SO}(K)$, one such witness is $\phi_i \in \mathrm{Sym}(K)$ with $\exp(\phi_i) = P$ (matrix logarithm of SPD; see e.g. Higham (2008), §6.4) and $\phi_j \in \mathfrak{so}(K)$ with $\exp(-\phi_j) = O$. The map $(\phi_i, \phi_j) \mapsto \exp(\phi_i)\exp(-\phi_j)$ is not injective — distinct pairs can produce the same Ω, the orbit $(\phi_i + aI, \phi_j + aI)$ for $a \in \mathbb{R}$ being a trivial example. Thus while individual gauge frames $\exp(\phi_i)$ do not exhaust

$GL^+(K)$, the transport operators Ω_{ij} between agents do. At the start of each training sequence, agent beliefs are initialized to match their token priors:

$$q_i(t=0) = p_{\mathrm{token}(i)}, \tag{129}$$

where token(i) denotes the token type at position i. Thus $\mu_i(0) = \mu_p^{\mathrm{token}(i)}$ and $\Sigma_i(0) = \Sigma_p^{\mathrm{token}(i)}$. All instances of the same token begin with identical beliefs regardless of position; context-dependent representations emerge through belief evolution during the E-step.

5.2 BERT Validation Results (Summary)

The full BERT validation results are reported in Supplementary Appendix E. As summarized in the protocol description above, the degenerate-limit predictions are confirmed across 105 passages, five architectures, and sequence lengths from 16 to 512 tokens. The key quantitative findings are reported in Section 5.1.1. We now turn to the $GL(K)$ language model, which provides the primary empirical test of the unreduced framework.

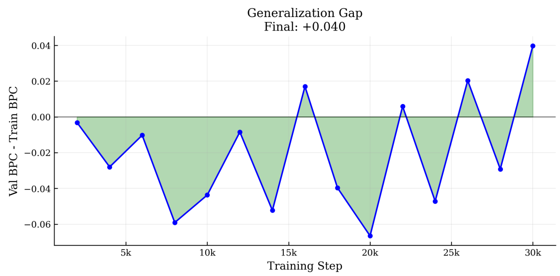
5.3 $GL(K)$ Language Modeling: The Full General Model

Unlike the BERT validation above, which tested the degenerate limits against an existing standard transformer across 105 diverse passages, our language modeling experiments implement the most general forms of the gauge-theoretic framework. These $GL(K)$ models retain full non-isotropic covariances Σ_i, non-trivial gauge transport $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$, and KL-divergence attention. This is a genuine test of whether the gauge-theoretic structure, in its unreduced form, can implement meaningful language modeling.

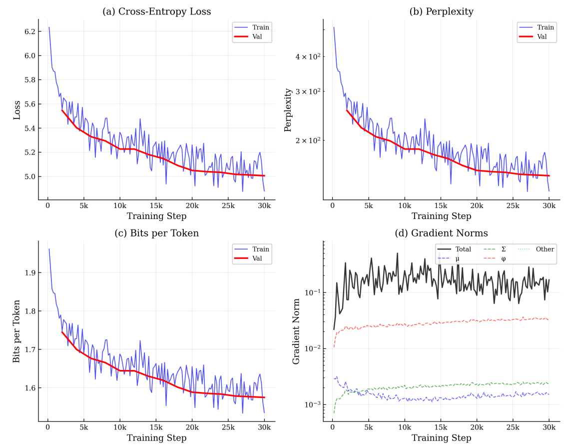
5.3.1 Training Dynamics

Figure 2 shows the training dynamics for our best $GL(K)$ gauge VFE model: a GL(10) configuration with $K = 90$ (9 heads of dimension 10). The model achieves stable convergence with perplexity dropping from 50,257 at initialization to $\sim$75 at convergence; a $\sim$671$\times$ improvement over random chance (vocabulary size 50,257). The gradient norms for μ (belief means), Σ (covariances), and ϕ (gauge frames) evolve with distinct dynamics; the gauge frame gradients (ϕ, red) exhibit characteristic oscillations as the model learns optimal frame alignments. Training and validation curves track closely throughout training, indicating the gauge-theoretic inductive bias provides effective regularization.

Figure 3 shows the learned KL-divergence attention patterns at convergence for four of the nine heads (H0, H2, H4, H8). Despite having no learned attention projections (W_Q, W_K, W_V), the gauge VFE model learns structured, non-uniform, and head-specialized attention patterns purely through gauge transport and KL-divergence dynamics. All four heads exhibit the causal mask and strong early-position attention (vertical stripes at key positions 0–10), reminiscent of the "attention sink" phenomenon observed in standard transformers (Xiao et al., 2024). Beyond these shared features, the heads develop qualitatively distinct attention strategies. Head 0 (Figure 3a) distributes attention broadly and diffusely across the available context, consistent with a global information-gathering role. Head 2 (Figure 3b) develops sharper vertical stripes at specific key positions corresponding to high-frequency or



(a) Generalization gap (val BPC − train BPC) for the GL(10), $K = 90$ gauge VFE model.



(b) Training curves for the GL(10), $K = 90$ gauge VFE model: loss, perplexity, bits per token, and gradient norms.

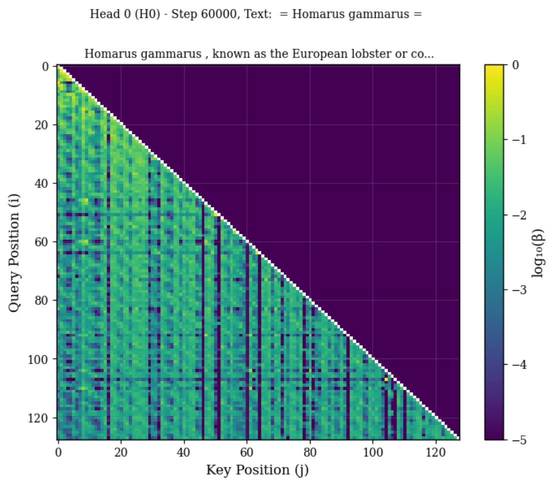
Figure 2: Training dynamics for the GL(10) gauge VFE model with $K = 90$ embedding dimensions (9 heads of dimension 10) on WikiText-103. (a) The generalization gap (validation BPC minus training BPC) remains small throughout training, indicating effective regularization from the gauge-theoretic inductive bias. (b) Training curves showing loss, perplexity, bits per token, and gradient norms; perplexity drops from >900 to ∼75 over 60,000 steps.

structurally important tokens, suggesting a token-selective pattern. Head 4 (Figure 3c) exhibits prominent horizontal bands at specific query positions (likely punctuation or sentence boundaries) where the query attends broadly to its entire available context, consistent with a positional or syntactic aggregation role. Head 8 (Figure 3d) concentrates attention in a narrow band near the diagonal, exhibiting a locality bias reminiscent of the sliding-window attention prior (Eq. 30), but learned rather than imposed. This functional diversification of broad context, token-selective, query-driven aggregation, and local recency emerges solely from the KL-divergence geometry on the statistical manifold, without any architectural inductive bias toward head specialization. The log-scale visualization reveals that attention weight variation spans approximately five orders of magnitude (10^{-5} to 10^{0}), confirming that the model learns highly selective attention from the gauge-theoretic dynamics alone.

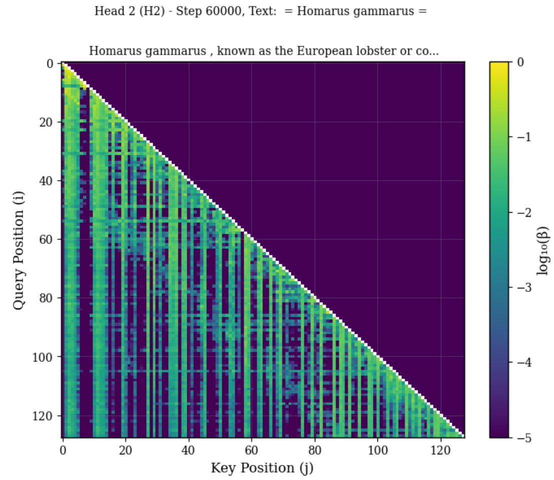
5.3.2 Results

Table 4 presents our best GL(K) gauge VFE model alongside a standard transformer baseline trained on the same data.

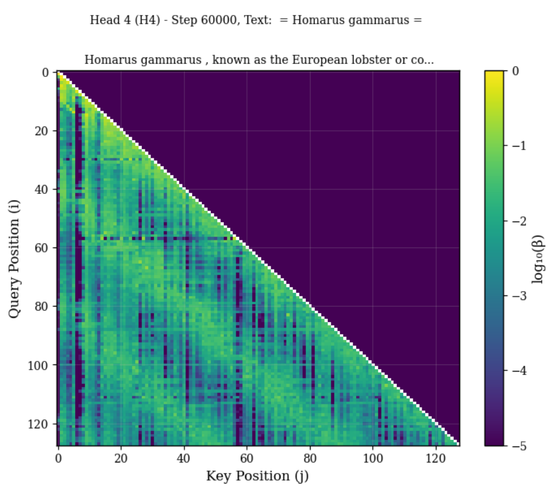
Our best gauge VFE model, a GL(15) configuration with $K = 90$ and 6 heads (81.4M params), achieves test perplexity 71.6 (∼702× improvement over random chance, i.e., vocabulary size 50,257), demonstrating that the unreduced gauge-theoretic framework, with KL-divergence attention, gauge transport, and natural gradient dynamics on the statistical manifold, is sufficient for non-trivial language modeling. The smaller GL(10) variant (58.8M params, 9 heads) achieves test PPL 76.4±1.05 (mean over two seeds; corresponding



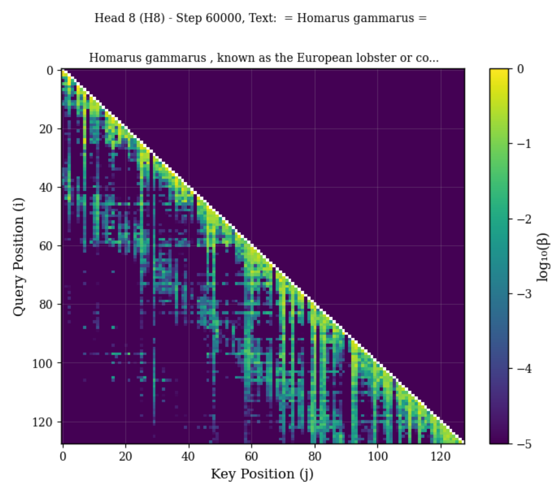
(a) Head 0: broad, diffuse contextual attention distributed across the full available context.



(b) Head 2: token-selective attention with vertical stripes at specific high-frequency key positions.



(c) Head 4: query-driven aggregation with prominent horizontal bands at structurally salient positions.



(d) Head 8: local attention concentrated near the diagonal, exhibiting a learned recency bias.

Figure 3: Learned per-head KL-divergence attention patterns ($\log_{10} \beta_{ij}$) for four of nine heads in the GL(10) gauge VFE model ($K = 90$) at convergence (step 60,000) on a representative WikiText-103 passage. All heads share the causal mask and early-position attention sinks, but develop qualitatively distinct strategies such as broad context (a), token-selective (b), query-driven aggregation (c), and local recency (d) purely from gauge transport and KL-divergence dynamics, without learned attention projections (W_Q, W_K, W_V).

validation PPL 75.5 ± 0.44). The improvement from GL(10) to GL(15) ($76.4 \rightarrow 71.6$) at fixed $K = 90$ indicates that larger gauge groups, which provide richer transport operators

$\Omega_{ij} \in \mathrm{GL}(15)$, improve language modeling performance at the cost of additional gauge frame parameters.

Embedding-matched vs. parameter-matched comparisons. The baselines in Table 4 reveal a parameter-vs-embedding tradeoff that clarifies the role of gauge structure. Each gauge VFE configuration is paired with two standard transformer baselines: one embedding-matched (same d_{model}) and one parameter-matched (comparable total parameters). The GL(15) gauge VFE (81.4M params, test PPL 71.6) is directly parameter-comparable to the 84.2M standard transformer ($d_{\mathrm{model}} = 1{,}280$), which achieves test PPL 48.5 but requires 14× the embedding dimension. At matched embedding dimension ($d_{\mathrm{model}} = 90$), a standard transformer achieves only PPL 118.6 with 4.6M parameters; the gauge VFE outperforms it by 1.66×, demonstrating that gauge transport and KL-divergence attention extract substantially more representational capacity from a given embedding dimension than dot-product attention with MLPs. The parameter-matched baselines for the $K = 10$ configuration show the same pattern at smaller scale: the embedding-matched standard transformer ($d_{\mathrm{model}} = 10$, 0.5M params) achieves only PPL 548, while the parameter-matched baseline ($d_{\mathrm{model}} = 116$, 6.0M params) reaches PPL 119.3. This consistent pattern across scales suggests that the gauge-theoretic structure provides an effective mechanism for compensating for low embedding dimensionality through geometric structure, at the cost of higher total parameter count invested in per-token covariance matrices Σ_i and gauge frame parameters ϕ_i.

Ablation baselines. Three additional baselines at $d_{\mathrm{model}} = 90$ with 9 heads isolate specific architectural contributions (Table 4, ablation rows). *Attention-only* (no MLP) achieves test PPL 142.8, confirming that removing the MLP from a standard transformer is not sufficient to match the gauge VFE's performance—the VFE's advantage stems from its geometric structure, not merely from operating without an MLP. *RoPE* at $d_{\mathrm{model}} = 90$ achieves test PPL 138.6, indicating that positional encoding alone provides only a modest improvement at this embedding dimension. *Parameter-equalized* (wider MLP, 9.2M params vs. 4.6M for the embed-matched baseline) achieves test PPL 145.8, showing that simply increasing MLP width does not recover the gauge VFE's performance.

An important caveat: the ablation baselines were trained for 15,000 gradient updates with batch size 64 ($\sim$123M tokens seen), whereas the gauge VFE models used 60,000 updates with batch size 16 ($\sim$123M tokens seen). While the total tokens seen are comparable, the training dynamics differ (more frequent updates with smaller batches for VFE vs. fewer updates with larger batches for ablations). The embed-matched and param-matched standard baselines in the upper portion of the table use training regimes matched to their respective VFE comparisons, ensuring controlled comparisons for the primary results.

To contextualize the gauge VFE result, we compare against the strongest purely non-neural baseline on WikiText-103: the modified Kneser-Ney 5-gram model (Chen and Goodman, 1998). Merity et al. (2017) report KN-5 test perplexity $\sim$153–156, but this uses word-level tokenization with a $\sim$267K vocabulary; since perplexity is computed per-token and a larger vocabulary makes next-token prediction harder, this figure is not directly commensurable with BPE-tokenized results. To obtain a controlled comparison, we trained a modified Kneser-Ney 5-gram model on WikiText-103 using the same GPT-2 BPE tokenization (50,257 vocabulary) as the gauge VFE model.[1] Under matched tokenization, the KN-5 baseline achieves test perplexity 134.8 (validation 135.9), as shown in Table 4. The best gauge VFE model (test PPL 71.6) surpasses this matched baseline by 63.2 PPL (1.88×), confirming under controlled conditions that gauge-covariant variational inference outperforms the strongest classical statistical method. The gauge VFE achieves this improvement through an entirely different mechanism: variational inference over gauge-transported Gaussian beliefs rather than count-based smoothing suggesting that the geometric structure of the framework captures statistical regularities inaccessible to n-gram models. For reference, neural language models on WikiText-103 range from PPL $\sim$49 for basic LSTMs (Grave et al., 2017) to PPL $\sim$18 for large optimized transformers (Dai et al., 2019), establishing the performance ceiling for architectures with full neural components.

Closing the gap to neural baselines is not the goal of this work; rather, these results establish that gauge-covariant variational inference is a viable computational substrate for language modeling (one that already outperforms the best classical statistical methods) supporting the explanatory thesis that transformers succeed because they approximate this underlying mathematical structure.

5.3.3 Emergent Semantic Structure

We analyze the learned belief embeddings $\mu \in \mathbb{R}^{90}$ and gauge frames $\phi \in \bigoplus_{a=1}^{9} \mathfrak{gl}(10) \cong \mathbb{R}^{9\times 100=900}$ for emergent categorical structure, using five linguistic categories (content word, digit, function word, letter, punctuation) across 500 representative tokens. Clustering metrics are computed in the full-dimensional spaces at every 10,000 training steps.

Belief Embeddings. The learned belief means μ develop statistically significant categorical structure over training. The inter-class to intra-class distance ratio grows from 1.00 at initialization to 1.06 at convergence, indicating that tokens of different linguistic categories become modestly more separated than tokens within the same category. Per-dimension ANOVA tests find 82% of the 90 embedding dimensions exhibit statistically significant category dependence (geometric mean $p = 1.4 \times 10^{-5}$, mean $F = 7.1$). The first three principal components capture 16.9% of the variance (50% at 20 components), and the Calinski-Harabasz index reaches 7.7. The silhouette score is essentially zero (+0.004), indicating that the categories approach the boundary of forming separable clusters in the full 90-dimensional space, a substantial improvement over the anti-clustering regime typical of random embeddings.

Gauge Frames. The gauge frame coefficients $\phi \in \mathbb{R}^{900}$ (9 heads × 100 dimensions per $\mathfrak{gl}(10)$ head) present a richer picture. PCA visualization reveals clear categorical separation in the top principal components: punctuation tokens separate sharply from content words along PC1, with function words, letters, and digits occupying distinct intermediate regions (Figure **??**). This low-dimensional separation is more visually striking than that of the belief embeddings, despite the first three principal components capturing only 6.8% of the total variance across 900 dimensions. Per-dimension ANOVA analysis confirms this structure:

1. Our matched-tokenization KN-5 implementation uses three discount parameters per order following Chen and Goodman (1998), trained on $\sim$119M BPE tokens ($\sim$212M unique n-grams across orders 1–5). The $\sim$18 PPL reduction from the word-level figure ($\sim$153–156 $\to$ 134.8) is consistent with the smaller BPE vocabulary making per-token prediction easier.

62% of 100 sampled gauge frame dimensions are statistically significant (geometric mean $p = 7.7 \times 10^{-4}$, mean $F = 4.7$). Yet the full-dimensional silhouette score is negative (-0.105), and the inter/intra class distance ratio is 1.055 (near unity), indicating that in the bulk of the 900-dimensional space, within-category distances exceed between-category distances. This apparent contradiction, with clear low-dimensional categorical separation coexisting with negative full-dimensional silhouette, reveals a dual structure: the gauge frames allocate a low-dimensional subspace to encoding token category (visible in the top PCs) while devoting the vast majority of their capacity to within-category diversification, equipping tokens of the same linguistic type with distinct transport operators $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ suited to their different contextual roles.

Quantitative Clustering Evidence. The clustering metrics for μ evolve progressively throughout training, strengthening monotonically rather than peaking early and declining. The Calinski-Harabasz index grows from 1.0 at initialization to 7.7 at convergence (peaking at 7.8 at step 50,000), and the ANOVA-significant fraction increases from 1% to 82% (peaking at 84% at step 50,000), indicating that category-driven organization strengthens as the model optimizes for language modeling. For ϕ: the metrics stabilize after step 20,000 (silhouette ≈ -0.105 to -0.131, Calinski-Harabasz ≈ 4.0–4.6, ANOVA fraction ≈ 49–66%), consistent with the gauge frames converging to a transport-optimized configuration and refining within that regime.

Implications. The contrast between μ and ϕ is theoretically informative. The belief means μ directly couple to the observation model $p(o \mid \mu)$ through the output projection, creating direct pressure to encode token identity; the progressively strengthening categorical structure (82% ANOVA significance, near-zero silhouette) reflects this coupling. The gauge frames ϕ parameterize transport operators $\Omega = \exp(\phi)$ that mediate inter-token communication, and their dual structure of categorical separation in a low-dimensional subspace, within-category diversification in the bulk reflects a dual functional requirement. The low-dimensional categorical component ensures that transport operators respect the coarse linguistic type of a token (e.g., punctuation tokens share a qualitatively different frame orientation from content words, visible in Figure **??**). The high-dimensional diversification ensures that tokens of the same category are equipped with distinct transport operators suited to their specific contextual roles, which drives the negative full-dimensional silhouette (-0.105). This suggests that the transport operators Ω_{ij} play a complementary role to the attention weights β_{ij}: while the attention distribution determines which tokens communicate, the gauge transport determines how information is geometrically transformed during that communication. This depends on both the categorical identity and the contextual role of each token.

Our results demonstrate that the gauge-theoretic framework scales to production language modeling tasks, with the theoretical constructs (μ, Σ, ϕ, KL-attention) functioning as intended during actual training at significantly lower embedding dimensions in exchange for larger gauge dimensions.

Representational Capacity vs. Output Capacity. The full representational capacity is $K + n_{\text{heads}} \cdot d_{\text{head}}^2$ dimensions per token (990 for $K = 90$ with 9 heads of GL(10)), yet only the K-dimensional belief means μ couple to the output. This is a structural consequence of gauge covariance: the observation model must depend only on gauge-covariant quantities, not on the internal frame parameters. The gauge frames influence predictions indirectly through the transport operators Ω_{ij} that mediate belief aggregation during VFE iterations. Increasing K (with multi-head $\mathrm{GL}(d_{\text{head}})$ to control matrix exponential cost) is the theoretically justified route to higher output capacity.

6 Discussion

6.1 Summary of Contributions

Standard attention as a degenerate limit. The BERT validation (Supplementary Appendix E) confirms quantitative agreement between gauge-aligned KL attention and standard dot-product attention across 105 diverse passages and five pretrained architectures (grand mean $\bar{r} = 0.804$ (hierarchical posterior mean 0.867, HDI $[0.808, 0.915]$) at the theory-predicted optimal temperature $\tau \approx 2\sqrt{d_k}$, with entropy matching $H(\beta)/H(\alpha) = 1.076$). We note that this high correlation follows in part from an algebraic identity under approximately constant key-norms; the non-trivial content lies in the quantitative predictions (temperature scaling, key-norm bias with Cohen's $d = 1.43$, per-head temperature dispersion explaining cross-model variation). We treat the BERT analysis as a consistency check; the $\mathrm{GL}(K)$ language model provides the stronger empirical test.

The unreduced framework as a language model. The GL(15) gauge VFE achieves test perplexity 71.6 on WikiText-103 (Table 4), without MLPs, pointwise activation functions, or learned attention projections. Embedding-matched and parameter-matched baselines (Table 4) clarify the nature of this result. At matched embedding dimension ($d_{\text{model}} = 90$), a standard transformer achieves only PPL 118.6 with 4.6M parameters, while the parameter-matched standard transformer (84.2M params, $d_{\text{model}} = 1{,}280$) achieves PPL 48.5 but requires 14× the embedding dimension. The gauge VFE (81.4M params) thus outperforms standard transformers at comparable embedding dimensions by 1.66×, although the comparison involves a 17.7× parameter overhead in the gauge VFE; against parameter-equalized ablation baselines at $d_{\text{model}} = 90$ the gauge VFE still outperforms by 1.87–1.91×. It underperforms when the standard transformer can instead invest its parameters in a much larger embedding space. This indicates that gauge transport and KL-divergence attention provide an effective mechanism for compensating for low embedding dimensionality through geometric structure, though the per-token covariance and gauge frame parameters incur a parameter overhead relative to shared weight matrices. That gauge-covariant variational inference alone, with no neural architecture beyond a linear output projection, produces a functioning language model supports the explanatory claim that transformers succeed in part because they approximate the underlying mathematical structure of distributed variational inference.

6.2 Architectural Implications for Standard Transformers

The framework provides theoretical explanations for empirical design choices in transformers. The framework derives a key-norm cancellation condition ($\|\mu_j\|^2 \approx C$) required for frame-independent inference (Section 4.2.1); layer normalization is one mechanism that achieves this condition, explaining why some form of norm regulation is empirically beneficial. The temperature scaling $1/\sqrt{d_k}$ is the natural normalization arising from dimensional concentration of measure. Multi-head attention corresponds to restricting the gauge algebra to a block-diagonal subalgebra $\bigoplus_a \mathfrak{gl}(d_{\text{head}})$, with each head implementing an independent irreducible representation (Section 4.9). The output projection W_O provides the only standard mechanism for post-attention cross-head information mixing.

These identifications also suggest alternatives to current practice. While layer normalization mitigates key-norm bias uniformly across all tokens, the gauge-theoretic perspective suggests learning optimal token norm profiles that balance representational capacity (large norms encode more information) against attention fidelity (low norms avoid bias). This trade-off is formally analogous to the rate-distortion problem: representational capacity corresponds to rate (bits stored per token), while attention bias corresponds to distortion (deviation from frame-independent inference). Soft norm constraints or adaptive normalization schedules that vary across layers or heads could implement position-dependent operating points along this trade-off curve.

Per-head covariance as a measurable architectural degree of freedom. In the Lie algebra decomposition $\mathfrak{gl}(d_{\text{head}}) = \mathfrak{sl}(d_{\text{head}}) \oplus \mathbb{R}$ (Section 4.4.3), the per-head temperature corresponds to the $\mathbb{R}$ (trace) component — the overall scaling degree of freedom within each head's gauge block. Standard transformers absorb this into the learned projections, entangling temperature with projection geometry. The gauge framework retains it as an explicit, learnable κ_a, and the empirical finding that per-head optimal temperatures span a wide range (RoBERTa: median = 25, std = 15.3) demonstrates that this covariance structure carries real information that a single global temperature discards. The heterogeneity of implicit per-head temperatures varies by a factor of 2.4× across architectures (CV = 0.331 for BERT-large to 0.787 for RoBERTa), indicating substantial variation in the degree to which different architectures exploit this degree of freedom.

6.3 The Flat Bundle Limit and an Open Question About Language

By Lemma 2, both the $\mathrm{GL}(K)$ gauge transformer and standard transformers operate in the flat bundle regime with trivially vanishing holonomy. For the gauge VFE, the transport $\Omega_{ij} = \exp(\phi_i)\exp(-\phi_j)$ satisfies the cocycle $\Omega_{ij}\Omega_{jk}\Omega_{ki} = I$ as an algebraic identity of the per-token-frame parameterization. For standard transformers, the learned projection $W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$ (Section 2.1.3) is position-independent and therefore trivially flat. In the most general discrete gauge theory on a token graph, the transport Ω_{ij} on each edge would be an independent group element with unconstrained holonomy around closed loops, giving path-dependent meaning transport in which relating token i to token k directly differs from relating them indirectly via an intermediate j. The framework's Regime II edge-relaxed cocycle (Lemma 2; companion paper (Dennis, 2025)), accessed in the codebase through the `connection.py` MLP mode, departs from the per-token-frame parameterization and admits non-trivial holonomy by construction; the present work analyzes only the cocycle-satisfying Regime I.

The structural matching between the two architectures in Regime I is parametric: both impose flat transport as a consequence of how Ω is parameterized, not as a learned property of language. Whether natural language's compositional-semantic content carries an independent functional pressure toward path-independent meaning transport—and, if so, whether that pressure is part of why transformer-style architectures succeed on language tasks—is an open question that the present work does not settle. Two specific gaps prevent making this a substantive empirical claim. First, the formal-semantics tradition treats compositionality as a homomorphism between syntactic derivations and meaning vectors (Frege; Montague; Partee); the categorical-distributional reading of Coecke, Sadrzadeh and Clark establishes that meanings of complete sentences live in a single space, which is codomain uniformity of the meaning map rather than path-independence on a discrete gauge bundle on tokens. The mapping between the homomorphism reading and the gauge-transport reading is a novel construction that we do not derive. Second, no operational criterion separates "compositional" from "pragmatic" token relations in a way that does not presuppose the architectural answer. The compositional-generalization benchmarks COGS (Kim and Linzen, 2020) and SCAN (Lake and Baroni, 2018) probe systematic generalization and report that vanilla transformers fail to generalize structurally without architectural augmentation, so this body of evidence sits on the side of empirical findings to interpret rather than on the side of established positive support for an architecture-language matching claim.

A future test that the Regime II construction makes possible is to train a non-flat gauge architecture (edge-relaxed cocycle, allowing $\Omega_{ij}\Omega_{jk}\Omega_{ki} \neq I$) and measure the learned holonomy H_{ijk} across token triples, comparing triples drawn from compositional-generalization probes against triples drawn from pragmatic-inference probes. The measurement has empirical content only after an independent operational labeling of compositional versus pragmatic triples is supplied; constructing that labeling is the prerequisite, not a downstream consequence, of the experiment, and bag-of-words counterexamples already show that flatness alone is insufficient to explain language-modeling success. We flag this empirical program as a direction for the companion paper (Dennis, 2025).

6.4 Conjecture: The Transformer as a Renormalization Group Fixed Point

The three-limit reduction of Section 4.2 can be understood as a *renormalization group (RG)-style coarse-graining flow* on the space of gauge-theoretic VFE models. Under an i.i.d. independence assumption on token-level perturbations, the central limit theorem delivers a conditional contraction result (Proposition 3 below). The Proposition is a direct application of the CLT to the coarse-graining definitions in Eqs. (130)–(132); its content lies in the i.i.d. premise, not in any gauge-theoretic structure. We separately conjecture—but do not claim to have validated on trained models—that the standard transformer occupies a stable infrared fixed point of this flow when the i.i.d. assumption is replaced by the empirical attention-graph statistics of trained models (Conjecture 4).

6.4.1 Setup and Scaling Dimensions.

Each model in the gauge VFE family is specified by three coupling constants measuring deviation from the transformer limit. We split the anisotropy into an *intrinsic* component (the deviation of the per-token covariance from isotropic) and an *emergent* component generated by within-cluster mean dispersion under coarse-graining:

$$g_1^{(\mathrm{orig})} = \|\Sigma_i - \sigma^2 I\|/\sigma^2 \qquad \text{(intrinsic anisotropy)}, \tag{130}$$

$$g_2 = \|\Omega_{ij} - \Omega\|/\|\Omega\| \qquad \text{(gauge variation)}, \tag{131}$$

$$g_3 = \|H_{ijk} - I\| \qquad \text{(linear holonomy)}, \tag{132}$$

where $H_{ijk} = \Omega_{ij}\Omega_{jk}\Omega_{ki}$. The transformer limit corresponds to the intrinsic-channel fixed point $g_1^{(\mathrm{orig}),*} = g_2^* = g_3^* = 0$. The total anisotropy is $g_1^{(\mathrm{tot})} = g_1^{(\mathrm{orig})} + g_1^{(\mathrm{emer})}$ with $g_1^{(\mathrm{emer})} = \|\mathrm{Var}_A(\mu)\|/\sigma^2$; only the intrinsic channel flows to zero under R_n, while the emergent component is generated by the coarse-graining map itself.

A coarse-graining transformation R_n groups n tokens into a meta-agent with beliefs:

$$\mu_A = \frac{1}{|A|}\sum_{i\in A}\mu_i, \qquad \Sigma_A = \frac{1}{|A|}\sum_{i\in A}\Sigma_i + \mathrm{Var}_A(\mu). \tag{133}$$

Under the assumption that perturbations Δ_i and transport variations $\delta\Omega_{ij}$ are approximately independent across tokens—an assumption that holds by construction but requires verification in trained models—the central limit theorem yields:

$$\big(g_1^{(\mathrm{orig})}\big)' = n^{-1/2} g_1^{(\mathrm{orig})}, \qquad g_2' = n^{-1} g_2, \qquad g_3' = n^{-1} g_3, \tag{134}$$

with all scaling dimensions negative ($y_1 = -\frac{1}{2}$, $y_2 = -1$, $y_3 = -1$ for the linear holonomy norm, and $y_3^{(\mathrm{action})} = -2$ for the squared holonomy $\|H - I\|^2$), classifying every operator as *irrelevant.*

Proposition 3 (Conditional CLT contraction of the gauge VFE family) *Let $\mathcal{T}(K, N, g_1^{(\mathrm{orig})}, g_2, g_3)$ denote the family of gauge VFE models, and assume that the perturbations Δ_i and transport variations $\delta\Omega_{ij}$ are independent and identically distributed across tokens. Then:*

(i) ***Intrinsic fixed point.*** *$g_1^{(\mathrm{orig}),*} = g_2^* = g_3^* = 0$ is a fixed point of R_n in the intrinsic-channel coordinates, for all $n \geq 2$ and $K \geq 1$.*

(ii) ***Stability under independence.*** *All scaling dimensions are negative ($y_1 = -\frac{1}{2}$, $y_2 = y_3 = -1$). Under the i.i.d. assumption, the intrinsic-channel origin meets the textbook definition of an infrared-stable fixed point (Cardy, 1996; Goldenfeld, 1992).*

(iii) ***Emergent anisotropy.*** *R_n generates total-channel anisotropy via $\mathrm{Var}_A(\mu)$ even at $g_1^{(\mathrm{orig})} = 0$, so the total $g_1^{(\mathrm{tot})}$ does not flow to zero. The augmented (μ_A, Σ_A) family contains this term, analogous to the effective-action augmented class in Wilsonian RG (Cardy, 1996).*

The three clauses are textbook CLT applied to the definitions in Eqs. (130)–(132); the gauge content of the framework does not enter the calculation. The empirical content—the application of these scaling dimensions to the learned representations of trained transformers, where the i.i.d. assumption is not known to hold—is logically separate from Proposition 3 and remains an open conjecture.

Conjecture 4 (Universality of the transformer limit on trained representations) *Under the coarse-graining map R_n applied to learned representations of trained transformer attention graphs:*

(a) ***Universality.*** *All finite-coupling gauge VFE models flow (in the intrinsic channel) to the transformer limit under repeated coarse-graining; transformers absorb the emergent anisotropy floor into W_Q, W_K while the gauge VFE tracks it in Σ_i.*

(b) ***Efficiency gap (heuristic).*** *The absorbed degrees of freedom scale as $O(K^2)$. We conjecture an $O(\sqrt{K})$-class sample-efficiency advantage for the gauge VFE relative to a parameter-matched standard transformer; the exponent is heuristic and is not derived from a quantitative sample-complexity bound (VC, PAC-Bayes, or Chinchilla-style power law) in this work. The prediction should be read as a class label for the expected scaling, to be falsified or refined by the experiments listed under Future Directions below.*

Scope and status. The scaling dimensions in Eq. (134) are textbook CLT averaging of independent perturbations and establish Proposition 3; this content does not itself require the gauge framework. The non-trivial empirical content—Conjecture 4—requires that the i.i.d. assumption hold approximately on the attention graphs of trained transformers, which is the open question. We have numerically reproduced the CLT exponents to three significant figures under direct averaging of synthetic i.i.d. perturbations; this is a sanity check that the analytic prediction matches the CLT on synthetic data, not validation of the framework on trained models. When the same predictions are tested on attention-graph coarse-graining with spectral clustering, the measured exponents deviate: $y_2 \approx -0.6$ and $y_3 \approx +0.2$ versus the linear-norm predictions -1 and -1 (the action-norm prediction $y_3^{(\text{action})} = -2$ applies to the squared holonomy $\|H - I\|^2$, not the linear form measured on the graph). The interpretation of these deviations is open. Two non-exclusive hypotheses are consistent with the data. (H1) Finite-size bias in the spectral coarse-graining map produces a correction of the form $y_3^{\text{meas}}(L) = -1 + cL^{-\omega} + \dots$ (Cardy, 1996); the measured shift would require the correction term to dominate the asymptotic at the system sizes tested, a regime standardly indicative of crossover behavior rather than small-L corrections to an infrared-stable fixed point. (H2) Genuine cross-token correlation in trained transformer attention produces an $O(1)$ floor on $\mathrm{Var}(\bar{X}_n)$ that breaks the CLT $n^{-1/2}$ rate; this would classify g_3 as a relevant operator in the standard Wilsonian sense (Cardy, 1996; Goldenfeld, 1992), falsifying part (a) of Conjecture 4. The data do not discriminate H1 from H2. Validation on trained models—with quantitative finite-size scaling across multiple system sizes for H1 and direct correlation measurements for H2—is needed before either form of the conjecture can be considered empirically supported. We defer this analysis to a companion paper where it can be developed with trained-model evidence.

The ordering of the three limits in Section 4.2 was determined algebraically by the structure of the Q-K-V derivation and is consistent with—though not derived from—the RG hierarchy implied by Proposition 3: gauge variation and linear holonomy contract under R_n as n^{-1} ($y_2 = y_3 = -1$) while anisotropy contracts more slowly as $n^{-1/2}$ ($y_1 = -\frac{1}{2}$), matching Limit 3 $\to$ Limit 2 $\to$ Limit 1 once one accounts for the action-norm intensification $y_3^{(\text{action})} = -2$. The emergent anisotropy from Proposition 3(iii) provides a geometric interpretation of LayerNorm as a projection back toward the isotropic submanifold at each layer.

6.5 Limitations

The BERT validation (Supplementary Appendix E) tested the flat bundle limit where all frames are globally aligned and the connection is trivial. This is the appropriate limit for comparison with standard transformers, which lack explicit gauge structure, but it does not probe the regime where the framework is most distinctive. The Regime II edge-relaxed cocycle of Lemma 2 (above) and the companion paper Dennis (2025) acquires direct dynamical content under bidirectional attention, where closed loops in the data-flow graph make the Wilson observable a property of the message-passing pattern rather than a purely formal index-space invariant. Extending to autoregressive architectures (GPT, LLaMA), non-64 head dimensions, larger model scales, non-English languages, and to non-trivial-holonomy training configurations on bidirectional attention all remain future work.

The gauge VFE language model uses 81.4M parameters yet achieves worse perplexity than a parameter-matched standard transformer (PPL 71.6 vs. 48.5 at 84.2M). The parameter overhead arises from per-token covariance matrices Σ_i and gauge frame parameters ϕ_i. At matched embedding dimension ($d_{\text{model}} = 90$), a standard transformer achieves only PPL 118.6 (4.6M params), and the gauge VFE outperforms it by 1.66×. This embedding-matched comparison consumes 17.7× as many parameters in the gauge VFE as in the standard transformer; the param-equalized ablation row (Std. wider MLP, 9.2M params, PPL 145.8) and the attention-only baseline (9.1M params, PPL 142.8) are the more honest controls and the gauge VFE still outperforms them by 1.91× and 1.87× respectively. Gauge structure thus compensates for low embedding dimensionality at the cost of higher total parameter count. Whether architectural modifications (parameter sharing across tokens, low-rank covariance parameterizations) can close this efficiency gap while preserving the gauge-theoretic structure is an open question.

The gauge VFE is computationally more expensive than standard attention, primarily due to matrix exponentials and per-pair KL divergences. We emphasize that this is an explanatory theory: the goal is not to produce state-of-the-art language models but to provide a mathematical foundation for understanding why attention-based architectures work. Just as statistical mechanics explains thermodynamics without replacing steam engines, the gauge-theoretic framework explains transformers without replacing them.

Scope qualifications on the "no neural networks" claim. Two qualifications attach to the architectural minimality stated in the abstract and §5.3. First, the implementation retains a single linear projection from the K-dimensional belief means to vocabulary logits; this projection is subsumed by the prior-bank decode logits $= -D_{\mathrm{KL}}(q\|\pi_v)/\tau$ when the prior bank is enabled, and the gauge VFE results in Table 4 are reported in the prior-bank configuration so that no learned $W_Q/W_K/W_V$/MLP/activation function appears in the model. Second, the codebase provides an opt-in non-flat transport variant (`connection.py` MLP mode) for research on curved bundles; this variant is not enabled in the experiments reported here. Both are documented in the project's `CLAUDE.md`.

Diagonal-covariance RoPE. When the diagonal-covariance approximation $\Sigma_i = \mathrm{diag}(\sigma_i^2)$ is combined with RoPE under the `rope_full_gauge='off'` configuration, the rotary phase $R(\theta_i)$ acts on μ_i but not on σ_i, so the downstream Mahalanobis norm $\mu^\top \mathrm{diag}(\sigma^{-2})\mu$ divides a rotated μ by an un-rotated σ. This breaks strict $\mathrm{SE}(K)$ covariance on the diagonal path; full-gauge alternatives that transport σ are available in the codebase but are not used in the headline results, where the diagonal approximation is selected for $\mathrm{GL}(K)$ scaling.

Seed disclosure. The headline GL(15) result in Table 4 is from a single seed; the GL(10) row reports the mean over n=2 seeds (std 1.05). Multi-seed runs at the K=90 scale exceed the GPU budget of the present work; companion-paper experiments will report $n \geq 5$ seed runs.

6.6 Future Directions

6.6.1 Cross-Head Gauge Mixing.

Standard multi-head attention restricts the gauge algebra to $\bigoplus_a \mathfrak{gl}(d_{\text{head}}) \subset \mathfrak{gl}(d_k)$, projecting out the majority of generators (87.5% for typical architectures). Restoring the off-diagonal generators (Section 4.4.3) would make attention scores themselves cross-head aware, so that head a's decision about what to attend to depends on the content of head b—qualitatively different from the post-attention mixing provided by W_O. For equivariant architectures with SO(3) irrep structure, these off-diagonal blocks correspond to Clebsch-Gordan couplings between angular momentum channels. Whether the computational cost of richer transport can be justified by improved expressiveness, and whether structured sparsity (coupling only selected head pairs) achieves most of the benefit, is a central experimental question.

6.6.2 Non-Uniform Head Dimensions.

Standard transformers use uniform head dimensions ($d_{\text{head}} = d/H$) rather than the geometrically natural dimensions dictated by irrep structure (e.g., $2\ell + 1$ for SO(3) irreps ℓ). Whether transformers with non-uniform head dimensions matching irrep sizes achieve better performance or sample efficiency on tasks with known symmetries is an open question with practical implications for equivariant architectures.

6.6.3 Gauge Symmetry as Inductive Bias.

By leveraging gauge theory coupled with information geometry, we can introduce geometric features that constrain model behavior according to symmetry principles, analogous to how convolutional neural networks impose translation equivariance or graph neural networks respect permutation invariance (Finzi et al., 2020; Weiler et al., 2018; Kondor and Trivedi, 2018). The GL(K) invariance theorem allows us to interpret standard transformer attention as a degenerate case of gauge-covariant KL-based attention, operating in the constant-gauge, isotropic, flat-bundle limit. Our framework shifts toward a landscape where specific subgroups (SO(N), SU(N), Lorentz group, etc.) can be imposed as inductive biases, potentially enabling transformers to learn from fewer examples in domains with known physical structure.

6.6.4 Fields of Transformers.

Standard attention transformers are a 0-dimensional gauge theory in which all geometric structure is absorbed into learned parameters. Lifting this restriction yields fields of transformers over an N-dimensional base manifold, coupled by gauge connections $A_\mu(c)$ that define how information is parallel-transported between transformers at neighboring points. Unlike the flat bundle of the token graph, this transport is generally path-dependent: the holonomy $\mathcal{P}\exp\left(-\oint A_\mu \mathrm{d}x^\mu\right)$ around a closed loop encodes curvature. Such architectures would be natural for data with intrinsic spatial or temporal extent—video, multi-agent robotics, or scientific data on manifolds—where geometric relationships between processing locations carry information that flat architectures discard.

6.6.5 Multi-Timescale Dynamics and Backpropagation-Free Learning.

The full variational free energy (Section 3.4) includes both the fast belief subsystem $\mathcal{F}_{\text{fast}}$ and the slow model subsystem $\mathcal{F}_{\text{slow}}$, with timescale separation $\eta_s \ll \eta_q$. Standard transformer training operates entirely within this separation: the forward pass corresponds to fast-variable dynamics, while gradient descent on weights corresponds to slow-variable learning. Current transformers are thus the adiabatic limit where slow variables are frozen during inference and updated only between forward passes via backpropagation.

The full VFE prescribes coupled dynamics at both timescales simultaneously, with the model-model alignment term $\gamma_{ij} D_{\text{KL}}(s_i \| \tilde{\Omega}_{ij} s_j)$ driving agents toward consensus on generative models as well as beliefs. Implementing both subsystems during inference would unify in-context learning, fine-tuning, and continual adaptation as limits of the same variational dynamics. A further extension toward backpropagation-free learning—updating all variational parameters via local VFE gradients combined with a delta rule for the output projection—would test whether the variational free energy principle provides a viable alternative learning algorithm. Both extensions remain theoretical and their practical feasibility is unestablished.

6.6.6 O(K) Gauge Transport and the Exact Isotropic Limit.

The isotropic covariance limit (Section 4.2) requires $\Omega_{ij} \in \mathrm{O}(K)$ for both the vanishing of the geometric bias $S(\Omega_{ij}) = 0$ and the preservation of isotropic covariance under transport. While the matrix exponential parameterization restricts to the identity component $\mathrm{SO}(K)$, the extension to full $\mathrm{O}(K)$ via per-agent diagonal reflections $R_i = \mathrm{diag}(s_i)$ (Eq. 56) introduces a controlled discrete degree of freedom without disrupting the continuous gauge dynamics. The sign vectors $s_i \in \{-1, +1\}^K$ are learnable via the straight-through estimator, and the factorization $\mu_i \mapsto s_i \odot \mu_i$ (Eq. 57) means that reflections are applied at the embedding level, leaving all downstream attention and VFE computations unchanged. Whether this additional discrete gauge freedom improves language modeling performance in the isotropic regime, and whether the learned sign vectors develop interpretable structure (e.g., clustering by semantic or syntactic category), are experimental questions that probe the role of the disconnected component of $\mathrm{O}(K)$ in natural language representation.

6.6.7 Renormalization Group Universality.

Conjecture 4 generates several falsifiable predictions that can be tested on trained models:

1. *Sample efficiency grows with K.* Train gauge VFE and standard transformers at matched parameter counts for $K \in \{8, 16, 32, 64\}$. The ratio of tokens needed to reach a target perplexity should grow as $R(K) \propto \sqrt{K}$.

2. *Same scaling exponent.* Both architectures should exhibit the same data-scaling exponent β in $\mathrm{PPL}(D) \sim D^{-\beta}$, differing only in the prefactor (same universality class, different efficiency frontiers).

3. *Coarse-graining exponents on trained models.* Extract attention matrices from trained checkpoints, apply spectral coarse-graining, and measure how g_1, g_2, g_3 scale with cluster size. Unlike the synthetic CLT validation (which tests pure mathematics),

this would test whether the independence assumptions underlying Eq. (134) hold for learned representations.

4. *LayerNorm as isotropy projector.* Measure the anisotropy of representation covariance before and after LayerNorm at each layer. The prediction is that LayerNorm systematically reduces anisotropy, consistent with its role as a projection toward the isotropic submanifold.

5. *Compute crossover.* At large scale, the transformer should match or exceed gauge VFE performance at $C > C^* = O(K^2 V)$ FLOPs.

7 Conclusion

We have presented a gauge-theoretic framework in which attention emerges as the communication mechanism of distributed variational inference. Modeling each token as a Gaussian agent on a statistical fiber bundle, we derived the generalized attention weight $\beta_{ij} = \mathrm{softmax}(-D_{\mathrm{KL}}[q_i \| \Omega_{ij} q_j]/\tau)$ from variational free energy minimization, and showed that standard transformer attention is recovered as a degenerate limit under successive restrictions (isotropic covariances, flat gauge connection).

The $\mathrm{GL}(K)$ invariance theorem (Theorem 1) permits interpreting standard transformer attention as a degenerate case of this gauge structure: in the isotropic, flat-bundle limit, the bilinear form $W_Q W_K^\top$ can be identified with a constant gauge transport via $W_Q W_K^\top = \sigma^{-2} \Omega^{-\top}$. The unreduced $\mathrm{GL}(K)$ gauge VFE achieves test perplexity 71.6 on WikiText-103 without MLPs, learned attention projections, or pointwise activation functions, surpassing classical baselines and outperforming standard transformers at matched embedding dimension by 1.66×. The model learns structured, head-specialized attention patterns through gauge transport and KL-divergence dynamics alone (Section 5.3), with learned gauge frames developing a dual structure of categorical separation in a low-dimensional subspace and within-category diversification in the bulk. A separate consistency check on frozen BERT confirms quantitative agreement between KL-based and dot-product attention in the degenerate limit (Supplementary Appendix E).

The framework further reveals a geometric interpretation of learning itself: the untrained network corresponds to a gauge-symmetric vacuum state whose degeneracy is broken by observations, with training acting as explicit symmetry breaking. We anticipate this perspective—that attention is gauge-covariant inference, and learning is symmetry breaking—will find application not only in machine learning but in understanding biological cognition, collective intelligence, and other domains where distributed agents must communicate under geometric constraints.

7.1 Code Availability

The complete simulation suite that implements the gauge-theoretic variational inference framework described in this work is publicly available at https://github.com/cdenn016/epistemic-geometry.

All experiments reported in the results section can be reproduced using the provided configuration files and random number generator seeds documented in the repository. The codebase requires Python 3.9+ with NumPy, PyTorch, SciPy, and Joblib dependencies as specified in `requirements.txt`.

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Supplementary Material. All appendices are provided in the Supplementary Material: Appendices A through D cover differential geometry, covariance dynamics, gauge frame gradients, and numerical methods; Appendices E through G present the BERT validation, RG universality conjecture, and symmetry breaking simulations; Appendix H proves the conditional uniqueness of the forward KL divergence via variational duality.

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Table 1: Summary of notation. Symbols are grouped by category: base geometry, agents and beliefs, gauge structure, variational quantities, and Transformer correspondences.

Symbol	**Domain**	**Description**
Base geometry and fibers		
$\mathcal{C}$	Manifold	Base manifold (spatial domain)
$\mathcal{B}_q$ ($\mathcal{E}_q$)	$\mathbb{R}^{K_q}$	Belief fiber
$\mathcal{B}_p$ ($\mathcal{E}_p$)	$\mathbb{R}^{K_p}$	Model fiber
K (K_q)	$\mathbb{Z}^+$	Belief fiber (embedding) dimension
N	$\mathbb{Z}^+$	Sequence length (number of agents)
Agents and beliefs		
i, j	$\{1, \ldots, N\}$	Agent (token) indices
$q_i = \mathcal{N}(\mu_{q,i}, \Sigma_{q,i})$	$\mathcal{P}(\mathbb{R}^K)$	Belief distribution of agent i (context-dependent)
$p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i})$	$\mathcal{P}(\mathbb{R}^K)$	Prior distribution (from token embedding)
s_i, r_i	$\mathcal{P}(\mathbb{R}^{K_p})$	Model-fiber belief and prior (generative model)
Gauge geometry		
$G = \mathrm{GL}(K)$	Lie group	Structure group (general linear group)
$\mathfrak{g} = \mathfrak{gl}(K)$	Lie algebra	Lie algebra of G
$\phi_i \in \mathfrak{g}$	$\mathbb{R}^{K \times K}$	Gauge frame parameters for agent i (belief fiber)
$\tilde{\phi}_i \in \mathfrak{g}$	$\mathbb{R}^{K_p \times K_p}$	Gauge frame parameters for agent i (model fiber)
$\Omega_{ij} = \exp(\phi_i) \exp(-\phi_j)$	$\mathrm{GL}(K)$	Belief-fiber transport from frame j to frame i
$\tilde{\Omega}_{ij}$	$\mathrm{GL}(K_p)$	Model-fiber transport from frame j to frame i
$\Phi_i, \tilde{\Phi}_i$	$\mathbb{R}^{K_p \times K_q}$, $\mathbb{R}^{K_q \times K_p}$	Cross-fiber morphisms (belief $\leftrightarrow$ model); introduced in Supplementary Appendix A for cross-reference with the companion paper (Dennis, 2025); not used in any equation of the present paper (see Section 3)
$\rho : G \to \mathrm{GL}(V)$		Representation of G on the fiber V
Variational quantities		
$\mathcal{F}$	$\mathbb{R}$	Gauge-covariant variational free energy
$D_{\mathrm{KL}}(q \Vert p)$	$\mathbb{R}_{\geq 0}$	Forward Kullback-Leibler divergence
β_{ij}	$[0, 1]$, $\sum_j \beta_{ij} = 1$	Belief attention weight (source-selection posterior)
γ_{ij}	$[0, 1]$	Model-fiber coupling weight
α (α_i)	$\mathbb{R}^+$	Prior precision; state-dependent variant (Sec. 3.7)
κ (τ)	$\mathbb{R}^+$	Attention temperature; $\kappa_a \propto \sigma_a^2$ in isotropic limit (Sec. 4.4)
λ	$\mathbb{R}^+$	Belief alignment coupling strength
π_j	$[0, 1]$, $\sum_j \pi_j = 1$	Attention prior (encodes masking, positional bias)
η_q, η_s	$\mathbb{R}^+$	Fast (belief) and slow (model) learning rates; $\eta_s \ll \eta_q$ in the present work
$\hat{\alpha}$	$\mathbb{R}^+$	Trained-from-loss scalar version of α used in Algorithm 1
Transformer correspondences (flat-bundle, isotropic limit)		
Q_i, K_j	$\mathbb{R}^{d_k}$	Query and key vectors ($\to \mu_i, \mu_j$ under reduction)
V_j	$\mathbb{R}^{d_v}$	Value vector
W_Q, W_K	$\mathbb{R}^{d_k \times d}$	Projection matrices ($\to$ gauge transformations)
d_k	$\mathbb{Z}^+$	Head dimension ($= K$ per irrep block)

Property	**Full Gauge Theory**	**Standard Transformer**
Gauge group G	$\mathrm{GL}(K)$ or subgroup	$\mathrm{GL}(d_k)$ (implicit)
Gauge transformation	$\Omega_{ij} = e^{\phi_i} e^{-\phi_j} \in \mathrm{GL}^+(K)$	$W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$ (learned, constant)
Frame field $\phi_i(c)$	Varies per agent/position	Absorbed into W_Q, W_K
Transport structure	Agent-pair dependent	Uniform across all pairs
Connection A_μ	$-\partial_\mu \phi_i + \mathcal{O}(\phi^2)$ (0D: absent)	0 (flat connection)
Curvature / Holonomy	Trivial ($\Omega_{ij}\Omega_{jk}\Omega_{ki} = I$; Sec. 6.3)	0 (trivial holonomy)
KL coupling	$D_{\mathrm{KL}}[q_i \| \Omega_{ij} q_j]$	$D_{\mathrm{KL}}[q_i \| \Omega q_j]$ with constant Ω
What is learned	Frame fields ϕ_i	Global gauge W_Q, W_K

Table 2: Comparison between full gauge-theoretic formulation and standard transformers.

VFE / Gauge Theory	**Limit / Mechanism**	**Transformer Architecture**	**Status**
Attention Mechanism			
$\beta_{ij} = \text{softmax}(-D_{\text{KL}}/\tau)$	Exact (Sec. 3.4)	$\text{softmax}(QK^\top/\sqrt{d_k})$	D
$\Omega_{ij} = e^{\phi_i}e^{-\phi_j} \in \text{GL}^+(K)$	Constant Ω (Limit 2)	$W_Q W_K^\top = \sigma^{-2}\Omega^{-\top}$	D$^\sharp$
Forward KL uniqueness	Thm. (Supp. App. H)	Why dot-product attention	D
$\hat{\mu}_i = \sum_j \beta_{ij}\Omega_{ij}\mu_j$	Ω absorbed into W_V	$\sum_j \alpha_{ij}V_j$ (value agg.)	D
$\text{GL}(d_{\text{head}})^H \subset \text{GL}(d_k)$	Irrep decomposition	Multi-head head-space kernel	D$^\sharp$
Rectangular U_Q^a, U_K^a, U_V^a (subspace embeddings)	Not derived (structural)	Multi-head subspace selection	S
Positional Structure			
$\pi_k = 0$ for $k > i$	Prior constraint	Causal mask	D
$\log \pi_k = -m\lvert i-k\rvert$	Distance-decaying prior	ALiBi	D
Learnable $\pi_k(\lvert i-k\rvert)$	Learnable prior	Relative pos. bias (T5)	D
$\phi(pos) \in \text{SO}(2)^{d_k/2}$	Pos.-dep. gauge frame	RoPE	D
$\pi_k = 0$ for $\lvert i-k\rvert > w$	Window prior	Sliding window attention	D
Normalization and Scaling			
$\tau = \sqrt{d_k}$ (dot-product form; σ^{-2} absorbed)	Concentration of measure	$1/\sqrt{d_k}$ scaling	D
$\lVert\mu_j\rVert^2 \approx C$ condition	Gauge-geometric necessity	Layer normalization	S
$\mathfrak{gl}(K) = \mathfrak{sl}(K) \oplus \mathbb{R}$	Remove trace component	Mean centering in LN	S
Training and Optimization			
$\mathcal{F}[\{q_i\}]$	Deterministic limit	Loss $\mathcal{L}(\theta)$	D
$-\mathbb{E}_q[\log p(o\vert k)]$	Point evaluation limit	Cross-entropy / MSE loss	D
$\alpha D_{\text{KL}}(q_i\Vert p_i)$	$\alpha = $ const, deterministic	L2 weight decay	D
$\alpha_i^* = c_0/(b_0 + D_{\text{KL}})$	State-dependent precision	Adaptive weight decay	D
$d\mu/dt = -\eta\Sigma\nabla\mathcal{F}$	Euler discretization	Gradient descent (backprop)	D
Chain rule through layers	Layered composition	Backpropagation	S
Vacuum (no observations)	Gauge-symmetric minimum	Untrained network	S
Observations break symmetry	Explicit breaking	Training / learning	S
Architecture			
$\mu^{(\ell+1)} = \mu^{(\ell)} - \tilde{\eta}\Sigma\nabla\mathcal{F}$	Natural gradient flow	Residual connection	S
$e \cdot \exp(-\lVert e\rVert^2/\tau)/Z$ (Eq. 120)	Boltzmann GLU gate	GELU/SiLU activation	I
VFE iterations (T inner, L outer)	Shared / fresh weights	Layers / depth; DEQ at $L = 1$	D
Prior bank $\{(\mu_p^c, \Sigma_p^c, \phi_c)\}$	Deterministic limit	Embedding matrix W_{embed}	S
$p(o\vert\mu) = \text{Cat}(\text{softmax}(W\mu))$	Observation model	Output proj. + softmax	D
Fast subsystem $\mathcal{F}_{\text{fast}}$	Frozen models	Forward pass (inference)	S
Slow subsystem $\mathcal{F}_{\text{slow}}$	Between-pass updates	Weight training	S

Table 3: Comprehensive correspondence between the gauge-theoretic VFE framework and transformer architecture. **Status column:** D = derived from the variational principle (mathematically exact or follows from explicit limits); D$^\sharp$ = derived up to a non-uniqueness equivalence class (the bilinear form $W_Q W_K^\top$ can be thin-SVD-lifted to a full-rank invertible per-head map $M_h \in \text{GL}(d_{\text{head}})$, with the lift fixed only up to right multiplication by elements of the isotropy group); S = *structural* correspondence (the framework explains the component's role but does not uniquely predict its specific form, as in any Euler discretization yielding $\mu^{(\ell+1)} = \mu^{(\ell)} + \Delta\mu^{(\ell)}$); I = *interpretive* correspondence (the framework and the architectural element belong to a common family but are not functionally identical; e.g., the Boltzmann gate of Eq. 120 uses the Gaussian PDF while GELU uses the Gaussian CDF).

Model	Layers	d_{model}	Params	Val PPL	Test PPL
Gauge VFE models (no MLPs, no $W_Q/W_K/W_V$, no activation functions)					
Gauge VFE (GL(15), $K = 90$, 6 heads)	1	90	81.4M	69.3	71.6
Gauge VFE (GL(10), $K = 90$, 9 heads)	1	90	58.8M	75.5	76.4*
Gauge VFE (GL(10), $K = 10$)	1	10	—	—	—
Standard transformer baselines (dot-product attention + MLP)					
Std. transformer (param-matched to $K = 90$)	1	1,280	84.2M	33.8	48.5
Std. transformer (embed-matched to $K = 90$)	1	90	4.6M	97.2	118.6
Std. transformer (param-matched to $K = 10$)	1	116	6.0M	109.9	119.3
Std. transformer (embed-matched to $K = 10$)	1	10	0.5M	535.1	548.0
Ablation baselines ($d_{model} = 90$, 9 heads, 15k steps†)					
Std. transformer + RoPE	1	90	9.1M	136.6	138.6
Std. attn-only (no MLP)	1	90	9.1M	137.4	142.8
Std. param-equalized (wider MLP)	1	90	9.2M	143.0	145.8
Non-neural baseline					
Modified KN-5 (matched BPE)	—	—	—	135.9	134.8

Table 4: Language modeling on WikiText-103 (vocab 50,257; random-chance PPL $\approx$ 50,000). Gauge VFE models use KL-divergence attention with no MLPs, pointwise activation functions, or learned attention projections (W_Q, W_K, W_V); only a linear output projection to vocabulary logits is retained. The $K = 10$ VFE results are forthcoming. Each gauge VFE configuration is compared against two standard transformer baselines: one *embedding-matched* (same d_{model}) and one *parameter-matched* (comparable total parameter count). Ablation baselines isolate the effect of RoPE, MLP removal, and parameter reallocation at $d_{\text{model}} = 90$. The modified Kneser-Ney 5-gram baseline (Chen and Goodman, 1998) is trained on the same data with matched GPT-2 BPE tokenization (50,257 vocab). All neural models use a single layer and sequence length 128. †Ablation baselines trained for 15,000 gradient updates (batch size 64) vs. 60,000 for gauge VFE and the embed/param-matched baselines (batch size 16); see text for discussion of training regime differences. *Mean over n=2 seeds with std 1.05, reproducible from `publication_outputs/scaling_analysis/aggregated_K_sweep.csv`.